The CSP Dichotomy Theorem

Abstract

The Constraint Satisfaction Problem over a finite constraint language is solvable in polynomial time if the language admits a weak near-unanimity polymorphism, and is NP-complete otherwise. This document is an exposition of the proof, following Zhuk's simplified argument [14] and supplying the prerequisites it imports.

It states what is being proved and separates the parts that mention a model of computation from the parts that do not (§1); fixes the vocabulary (§§2–5); states the theory of strong subuniverses and proves it (§§6–7); states the main induction and the three theorems the algorithm consumes (§8); and states the algorithm and the hard half (§§9–10).

§11 says, for every statement here, whether it is proved, imported with an exact citation, outlined, stated only, or open, and lists the six items that block the rest. It should be read before any part of this document is relied on: the theory of §7 is largely complete, and §8 is not.

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Introduction

The dichotomy conjecture of Feder and Vardi [7] was settled independently by Bulatov [6] and Zhuk [13] in 2017. Both proofs are long. Zhuk's, in the form published in 2020, is seventy-eight journal pages, and its central induction alternates between global and local information in a way its author later described as forcing "a very complicated induction to prove most of the claims".

In 2024 Zhuk published a simplified proof [14], and it is the argument followed here. The simplification is a redesign of the theory of strong subalgebras so that *every reduction is either strong or global*: for each domain D_x one can build a chain $D_x \supsetneq D_x^{(1)} \supsetneq \cdots \supsetneq \{a\}$ in which every step is either a strong subset or the intersection with a block of an equivalence relation with very strong properties. The types of the intermediate sets need not be tracked — only that such a chain exists, written $C \lhd\lhd\lhd B$. That is what makes the argument tractable, and it is what this exposition is organised around.

What is written out, and what is not. This is a corrected exposition and not a survey: where the argument as usually presented is wrong, incomplete, or ambiguous, the corrected statement is given here and proved, without marginal notes about what other presentations have. A companion audit document records those departures, with the source's text quoted and the reason for each change, so that the corrections can be checked rather than taken on trust.

It is also not complete. §11 is where the incompleteness is recorded: one statement is *open*, several are proved only in outline, and the spine of §8 — the lemma that produces bridges from an instance — is stated without proof. Read that table before relying on anything here.

Route and sources. [14] is the spine. Several prerequisites are imported rather than proved, chiefly the Absorption Theorem in Brady's form [2] and the strong-subalgebra calculus of [15]; §11.4 lists all of them with their exact hypotheses. [13] supplies the algorithm itself, which [14] does not contain. Section 4 of [14], on XY-symmetric operations, is an independent second result and is not needed for the dichotomy; it is omitted.

Caveats. Marked environments throughout record places where the informal argument has latitude that a careful one does not: a suppressed side condition, a quantifier whose scope is ambiguous, an object asserted to exist without an argument, or a step whose well-foundedness is left to the reader. They are part of the mathematics rather than commentary on it — several of the corrections below were found by writing one down. A sample of what they turn up: that $\mathbf{Z}_p$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Caveat 3.3); that σ^* need not be a congruence (Caveat 3.24); that images do not commute with intersection, and that one proof needs them to (Caveat 3.17); that "we cannot weaken forever" needs a measure that is not the obvious one, and that the obvious one points the wrong way (Caveat 5.16); and that the main induction quantifies over the instance *inside* (Caveat 8.2).

1 What is being proved

1.1 The theorem

Fix a finite set A and a finite set Γ of relations on A, the *constraint language*. An instance of $\mathrm{CSP}(\Gamma)$ is a conjunction $R_1(\bar{x}_1) \wedge \cdots \wedge R_s(\bar{x}_s)$ with each $R_i \in \Gamma$; the question is whether it is satisfiable.

Definition 1.1 (Core, rigid core)**.** Γ is a *core* if every endomorphism of Γ — every unary polymorphism — is an automorphism, and a *rigid core* if the identity is its only endomorphism. Write Γ^{rig} for Γ together with the unary relation $\{a\}$ for every $a \in A$.

The theorem this document is about is the rigid-core one.

Theorem 1.2 (Dichotomy for a rigid core; Bulatov, Zhuk)**.** *Let Γ be a finite constraint language on a finite set A containing the singleton relation $\{a\}$ for every $a \in A$. If some weak near-unanimity operation preserves Γ then $\mathrm{CSP}(\Gamma)$ is solvable in polynomial time; otherwise $\mathrm{CSP}(\Gamma)$ is NP-complete.*

The hypothesis is exactly what makes $\mathrm{Pol}(\Gamma)$ idempotent, since a polymorphism preserving $\{a\}$ satisfies $f(a, \ldots, a) = a$; so Convention 2.1 applies to it, and §10 may use the singletons as relations. It also makes Γ a rigid core, and conversely: by [2, Thm. 1.4.6], Γ is a rigid core if and only if every $\{a\}$ is pp-definable from Γ, so requiring the singletons outright costs nothing beyond pp-definability.

Two imported statements carry Theorem 1.2 to an arbitrary Γ.

Theorem 1.3 (Cores and idempotent reducts; imported)**.** (a) *Every finite Γ is homomorphically equivalent to a core Γ° on a subset of A, unique up to isomorphism, and $\mathrm{CSP}(\Gamma)$ and $\mathrm{CSP}(\Gamma^\circ)$ have the same satisfiable instances up to a syntactic translation [2, Prop. 1.4.3].*

(b) *If Γ is a core then $\mathrm{CSP}(\Gamma)$ and $\mathrm{CSP}(\Gamma^{\mathrm{rig}})$ are equivalent under logspace reductions [2, Thm. 1.4.7].*

(c) *If Γ is a core then $\mathrm{Pol}(\Gamma^{\mathrm{rig}})$ is exactly the set of idempotent members of $\mathrm{Pol}(\Gamma)$, and every $f \in \mathrm{Pol}(\Gamma)$ factors as $f = f_{\mathrm{un}} \circ f_{\mathrm{id}}$ with $f_{\mathrm{un}}(x) = f(x, \ldots, x)$ an automorphism and $f_{\mathrm{id}} = f_{\mathrm{un}}^{-1} \circ f$ idempotent [2, Prop. 1.4.10, 1.4.11].*

Lemma 1.4 (The criterion descends to the rigid core)**.** *Let Γ be finite and Γ° a core of it. Then some weak near-unanimity operation preserves Γ if and only if some weak near-unanimity operation preserves $(\Gamma^\circ)^{\mathrm{rig}}$.*

Proof. Let $e\colon \Gamma^\circ \to \Gamma$ and $r\colon \Gamma \to \Gamma^\circ$ be the homomorphisms of Theorem 1.3(a). If w is a weak near-unanimity polymorphism of Γ then $\bar{w} = r \circ w|_{(A^\circ)^k}$ is one of Γ°: it preserves the relations, being a composite of homomorphisms, and post-composition with r preserves each identity $w(y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$. Conversely if w is one of Γ° then $e \circ w \circ (r, \ldots, r)$ is one of Γ, by the same two observations.

So we may assume $\Gamma = \Gamma^\circ$ is a core, and must pass between Γ and Γ^{rig}. Since $\Gamma \subseteq \Gamma^{\mathrm{rig}}$, every polymorphism of Γ^{rig} is one of Γ. In the other direction let $w \in \mathrm{Pol}(\Gamma)$ be a weak near-unanimity operation and factor it as in Theorem 1.3(c). Applying the automorphism w_{un}^{-1} to each of the identities $w(y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ leaves them equalities, so w_{id} is again a weak near-unanimity operation; it is idempotent, hence a polymorphism of Γ^{rig} by Theorem 1.3(c). $\square$

Corollary 1.5 (Dichotomy; Bulatov, Zhuk)**.** *Let Γ be a finite constraint language. If some weak near-unanimity operation preserves Γ then $\mathrm{CSP}(\Gamma)$ is solvable in polynomial time; otherwise $\mathrm{CSP}(\Gamma)$ is NP-complete.*

Proof. Let Γ° be a core of Γ and $\Delta = (\Gamma^\circ)^{\mathrm{rig}}$. By Theorem 1.3(a),(b), CSP(Γ) and CSP(Δ) are equivalent under logspace reductions, so each is in polynomial time exactly when the other is, and each is NP-complete exactly when the other is. By Lemma 1.4 the hypothesis holds for Γ exactly when it holds for Δ, and Δ satisfies the hypothesis of Theorem 1.2. □

Caveat 1.6 (What the reduction is not)**.** Corollary 1.5 is the form the criterion is usually quoted in, and it is correct, but the route to it is not a formality and two points are worth isolating.

Idempotence is the whole of it. What Lemma 1.4 has to supply, once one is at a core, is an *idempotent* weak near-unanimity operation, and it gets one only from the factorisation of Theorem 1.3(c) — available because a core's unary polymorphisms form a group. The corresponding statement for arbitrary nontrivial systems of identities is **false**: Willard's example [2, Ex. 1.4.5] is a core whose polymorphisms satisfy identities unsatisfiable by projections while its rigidification has no such polymorphisms at all, and is NP-complete. Weak near-unanimity survives the passage; "satisfies some nontrivial identities" does not.

The special operation. Everything after §2 works not with an arbitrary weak near-unanimity operation but with a *special* one of some arity n, supplied by Lemma 6.32. That is a change of hypothesis and needs one sentence: the special operation is a *term* operation of Pol(Γ), so it is itself a polymorphism and preserves every relation of Γ; and it is idempotent because Pol(Γ) is. So $\mathbf{A} = (A; w)$ satisfies Convention 2.1 and every relation of Γ is a subuniverse of a power of $\mathbf{A}$, which is what §§5 onwards assume.

Theorem 1.3 is imported and not proved here.

1.2 The five layers

Both halves of Theorem 1.2 are conditional on a model of computation, and that is not a formality to wave through. "Solvable in polynomial time" quantifies over an encoding of instances as strings, a machine, and a polynomial bound on its step count; "NP-complete" additionally quantifies over all problems in NP and over polynomial-time many-one reductions. The theorem therefore splits into statements that mention a machine and statements that do not, and the split does not agree with the tractable/hard split.

Write $\mathbf{A} = (A; w)$ for a finite idempotent algebra whose basic operation w is a special weak near-unanimity operation, and $\mathcal{V}_n$ for the class of such algebras with w of arity n (Convention 2.1).

Tractable half. Three layers.

- **(T0)** ***The algebraic core.*** The three theorems of §8 — the existence of a strong subuniverse or a linear congruence, the safety of a reduction to a strong subuniverse, and the codimension-one theorem — together with everything they rest on. These are statements about finite algebras, congruences, and CSP instances regarded as finite combinatorial objects. No computation appears in them.
- **(T1)** ***Correctness of the algorithm.*** Zhuk's algorithm written as a total function Solve from instances to Booleans, and the proposition

$$\textsc{Solve}(\mathcal{I}) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

  This is a statement about a recursive function and is provable from (T0) with no cost model.
- **(T2)** ***The complexity wrapper.*** That Solve runs in polynomial time in the size of the instance, for fixed Γ. This is the one place a cost model is needed on this side. It is stated here and not proved.

Hard half. Two layers.

(H0) ***The algebraic core.*** If no weak near-unanimity operation preserves Γ then Γ primitively positively interprets every finite constraint language — in particular one whose CSP is NP-hard. This is again pure algebra: the Galois connection between polymorphisms and primitive positive definability, the Maróti–McKenzie theorem that a Taylor term yields a weak near-unanimity term, and the Bulatov–Jeavons–Krokhin reduction.

(H1) ***The complexity wrapper.*** That a primitive positive interpretation yields a polynomial-time many-one reduction, and hence NP-hardness. This does need a program, but the program performs a syntactic substitution of formulas whose size is linear in the instance, so it is genuinely small.

(T0), (T1) and (H0) are the mathematics, and are what the literature actually proves. This document is written for them. (T2) and (H1) are stated precisely in §§9–10 and are not proved here; recovering the word *complete* in Theorem 1.2 also needs the standard fact that a fixed finite-template CSP lies in NP, which belongs with them and is Theorem 10.5.

Remark 1.7. The separation is not a dodge. Layer (T1) is strictly stronger than " CSP(Γ) is decidable", which is trivial for a finite template: it says that one particular, highly non-obvious algorithm — the one whose existence is the content of the dichotomy — is correct. Everything that makes the dichotomy hard lives in (T0). Conversely, no part of (T0)–(T1) becomes easier if a cost model is available.

2 Standing conventions

The standing hypotheses of this theory are usually carried in prose. Carried that way, a reader cannot audit which results actually consume which hypothesis, and several statements below turn on exactly that. We therefore fix the hypotheses as a labelled convention and record, for each one, where it is consumed.

Convention 2.1 (Standing hypotheses)**.** Throughout, $n \geq 3$ is a fixed integer and:

(a) every algebra is *finite* and has a *nonempty* universe;
(b) every algebra has exactly one basic operation, of arity n, written w;
(c) w is *idempotent*: $w(x, \ldots, x) = x$;
(d) w is a *weak near-unanimity* operation: $w(y, x, \ldots, x) = w(x, y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ for all x, y;
(e) w is *special*: $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$ for all x, y.

The class of algebras satisfying a–e is written $\mathcal{V}_n$. Since only finite algebras are considered, $\mathcal{V}_n$ is not a variety.

Remark 2.2 (Why one basic operation)**.** Restricting to a single basic operation is not a loss. The dichotomy hypothesis is that *some* weak near-unanimity operation preserves Γ; one may then discard the original signature and work in the algebra $(A; w)$ whose only basic operation is that one, because every relation of Γ is a subuniverse of a power of $(A; w)$ and that is all the argument uses. Passing from a WNU w to a *special* one is Lemma 6.32, and costs an increase of arity from n to $n^{n!}$.

The gain is that a term is then a finite tree with n-ary branching over a single symbol, so "term operation" needs no signature bookkeeping, and $\mathrm{Clo}(\mathbf{A})$ is the clone generated by one operation.

Caveat 2.3 (Where each hypothesis is consumed)**.** a is used in every induction on $|A|$ and in every minimality argument (a minimal element of a nonempty family of subsets exists); it cannot be dropped anywhere. c is what makes singletons subuniverses and makes Sg of a nonempty set nonempty. d is consumed only through the results it implies — the existence of absorption-theoretic structure — and never directly in §§5–8. e is used where a unary polynomial must be idempotent under iteration, chiefly in the theory of Sg-closures of two-element sets, and it is what forces $p \mid n - 1$ in Proposition 2.7.

Convention 2.4 (The empty set)**.** A *subuniverse* of $\mathbf{A}$ is a subset closed under w; the empty set is a subuniverse under this definition, and we allow it. A *subalgebra* is a nonempty subuniverse; $\mathbf{B} \leq \mathbf{A}$ always carries $B \neq \varnothing$.

The relations $<_T$, $\leq_T$ and $\lhd\lhd\lhd$ of §4 are defined only between *nonempty* sets; in particular $\varnothing \lhd\lhd\lhd A$ never holds. Where a construction can produce the empty set — a projection of an intersection, most often — the dotted variants $\dot{<}_T$, $\dot{\leq}_T$, $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ are used, and they mean "the stated relation holds, or the left-hand side is empty".

Caveat 2.5 (The dot is not decoration)**.** Conflating $\lhd\lhd\lhd$ with $\dot{\lhd}\dot{\lhd}\dot{\lhd}$ is the single most common way to misread §4. Every propagation statement of §6 that projects or intersects produces a dotted conclusion, and every statement that consumes a $\lhd\lhd\lhd$ hypothesis needs the undotted one. The two are different relations, and the case split is part of every use.

Convention 2.6 (Relations, and indices)**.** A relation is a subset of a product $A_1 \times \cdots \times A_m$ of universes, indexed by $[m] = \{1, \ldots, m\}$; $\mathbf{R} \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ means additionally that R is a subuniverse of the product algebra. We write $\mathrm{pr}_{i_1,\ldots,i_s}(R)$ for the projection onto the listed coordinates and $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ when every $\mathrm{pr}_i(R) = A_i$.

Products are indexed by *arbitrary finite index sets*, not only by initial segments of $\mathbb{N}$. Several constructions in §8 form a product over the variable set of an instance, or over a set of subuniverses, and reindexing those to $[m]$ by hand buys nothing.

2.1 Ten readings fixed

Ten notations and definitions below admit more than one reading. The choice is invisible in prose and decisive in the proofs, so all ten are fixed here, once, and the fixed reading is used everywhere.

(i) **The algebra $\mathbf{Z}_p$.** The two roles of $\mathbf{Z}_p$ are kept apart. The abelian *group* $(\mathbb{Z}_p; +, -)$ supplies the relation $x_1 - x_2 = x_3 - x_4$ in Definition 4.10 and carries no arity constraint. The *algebra* $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n)$ lies in $\mathcal{V}_n$ only when $p \mid n-1$ (Corollary 2.8), and it is the algebra that occurs in Definition 3.40, where $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ is a subuniverse of a product and therefore needs a common signature. Every "for some prime p" below silently carries $p \mid n-1$.

(ii) **σ^* is a tolerance.** It is a reflexive symmetric subuniverse of $\mathbf{A}^2$ and *not* in general a congruence (Proposition 3.25). That σ^* is a congruence is item (b) of the definition of a linear congruence and conclusion (a) of Lemma 7.47; typing it as a congruence would make the first vacuous and collapse the linear/PC distinction.

(iii) **The empty set.** As in Convention 2.4: subuniverses may be empty, subalgebras may not, and the typed relations hold only between nonempty sets. Read syntactically, $\varnothing$ is vacuously absorbing and vacuously central, so two clauses need the word *nonempty* written into them: the clause defining $<_{\mathrm{s}}$ requires its witness D nonempty (Definition 4.6), and Lemma 7.20 carries the explicit alternative "$C = \varnothing$" — without which it is false, as $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x+1)\}$ shows. For the same reason Corollary 7.2 concludes a *proper nonempty* subuniverse: without those words it is vacuous, since every algebra is a binary absorbing subuniverse of itself.

(iv) **The empty family is allowed.** In Definition 3.23 the family $S_1, \ldots, S_k$ may be empty, the empty intersection being A^2. This is not a free choice: the two formulations given there agree *only* under it, and under it an irreducible congruence is automatically proper (Proposition 3.25(a)) and σ^* exists.

(v) **$\lhd\lhd\lhd$ carries its chain.** $C \lhd\lhd\lhd^A B$ is *data*: a list of intermediate sets with their typed steps and witnessing congruences, not merely the proposition that such a list exists. Statements below speak of "the dividing congruences coming from $C \lhd\lhd\lhd^A B$", which is a function of the chain and not of the pair. See Caveat 4.26 for the two places where the choice of chain is load-bearing.

(vi) **"Dimension".** $\prod_i \mathbb{Z}_{q_i}$ with distinct primes is not a vector space over anything and has no dimension. We use the composition length of the finite abelian group, together with its additivity along short exact sequences. Nothing in §§4–6 uses the notion; it appears only in §8.3, where the correct statement is that such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces.

(vii) **"Linked instance".** We use the per-variable condition of Definition 5.11, *not* global connectivity of the value graph. The two are inequivalent — a fragmented instance is per-variable linked in each component — so non-fragmentation is carried as a separate hypothesis wherever it is needed (Caveat 3.15).

(viii) **Quotients of subsets.** B/δ always means the *image* $\{b/\delta : b \in B\}$ in $\mathbf{A}/\delta$, for B any subset of A; never the quotient of B by a congruence of $\mathbf{B}$. Consequently $(C_1 \cap \cdots \cap C_t)/\delta \subseteq \bigcap_i C_i/\delta$ always, with equality false in general (Caveat 3.17). The one place the reverse inclusion is needed is Lemma 6.5(fm), which establishes it from an explicit saturation hypothesis.

(ix) **Four small results, stated rather than assumed.** Each is used repeatedly below and each is easy, which is exactly why it is easy to leave unstated: Lemma 4.12 (the relation S of Definition 4.10(c) is a bridge with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$), Lemma 3.27 ($\sigma$ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is), Lemma 4.8 (the two formulations of S-freeness agree), and Lemma 3.19 (the parallelogram property implies rectangularity, and the rectangular closure exists).

(x) **The three alternatives of the central-relation lemma.** Lemma 7.20 is stated with three alternatives, the third being a proper nonempty binary absorbing subuniverse. A fourth — a nontrivial *projective* subuniverse — appears in the form of this result that holds without Convention 2.1, and is eliminated here by Lemma 7.19. Projectivity is used nowhere else in this document.

Proposition 2.7 (Where $p \mid n-1$ comes from)**.** *Let* $\mathbf{U}$ *be a finite algebra with a single basic operation* w *of arity* n *that is idempotent, weak near-unanimity and special, and suppose* $\mathbf{U}$ *is affine over* $\mathbb{Z}_p$ *with* $|U| > 1$*. Then* $p \mid n-1$*, and* w *acts on* $\mathbf{U}$ *as* $x_1 + \cdots + x_n$*.*

Proof. Being affine, $\mathbf{U}$ is polynomially equivalent to a $\mathbb{Z}_p$-module, so w acts as $w(\bar{x}) = \sum_i c_i x_i + d$ for scalars $c_i \in \mathbb{Z}_p$ and a constant d. Idempotence gives $(\sum_i c_i)x + d = x$ for all x, hence $\sum_i c_i = 1$ and $d = 0$.

The weak near-unanimity identities $w(y, x, \ldots, x) = w(x, y, x, \ldots, x) = \cdots = w(x, \ldots, x, y)$ read $c_1 y + (1 - c_1)x = c_2 y + (1 - c_2)x = \ldots$, and comparing coefficients of y gives $c_1 = \cdots = c_n =: c$, so $nc = 1$.

Specialness, $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$, reads

$$(n-1)c\,x + cy \;=\; (n-1)c\,x + c((n-1)c\,x + cy),$$

and comparing coefficients of y gives $c = c^2$, so $c \in \{0, 1\}$. Now $c = 0$ contradicts $nc = 1$, so $c = 1$; then $n \cdot 1 = 1$ in $\mathbb{Z}_p$, i.e. $p \mid n-1$, and w acts as $x_1 + \cdots + x_n$. □

Corollary 2.8. $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n)$ *lies in* $\mathcal{V}_n$ *if and only if* $p \mid n-1$*; and whenever a prime* p *arises from a linear congruence on some* $\mathbf{A} \in \mathcal{V}_n$ *— so that some subquotient of* $\mathbf{A}$ *is affine over* $\mathbb{Z}_p$ *and nontrivial — one has* $p \mid n-1$*. So the standing identification of* $\mathbf{Z}_p$ *as an object of* $\mathcal{V}_n$ *is legitimate wherever this document makes it.*

Proof. For the first claim, $\mathbf{Z}_p$ is affine over $\mathbb{Z}_p$ with $|\mathbb{Z}_p| > 1$, so Proposition 2.7 gives $p \mid n-1$ if $\mathbf{Z}_p \in \mathcal{V}_n$; conversely $p \mid n-1$ makes $x_1 + \cdots + x_n$ idempotent, and it is then a WNU and special, both identities reducing to $(n-1)x + y = y$. For the second, apply Proposition 2.7 to the nontrivial affine subquotient. □

Caveat 2.9 (The proposition is not decoration)**.** Proposition 2.7 is needed twice, and in both places its absence would leave a statement that does not typecheck. It is what makes $\mathbf{Z}_p$ an object of $\mathcal{V}_n$ at all, hence what makes Definition 3.40 well formed; and it is what upgrades the group isomorphism of Lemma 4.18 to an isomorphism of algebras.

3 Vocabulary

This section fixes the universal-algebraic vocabulary. Nothing in it is new, and a reader who knows the subject should skim to Definition 3.23 (irreducible congruences) and Definition 3.29 (bridges), which are the two places where the conventions genuinely matter.

3.1 Algebras, subuniverses, terms

Definition 3.1 (Algebra)**.** An algebra is a pair $\mathbf{A} = (A; w^{\mathbf{A}})$ with A a finite nonempty set and $w^{\mathbf{A}} \colon A^n \to A$, subject to Convention 2.1. We write A for the universe of $\mathbf{A}$ and drop the superscript on w when no confusion arises.

Definition 3.2 (The algebras $\mathbf{Z}_p$)**.** For a prime p with $p \mid n-1$, $\mathbf{Z}_p$ is the algebra on $\{0, 1, \dots, p-1\}$ with $w^{\mathbf{Z}_p}(x_1, \dots, x_n) = x_1 + \cdots + x_n \bmod p$.

Caveat 3.3 ($\mathbf{Z}_p$ is only sometimes in $\mathcal{V}_n$)**.** The operation $x_1 + \cdots + x_n \bmod p$ is idempotent exactly when $n \equiv 1 \pmod{p}$: idempotence says $nx \equiv x$ for all x, i.e. $p \mid n-1$. Given that, it is also a WNU (all the identities read $(n-1)x + y = y$) and special (both sides reduce to y).

So $\mathbf{Z}_p \in \mathcal{V}_n$ if and only if $p \mid n-1$, and every statement of the form "$\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for some prime p" (Lemma 4.18, Definition 3.40) silently produces a p dividing $n-1$. The side condition must stay visible, because "for some prime p" otherwise reads as an unconstrained existential; Corollary 2.8 is what discharges it.

Definition 3.4 (Subuniverse, subalgebra)**.** $B \subseteq A$ is a *subuniverse* of $\mathbf{A}$ if $w(b_1, \dots, b_n) \in B$ for all $b_1, \dots, b_n \in B$. If moreover $B \neq \varnothing$ then $\mathbf{B} = (B; w{\restriction}_{B^n})$ is a *subalgebra*, written $\mathbf{B} \leq \mathbf{A}$.

Definition 3.5 (Term operations)**.** $\mathrm{Clo}(\mathbf{A})$ is the least set of finitary operations on A containing all projections and closed under composition with w. Its m-ary part is written $\mathrm{Clo}_m(\mathbf{A})$. Elements of $\mathrm{Clo}(\mathbf{A})$ are the *term operations* of $\mathbf{A}$.

Caveat 3.6 (Terms versus term operations)**.** A term is a finite tree, and distinct terms may induce the same operation. Every statement below that quantifies over $\mathrm{Clo}(\mathbf{A})$ is a statement about *operations*. The syntactic layer nevertheless cannot be dispensed with, because Sg is characterised through terms (Lemma 3.8) and because the arity bookkeeping in the absorption arguments is syntactic.

Definition 3.7 (Generation)**.** For $X \subseteq A^m$, $\mathrm{Sg}_{\mathbf{A}}(X)$ is the least subuniverse of $\mathbf{A}^m$ containing X.

Lemma 3.8 (Generation by terms)**.** *Let $X = \{x_1, \dots, x_k\} \subseteq A^m$ be nonempty. Then*

$$\mathrm{Sg}_{\mathbf{A}}(X) = \{\, t^{\mathbf{A}^m}(x_1, \dots, x_k) \;:\; t \in \mathrm{Clo}_k(\mathbf{A}) \,\}.$$

The generator list is fixed*: the same list $x_1, \dots, x_k$ in the same order serves every element of the closure, with t varying.*

Caveat 3.9. The "fixed list" phrasing is what makes Lemma 3.8 usable. The weaker reading — for each y in the closure there are *some* generators and *some* term — is also true but useless, because every later application applies one term simultaneously to several closure elements and needs their generator lists to coincide.

3.2 Relations

Definition 3.10 (Relational vocabulary)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) R is *subdirect* if $\mathrm{pr}_i(R) = A_i$ for every i; $\mathbf{R} \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ abbreviates "R is a subdirect subuniverse".
(b) For $R \subseteq A^m$: R is *reflexive* if $(a, \ldots, a) \in R$ for every $a \in A$.
(c) For binary $\delta_1 \subseteq A_1 \times A_2$, $\delta_2 \subseteq A_2 \times A_3$: $\delta_1 \circ \delta_2 = \{(a, c) : \exists b\ (a, b) \in \delta_1 \wedge (b, c) \in \delta_2\}$, and $\delta^{-1} = \{(b, a) : (a, b) \in \delta\}$.
(d) For $B \subseteq A_1$ and $\delta \subseteq A_1 \times A_2$: $B \circ \delta = \{c : \exists b \in B\ (b, c) \in \delta\}$.
(e) For binary σ and $m \geq 2$, $\sigma^{[m]} = \{(a_1, \ldots, a_m) : (a_i, a_j) \in \sigma\ \forall i, j\}$.
(f) $R \subseteq A \times B$ is *bijective* if $|R| = |A| = |B|$.

3.3 Linkedness

Two operators run through §7 and are worth pinning down before anything uses them. They split a relation at its *first* coordinate and at its *last*, and which of the two is meant is load-bearing.

Definition 3.11 (The linking equivalences)**.** Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ with $m \geq 2$. $\mathrm{LeftLinked}(R)$ is the least equivalence relation on $\mathrm{pr}_1(R)$ such that

$$(a_1, a_2, \ldots, a_m) \in R \text{ and } (b_1, a_2, \ldots, a_m) \in R \implies (a_1, b_1) \in \mathrm{LeftLinked}(R),$$

and $\mathrm{RightLinked}(R)$ is the least equivalence relation on $\mathrm{pr}_m(R)$ such that

$$(a_1, \ldots, a_{m-1}, a_m) \in R \text{ and } (a_1, \ldots, a_{m-1}, b_m) \in R \implies (a_m, b_m) \in \mathrm{RightLinked}(R).$$

Equivalently: view R as a binary relation $\hat{R} \subseteq A_1 \times (A_2 \times \cdots \times A_m)$; then $\mathrm{LeftLinked}(R)$ is the transitive closure of $\hat{R} \circ \hat{R}^{-1}$, and $\mathrm{RightLinked}(R)$ is the transitive closure of $\check{R}^{-1} \circ \check{R}$ for the corresponding split at the last coordinate.

R is *linked* if $\mathrm{LeftLinked}(R) = \mathrm{pr}_1(R)^2$.

Lemma 3.12 (Basic properties)**.** *Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and write $P = \mathrm{pr}_1(R)$, $Q = \mathrm{pr}_m(R)$.*

(a) $\mathrm{LeftLinked}(R)$ *is a congruence of* $\mathbf{P}$*, and* $\mathrm{RightLinked}(R)$ *one of* $\mathbf{Q}$.
(b) $\hat{R} \circ \hat{R}^{-1} \subseteq \mathrm{LeftLinked}(R)$*, and if R is reflexive as a binary relation on A then* $R \subseteq \mathrm{LeftLinked}(R)$.
(c) *For $m = 2$ and $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$: R is linked* $\iff$ $\mathrm{LeftLinked}(R) = A_1^2$ $\iff$ $\mathrm{RightLinked}(R) = A_2^2$ $\iff$ *the bipartite graph on $A_1 \sqcup A_2$ with edge set R is connected.*

Proof. (a) $\hat{R} \circ \hat{R}^{-1}$ is pp-definable from R, hence a subuniverse of $\mathbf{A}_1^2$; it is symmetric, and reflexive on P. Its powers increase and so stabilise, and each power is again a subuniverse, so the transitive closure is a subuniverse of $\mathbf{A}_1^2$ contained in P^2 — that is, a congruence of $\mathbf{P}$. It satisfies the closure condition of Definition 3.11 and any equivalence satisfying that condition contains $\hat{R} \circ \hat{R}^{-1}$ and hence its transitive closure, so it is the least one. Symmetrically for RightLinked.

(b) The first is immediate from (a). For the second, if $(a, b) \in R$ then $(a, b), (b, b) \in R$, so $(a, b) \in \hat{R} \circ \hat{R}^{-1}$.

(c) A path in the bipartite graph from a to a' alternates between the two sides, and consecutive A_1-vertices on it share an A_2-neighbour, so they are $\mathrm{LeftLinked}(R)$-related; hence connectedness gives $\mathrm{LeftLinked}(R) = A_1^2$, using subdirectness for $\mathrm{pr}_1(R) = A_1$. Conversely $\mathrm{LeftLinked}(R) = A_1^2$ joins any two A_1-vertices by a path, and subdirectness gives every A_2-vertex an A_1-neighbour, so the graph is connected. The same argument on the other side gives the third equivalence. □

Caveat 3.13 (Which coordinate is split)**.** For $m > 2$ the two operators are not symmetric variants of one notion: they split R at opposite ends, and the intermediate coordinates go with the *other* side. So $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\delta))$ concerns the split $(1,2) \mid (3)$, and is a relation on $\mathrm{pr}_3(\delta)$ — not on pairs. A linkedness statement for the coarser split $(1,2) \mid (3,4)$ does not follow from it and is not implied by it: a path in the coarser bipartite graph need not lift, because consecutive edges through a common third coordinate may use different fourth coordinates. Caveat 7.18 is where that bites.

Caveat 3.14 (Composition is diagrammatic)**.** $\delta_1 \circ \delta_2$ applies δ_1 *first*, the opposite of the convention for composition of functions. Every composite of bridges and of congruences below is written left to right, and silently reversing it produces statements that are true-looking and wrong — in Corollary 6.19(l) the composite $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$ is asymmetric in k and ℓ and the order is the content.

Caveat 3.15 ("Linked" is used for two different things)**.** The word is used both for a binary relation (Definition 3.10(e)) and for a CSP instance (Definition 5.11). They are related but not the same predicate, and both occur in one proof in §8, where the passage from one to the other is the crux; Caveat 8.18 marks the step.

Definition 3.16 (Quotients of relations)**.** For an equivalence σ on A and $a \in A$, a/σ is the class of a; for $B \subseteq A$, $B/\sigma = \{b/\sigma : b \in B\}$; for $R \subseteq A^m$, $R/\sigma = \{(b_1/\sigma, \dots, b_m/\sigma) : \bar{b} \in R\}$.

Caveat 3.17 (Images do not commute with intersection)**.** B/σ and R/σ are *images.* Three consequences, all of them easy to pass over:

(a) B/σ is a subset of A/σ, not the quotient of B by $\sigma \cap B^2$. For B a subuniverse and σ a congruence the two are canonically isomorphic as algebras, and the two are used interchangeably below; B/σ always denotes the image, and the isomorphism is what licenses the interchange.
(b) In general $(C_1 \cap C_2)/\sigma \subsetneq C_1/\sigma \cap C_2/\sigma$. Lemma 6.5(fm) needs equality for a family $C_1, \dots, C_t$, and gets it from the extra hypothesis $(C_i \circ \sigma) \cap B = C_i$, i.e. that each C_i is σ-saturated inside B. That is a content step, not notation.
(c) R/σ need not be stable under anything, and need not be a subuniverse of $(\mathbf{A}/\sigma)^m$ unless σ is a congruence.

3.4 Rectangularity

Definition 3.18 (Parallelogram property, rectangularity)**.** Let $R \subseteq A_1 \times \cdots \times A_m$.

(a) The i-th variable of R is *rectangular* if for all $\bar{a}, \bar{b}$,

$$\bar{a}[i \mapsto b_i] \in R, \quad \bar{b}[i \mapsto a_i] \in R, \quad \bar{b} \in R \implies \bar{a} \in R,$$

where $\bar{a}[i \mapsto c]$ is $\bar{a}$ with its i-th entry replaced by c. R is *rectangular* if every variable is.
(b) *R has the parallelogram property* if for every permutation of the coordinates giving R' and every $\ell \in [m-1]$,

$$(a_1, \dots, a_\ell, b_{\ell+1}, \dots, b_m) \in R', \quad (b_1, \dots, b_\ell, a_{\ell+1}, \dots, a_m) \in R', \quad \bar{b} \in R' \implies \bar{a} \in R'.$$

(c) The *rectangular closure* of R is the least rectangular relation containing R.

Lemma 3.19 (Rectangularity basics)**.** (a) *A relation with the parallelogram property is rectangular.*
(b) *The rectangular closure exists.*

Proof. (a) Apply the parallelogram property to the permutation carrying coordinate i to position 1, with $\ell = 1$. (b) Rectangularity at coordinate i is a Horn implication, so the rectangular relations containing R are closed under arbitrary intersection and the family is nonempty (it contains $A_1 \times \cdots \times A_m$); the intersection of all of them is the least one. $\square$

Caveat 3.20 (Quantifying over permutations)**.** Definition 3.18(b) quantifies over $\mathfrak{S}_m$ and over the split point ℓ. Equivalently, and more usably, it quantifies over all ways of splitting $[m]$ into two nonempty blocks; the second formulation avoids transporting R along a permutation and is the one used below.

3.5 Congruences

Definition 3.21 (Stability)**.** Let $R \subseteq A_1 \times \cdots \times A_m$ and let σ be a binary relation on A_i. The i-th variable of R is *stable under* σ if $\bar{a} \in R$ and $(a_i, b) \in \sigma$ imply $\bar{a}[i \mapsto b] \in R$. R is *stable under* σ if every variable is (each with respect to σ on the corresponding factor, when the factors agree).

Definition 3.22 (Congruence)**.** A *congruence* on $\mathbf{A}$ is an equivalence relation on A that is a subuniverse of $\mathbf{A} \times \mathbf{A}$. It is *nontrivial* if it is neither the equality relation $0_{\mathbf{A}}$ nor A^2. The quotient $\mathbf{A}/\sigma$ is the algebra on A/σ with $w(\bar{a}/\sigma) = w(\bar{a})/\sigma$.

Definition 3.23 (Irreducible congruence and its cover)**.** A congruence σ on $\mathbf{A}$ is *irreducible* if it cannot be written as an intersection of binary subuniverses of $\mathbf{A} \times \mathbf{A}$ each stable under σ and each properly containing σ. Equivalently: there are no $\mathbf{S}_1, \ldots, \mathbf{S}_k \leq \mathbf{A}/\sigma \times \mathbf{A}/\sigma$ with $0_{\mathbf{A}/\sigma} = S_1 \cap \cdots \cap S_k$ and $S_i \neq 0_{\mathbf{A}/\sigma}$ for every i.

For irreducible σ, σ^* denotes the least $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ.

Caveat 3.24 (Three traps in Definition 3.23)**.** (a) Irreducibility is *not* meet-irreducibility in the congruence lattice. The $\mathbf{S}_i$ range over binary *subuniverses* stable under σ, which need not be equivalence relations. The two notions differ, and every use below is the stated one.

(b) σ^* need not be a congruence. It is a subuniverse of $\mathbf{A}^2$ stable under σ, nothing more; that σ^* *is* a congruence is an extra hypothesis in the definition of a linear congruence (Definition 4.10), which would be redundant otherwise.

(c) Existence and uniqueness of σ^* need an argument, given in Proposition 3.25. Uniqueness is exactly what irreducibility buys; without it there is no "the" cover, and σ^* is meaningless for a congruence not known to be irreducible.

(d) The family $S_1, \ldots, S_k$ may be *empty* (item iv of §2.1). This is not a free choice: the two formulations above agree only under it.

Proposition 3.25 (The cover exists, and is a tolerance)**.** *Let σ be an irreducible congruence on $\mathbf{A}$. Then*

(a) $\sigma \neq A^2$*;*

(b) *the family $\mathcal{F}$ of $\delta \leq \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ is nonempty and closed under intersection, so it has a least element σ^*;*

(c) σ^* *is reflexive and symmetric.*

Proof. (a) The empty family is allowed, and its intersection is A^2; so if $\sigma = A^2$ then σ is an intersection of a family of relations each properly containing it, vacuously, and σ is reducible.

(b) $\mathcal{F} \ni A^2$ by (a). If $\delta_1, \delta_2 \in \mathcal{F}$ then $\delta_1 \cap \delta_2$ is again a subuniverse of $\mathbf{A}^2$ stable under σ and contains σ; it contains σ *properly*, since otherwise $\sigma = \delta_1 \cap \delta_2$ exhibits σ as an intersection of two members of $\mathcal{F}$, contradicting irreducibility. So $\mathcal{F}$ is closed under binary, hence finite, intersection, and $\sigma^* = \bigcap \mathcal{F} \in \mathcal{F}$ is its least element.

(c) $\sigma^* \supseteq \sigma \supseteq 0_{\mathbf{A}}$ is reflexive. For symmetry, $(\sigma^*)^{-1}$ is a subuniverse of $\mathbf{A}^2$, is stable under σ because σ is symmetric, and properly contains $\sigma^{-1} = \sigma$; so $(\sigma^*)^{-1} \in \mathcal{F}$ and therefore $\sigma^* \subseteq (\sigma^*)^{-1}$, whence equality. ☐

Remark 3.26 (Why not a congruence)**.** Nothing in Proposition 3.25 gives transitivity, and none is available: that σ^* is a congruence is item (b) of Definition 4.10 and conclusion (a) of Lemma 7.47, both of which would be redundant otherwise. Symmetry, by contrast, comes free from minimality, and is what licenses the "switching x_1 and x_2" step in the proof of Lemma 4.16.

Lemma 3.27 (Passing to the quotient)**.** *Let σ be a congruence on $\mathbf{A}$. Then $\delta \mapsto \delta/\sigma$ is an inclusion-preserving bijection, commuting with intersections, from the subuniverses of $\mathbf{A}^2$ that contain σ and are stable under σ onto the subuniverses of $(\mathbf{A}/\sigma)^2$, sending σ to $0_{\mathbf{A}/\sigma}$. Consequently σ is irreducible, linear or PC exactly when $0_{\mathbf{A}/\sigma}$ is, and then $\sigma^*/\sigma = (0_{\mathbf{A}/\sigma})^*$.*

Proof. A subuniverse δ of $\mathbf{A}^2$ containing σ and stable under σ is a union of products of σ-blocks, so it is the preimage of δ/σ; conversely the preimage of a subuniverse of $(\mathbf{A}/\sigma)^2$ is a subuniverse containing σ and stable under σ. The two constructions are mutually inverse and preserve inclusion and intersection, and σ is the preimage of $0_{\mathbf{A}/\sigma}$. The statement of Definition 3.23 is a statement about that family alone, so it transports; and both the least element in Proposition 3.25(b) and the data of Definition 4.10 transport with it. ☐

Caveat 3.28 (This is the licence for "assume $\sigma = 0$")**.** Three proofs in §7 open by factoring out σ and assuming it is the equality relation. Lemma 3.27 is what licenses that (§2.1, item ix).

3.6 Bridges

Definition 3.29 (Bridge)**.** Let σ_1, σ_2 be congruences on $\mathbf{D}_1, \mathbf{D}_2$. A relation $\delta \leq \mathbf{D}_1^2 \times \mathbf{D}_2^2$ is a *bridge from σ_1 to σ_2* if

(a) the first two variables of δ are stable under σ_1;
(b) the last two variables of δ are stable under σ_2;
(c) $\mathrm{pr}_{1,2}(\delta) \supsetneq \sigma_1$ and $\mathrm{pr}_{3,4}(\delta) \supsetneq \sigma_2$;
(d) $(a_1, a_2, a_3, a_4) \in \delta$ implies $((a_1, a_2) \in \sigma_1 \iff (a_3, a_4) \in \sigma_2)$.

For a bridge δ put $\widetilde{\delta} = \{(x, y) : (x, x, y, y) \in \delta\}$. A bridge $\delta \subseteq D^4$ is *reflexive* if $(a, a, a, a) \in \delta$ for all $a \in D$.

Remark 3.30 (The motivating example)**.** Let $\mathbf{D}$ be the group $\mathbb{Z}_4$ with any compatible idempotent operation — for instance $x_1 - x_2 + x_3$, extended to arity n by dummy variables — so that every subgroup of a power of $\mathbb{Z}_4$ is a subuniverse. Then $\delta = \{(a_1, a_2, a_3, a_4) : a_1 - a_2 = 2a_3 - 2a_4\}$ is a bridge from $0_{\mathbf{D}}$ to congruence mod 2. Condition (d) is the content: $a_1 = a_2$ exactly when $a_3 \equiv a_4 \pmod 2$. Bridges are the formal shadow of a centralizer relation; see [12].

Definition 3.31 (Composition of bridges)**.** For bridges δ_1 from σ_0 to σ_1 and δ_2 from σ_1 to σ_2, set

$$(\delta_1 * \delta_2)(x_1, x_2, z_1, z_2) = \exists y_1 \exists y_2\ \delta_1(x_1, x_2, y_1, y_2) \wedge \delta_2(y_1, y_2, z_1, z_2).$$

Lemma 3.32 (Bridge composition; [13, Lem. 6.3])**.** *If $\sigma_0, \sigma_1, \sigma_2$ are irreducible then $\delta_1 * \delta_2$ is a bridge from σ_0 to σ_2, and $\widetilde{\delta_1 * \delta_2} = \widetilde{\delta_1} \circ \widetilde{\delta_2}$.*

Lemma 3.33 (Rectangularisation)**.** *Let σ_1, σ_2 be irreducible congruences on $\mathbf{A}_1, \mathbf{A}_2$ and let δ be a bridge from σ_1 to σ_2, with δ^{-1} the bridge $(x_1, x_2, x_3, x_4) \mapsto \delta(x_3, x_4, x_1, x_2)$ from σ_2 to σ_1. Then $\omega = \delta * (\delta^{-1} * \delta)^k$, for any $k \geq |A_1|$, is a bridge from σ_1 to σ_2 with*

(a) $\widetilde{\omega} = \mathrm{LeftLinked}(\widetilde{\delta}) \circ \widetilde{\delta} \supseteq \widetilde{\delta}$;
(b) $\widetilde{\omega} = \bigcup_L L \times \widetilde{\delta}(L)$, *where* L *ranges over the classes of* $\mathrm{LeftLinked}(\widetilde{\delta})$ *and* $\widetilde{\delta}(L) = \{y : (x,y) \in \widetilde{\delta}$ *for some* $x \in L\}$. *In particular* $\widetilde{\omega}$ *is* rectangular, *and* $(a,c) \in \widetilde{\omega}$ *with* $(a,b) \in \mathrm{LeftLinked}(\widetilde{\omega})$ *forces* $(b,c) \in \widetilde{\omega}$;
(c) $\mathrm{LeftLinked}(\widetilde{\omega}) = \mathrm{LeftLinked}(\widetilde{\delta})$ *and* $\mathrm{RightLinked}(\widetilde{\omega}) = \mathrm{RightLinked}(\widetilde{\delta})$;
(d) $\omega \subseteq \mathrm{pr}_{1,2}(\delta) \times \mathrm{pr}_{3,4}(\delta)$; *in particular* ω *inherits any containment* $\delta \subseteq \sigma_1^* \times \sigma_2^*$.

Proof. Each $*$ composes bridges whose three congruences are drawn from $\{\sigma_1, \sigma_2\}$, both irreducible, so Lemma 3.32 applies at every step: ω is a bridge from σ_1 to σ_2 and $\widetilde{\omega} = \widetilde{\delta}\circ(\widetilde{\delta}^{-1}\circ\widetilde{\delta})^k = (\widetilde{\delta} \circ \widetilde{\delta}^{-1})^k \circ \widetilde{\delta}$.

The relation $\widetilde{\delta}\circ\widetilde{\delta}^{-1}$ is reflexive on $\mathrm{pr}_1(\widetilde{\delta})$ and symmetric, so its powers increase and stabilise at its transitive closure, which by Definition 3.11 is $\mathrm{LeftLinked}(\widetilde{\delta})$; $k \geq |A_1|$ steps suffice. That is (a), and the containment holds because $\mathrm{LeftLinked}(\widetilde{\delta})$ is reflexive there.

(b) is (a) rewritten: $(a,c) \in \mathrm{LeftLinked}(\widetilde{\delta}) \circ \widetilde{\delta}$ says $c \in \widetilde{\delta}(L)$ for the class L of a. For rectangularity, let $(a,c),(a,d),(b,c) \in \widetilde{\omega}$ with $a \in L$ and $b \in L'$. Then c lies in both $\widetilde{\delta}(L)$ and $\widetilde{\delta}(L')$, so some $x \in L$ and $x' \in L'$ have $(x,c),(x',c) \in \widetilde{\delta}$, whence x and x' are $\mathrm{LeftLinked}(\widetilde{\delta})$-equivalent and $L = L'$; now $d \in \widetilde{\delta}(L)$ and $b \in L$ give $(b,d) \in \widetilde{\omega}$. The last clause is the same computation with (c).

(c) holds because $\widetilde{\delta} \subseteq \widetilde{\omega} \subseteq \mathrm{LeftLinked}(\widetilde{\delta}) \circ \widetilde{\delta}$, so the two relations have the same connected components: every edge of $\widetilde{\omega}$ joins vertices already connected in $\widetilde{\delta}$.

(d) By the definition of $*$, the first two coordinates of a tuple of a composite come from the first factor and the last two from the last factor, and both factors here are δ. $\square$

Caveat 3.34. Both conclusions of Lemma 3.32 are used: the first in every chain of bridges, the second — that $\widetilde{\cdot}$ commutes with composition — in Lemma 6.34, where a bridge with $\widetilde{\delta}$ linked is powered up until $\widetilde{\delta} = A^2$. The irreducibility hypothesis on *all three* congruences is needed and is easy to drop by accident. Lemma 3.38 has no such hypothesis and does not cite this lemma: it proves its own collapse identity by unfolding, which is three lines. The general composition result is used only where all three congruences really are irreducible, namely in Lemma 6.34, where a bridge with $\widetilde{\delta}$ linked is powered up until $\widetilde{\delta} = A^2$.

Definition 3.35 (Symmetric and pair-reflexive bridges)**.** Let δ be a bridge from σ to σ on $\mathbf{D}$. Call δ *symmetric* if $\delta(x_1,x_2,x_3,x_4) \iff \delta(x_3,x_4,x_1,x_2)$, and *pair-reflexive* if

$$(x_1,x_2) \in \mathop{\mathrm{pr}}_{1,2}(\delta) \implies (x_1,x_2,x_1,x_2) \in \delta.$$

Write $\delta^{-1}(x_1,x_2,x_3,x_4) = \delta(x_3,x_4,x_1,x_2)$.

Lemma 3.36 (Reflexivity is free)**.** *Let* δ *be a bridge from* σ *to* σ *on* $\mathbf{D}$ *with* $\widetilde{\delta} \supseteq \sigma$. *Then* δ *is reflexive.*

Proof. σ is a congruence, hence reflexive, so $(a,a) \in \sigma \subseteq \widetilde{\delta}$ for every $a \in D$; and $(a,a) \in \widetilde{\delta}$ says $(a,a,a,a) \in \delta$, which is reflexivity. $\square$

Caveat 3.37 (There is no reflexivisation problem)**.** Every consumer of a bridge below hypothesises $\widetilde{\delta} \supsetneq \sigma$ — that is what makes the bridge *nontrivial* and is the whole content of Lemma 4.13(b) — and that hypothesis contains $\widetilde{\delta} \supseteq \sigma$. So Lemma 3.36 applies at every such site and the reflexivity Lemma 3.38 needs is already there.

This is worth stating because it is easy to miss, and missing it is expensive: an earlier draft of this document, and an external review of it, both treated "which we may take reflexive" as an unproved normalisation requiring a reflexivisation lemma. There is nothing to prove. The step is not a normalisation at all — one does not *replace* δ by anything — it is an observation about the hypothesis one already has.

Lemma 3.38 (Symmetrisation)**.** *Let δ be a reflexive bridge from σ to σ on $\mathbf{D}$ and put $\delta^\circ = \delta * \delta^{-1}$, that is*

$$\delta^\circ(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6 \; \delta(x_1, x_2, x_5, x_6) \wedge \delta(x_3, x_4, x_5, x_6).$$

Then δ° is a reflexive, symmetric, pair-reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$ and $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta}^{-1} \supseteq \widetilde{\delta}$. Moreover any pair-reflexive bridge ε satisfies $\varepsilon \subseteq \varepsilon * \varepsilon$.

Proof. Symmetry is visible in the formula. For pair-reflexivity, if $(x_1, x_2) \in \mathrm{pr}_{1,2}(\delta^\circ)$ then $\delta(x_1, x_2, x_5, x_6)$ for some x_5, x_6, and taking the same witnesses twice gives $(x_1, x_2, x_1, x_2) \in \delta^\circ$; the same computation shows $\mathrm{pr}_{1,2}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta)$, the inclusion $\subseteq$ being immediate, and symmetry then gives $\mathrm{pr}_{3,4}(\delta^\circ) = \mathrm{pr}_{1,2}(\delta^\circ)$. Clause (c) of Definition 3.29 follows, and clause (d) because $(x_1, x_2) \in \sigma \iff (x_5, x_6) \in \sigma \iff (x_3, x_4) \in \sigma$. Stability is inherited from δ, and reflexivity from $(a, a, a, a) \in \delta$.

For the collapse identity $\widetilde{\delta^\circ} = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$, unfold both sides rather than citing Lemma 3.32, whose irreducibility hypothesis is not available here. By definition

$$\begin{aligned} \widetilde{\delta^\circ} &= \{(x, y) : \exists x_5 \exists x_6 \; \delta(x, x, x_5, x_6) \wedge \delta(y, y, x_5, x_6)\}, \\ \widetilde{\delta} \circ \widetilde{\delta}^{-1} &= \{(x, y) : \exists z \; \delta(x, x, z, z) \wedge \delta(y, y, z, z)\}. \end{aligned}$$

The inclusion $\supseteq$ is the case $x_5 = x_6 = z$. For $\subseteq$: from $\delta(x, x, x_5, x_6)$, clause (d) of Definition 3.29 and $(x, x) \in \sigma$ give $(x_5, x_6) \in \sigma$; stability of the fourth variable under σ then replaces x_6 by x_5, giving $\delta(x, x, x_5, x_5)$, and the same for $\delta(y, y, x_5, x_6)$. Take $z = x_5$. So clause (d) does not force the two intermediate coordinates *equal* — only σ-related — and it is stability that identifies them.

The identity contains $\widetilde{\delta}$ because $\widetilde{\delta}$ is reflexive, so $(x, y) \in \widetilde{\delta}$ gives $(x, y) \in \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ by taking y as the intermediate point. Finally, if ε is pair-reflexive and $\varepsilon(x_1, x_2, x_3, x_4)$, then taking (x_1, x_2) itself as the intermediate pair — legitimate because $\varepsilon(x_1, x_2, x_1, x_2)$ — gives $(\varepsilon * \varepsilon)(x_1, x_2, x_3, x_4)$. ☐

Caveat 3.39 (Pair-reflexivity is the hypothesis three lemmas need)**.** Pair-reflexivity is a hypothesis of Lemmas 7.44, 7.47 and 7.49, and it is not optional in any of them: Remark 7.45 exhibits a bridge on $\mathbf{Z}_3$ satisfying every other hypothesis of Lemma 7.44 and none of its conclusion. Lemma 3.38 is what makes the hypothesis costless: every bridge these statements are applied to below is a δ°.

Definition 3.40 (Perfect linear congruence)**.** A congruence σ on $\mathbf{A}$ is a *perfect linear congruence* if it is irreducible and there are a prime p and $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ with $\mathrm{pr}_{1,2}(\zeta) = \sigma^*$ and

$$(a_1, a_2, b) \in \zeta \implies ((a_1, a_2) \in \sigma \iff b = 0).$$

Perfect linear congruences are the payoff of the whole bridge machinery: they let the passage from σ to σ^* be tracked by a single element of $\mathbf{Z}_p$, which is what turns a CSP instance into a system of linear equations in §8.3.

4 The six types of strong subuniverse

This section defines six binary relations $C <_T^A B$ between subsets of a fixed algebra $\mathbf{A}$, one for each type T, and the transitive-closure relation $C \lhd\lhd\lhd^A B$. Everything in §§6–8 is expressed in this language, so the definitions repay care.

The informal reading is: $C <_T^A B$ says that C is a *strong* subset of B — strong enough that the CSP algorithm may reduce a variable's domain from B to C without losing all solutions — and T records *why* it is strong. Three of the reasons are absorption-theoretic (BA, C, S) and three are congruence-theoretic (D, L, PC).

4.1 Absorption

Definition 4.1 (Absorbing subuniverse)**.** A subuniverse B of $\mathbf{A}$ is *absorbing* if there is $t \in \mathrm{Clo}_k(\mathbf{A})$, for some k, such that for every $i \in [k]$

$$t(B, \ldots, B, \underbrace{A}_{i}, B, \ldots, B) \subseteq B.$$

We then say B *absorbs* $\mathbf{A}$ *with the term* t. If t may be chosen binary, B is a *binary absorbing* subuniverse (BA); if ternary, a *ternary absorbing* subuniverse.

Caveat 4.2 (The tuple form, and $k \geq 1$)**.** Definition 4.1 is a condition on *tuples constrained coordinatewise*: for every $\bar{a} \in A^k$ with $a_j \in B$ for all $j \neq i$, one has $t(\bar{a}) \in B$. The equivalent-looking phrasing "for all $b_1, \ldots, b_k \in B$ and $a \in A$, $t(b_1, \ldots, b_{i-1}, a, b_{i+1}, \ldots, b_k) \in B$" invites the variant in which b_i is quantified and then discarded, which is vacuously true at $k = 1$, where the intended content is not vacuous. The arity is required to satisfy $k \geq 1$.

Definition 4.3 (Central subuniverse)**.** A subuniverse C of $\mathbf{A}$ is *central* (C) if it is absorbing and for every $a \in A \setminus C$,

$$(a, a) \notin \mathrm{Sg}_{\mathbf{A}}\big((\{a\} \times C) \cup (C \times \{a\})\big).$$

Lemma 4.4 ([15, Cor. 6.11.1])**.** *A central subuniverse is a ternary absorbing subuniverse.*

Remark 4.5 (Zhuk's centre theorem)**.** Lemma 4.4 is the deepest import of this section. It is proved from the definitions via the Barto–Kazda relational description of absorption (Lemma 7.6) and a doubling argument, and it is the step that converts absorption at some arity into absorption at arity three.

Definition 4.6 (BA and centre free; S)**.** $\mathbf{A}$ is *BA and centre free* if it has no proper nonempty binary absorbing subuniverse and no proper nonempty central subuniverse.

For $\varnothing \neq C \lneq B \leq A$ write $C <_{\mathrm{S}}^A B$ if there is a *nonempty* subuniverse $D \subseteq C$ of $\mathbf{B}$ that is simultaneously binary absorbing and central in $\mathbf{B}$. $\mathbf{A}$ is S-*free* if there is no $C <_{\mathrm{S}} \mathbf{A}$.

Caveat 4.7 (The witness must be required nonempty)**.** $\varnothing$ is vacuously binary absorbing and vacuously central (§2.1, item iii), so without the word "nonempty" the clause defining $<_{\mathrm{S}}$ is universally true and the type S collapses. The global convention that subalgebras are nonempty does not repair it on its own, because D is introduced by an existential inside the clause rather than as a subalgebra of the ambient discourse.

Lemma 4.8 (The two readings of S-freeness agree)**.** $\mathbf{A}$ *is* S*-free if and only if no proper nonempty* $D \leq A$ *is simultaneously binary absorbing and central in* $\mathbf{A}$*.*

Proof. If such a D exists then $D <_{\mathrm{S}} \mathbf{A}$, taking $C = D$. Conversely if $C <_{\mathrm{S}} \mathbf{A}$ then by definition some nonempty $D \subseteq C \lneq A$ is binary absorbing and central in $\mathbf{A}$, and D is proper. □

Caveat 4.9 (S is not the conjunction of BA and C)**.** $C <_{\mathrm{S}} B$ does not say that C is binary absorbing and central in $\mathbf{B}$; it says that C *contains* some D that is. The type S exists precisely so that it is closed under enlarging C, which BA and C are not. Note also the asymmetry with S-freeness, which *is* stated as a conjunction. Getting this backwards makes Lemma 6.5(ft) unprovable.

Distinct from all of these is the notation $C <^{A}_{\mathrm{BA,C}} B$, which is a *conjunction*: it says that $C <^{A}_{\mathrm{BA}} B$ and $C <^{A}_{\mathrm{C}} B$ both hold. It is not a seventh type, and it is not the disjunction.

4.2 Linear and PC congruences

Definition 4.10 (Linear congruence)**.** A congruence σ on $\mathbf{A}$ is *linear* if

(a) σ is irreducible;
(b) σ^* is a congruence;
(c) there are a prime p and $S \leq (\sigma^*)^{[4]}$ such that for every block B of σ^* there is $m \geq 0$ with
$$(B/\sigma;\ (S \cap B^4)/\sigma) \ \cong\ (\mathbb{Z}_p^m;\ x_1 - x_2 = x_3 - x_4).$$

An irreducible congruence that is not linear is a *PC congruence*.

Caveat 4.11 (Cutting down before imaging, not after)**.** Clause (c) must read $(S \cap B^4)/\sigma$ and not $S \cap (B/\sigma)^4$. S is a relation on elements of A, while $(B/\sigma)^4$ is a set of 4-tuples of σ-classes; the two do not intersect, because they are sets of different things. The operation intended is: cut S down to the block, then take the image (§2.1, item viii).

The alternative repair — restrict the image relation S/σ to $(B/\sigma)^4$ — gives the same relation here, because S turns out to be stable under σ in each variable (Lemma 4.12) and tuples of S meet no other block. But that is a consequence of (c), not something available while (c) is being read, so the definition uses the image, which is well formed for any S.

Here $(\sigma^*)^{[4]}$ denotes the set of 4-tuples whose entries lie in a single block of σ^*, which is meaningful because (b) makes σ^* a congruence. The following is used repeatedly below (§2.1, item ix).

Lemma 4.12 (The witness of (c) is a bridge)**.** *Let σ be a linear congruence and S as in Definition 4.10(c). Then S is a reflexive pair-reflexive bridge from σ to σ with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$.*

Proof. Work inside a block B of σ^* and identify B/σ with $\mathbb{Z}_p^m$ so that $(S \cap B^4)/\sigma$ becomes $x_1 - x_2 = x_3 - x_4$; tuples of S meet no other block, since $S \leq (\sigma^*)^{[4]}$. Each variable is stable under σ, since the defining equation is an equation between σ-classes. Clause (d) of Definition 3.29 holds because $x_1 - x_2 = 0$ if and only if $x_3 - x_4 = 0$, which says $(x_1, x_2) \in \sigma \iff (x_3, x_4) \in \sigma$. For the projections, any (x_1, x_2) inside a common block satisfies $(x_1, x_2, x_1, x_2) \in S$, so $\mathrm{pr}_{1,2}(S) = \sigma^*$ — which is at once clause (c), since $\sigma^* \supsetneq \sigma$, and pair-reflexivity; symmetry of the defining equation in the two pairs gives $\mathrm{pr}_{3,4}(S) = \sigma^*$. Finally $(x, x, y, y) \in S$ reads $0 = 0$, so $\widetilde{S}$ is the whole of σ^*, and $(a, a, a, a) \in S$ gives reflexivity. □

This is the useful reformulation:

Lemma 4.13 ([14])**.** *For an irreducible congruence σ the following are equivalent:*

(a) *σ is linear;*
(b) *there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$.*

Caveat 4.14 (Definition (c) versus criterion (b))**.** Definition 4.10(c) is a per-block isomorphism statement with an existential m *depending on the block*, and it quantifies over blocks of σ^*, not of σ. Lemma 4.13 replaces all of that by one bridge, and the criterion is what every consumer below actually uses; the block description is better read as a structure theorem than as a definition.

The dichotomy "linear or PC" is *by definition* exhaustive on irreducible congruences. Note that PC-ness is a negative condition: every lemma about PC congruences is proved by contradiction against Lemma 4.13(b).

Lemma 4.15 (No bridge crosses the types)**.** *If σ_1 is linear, σ_2 is irreducible, and there is a bridge from σ_1 to σ_2, then σ_2 is linear.*

Lemma 4.16 (Bridges from PC congruences are trivial)**.** *Let σ be a PC congruence on A and δ a reflexive bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = \sigma^*$. Then*

$$\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4) \quad \textit{or} \quad \delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_4) \wedge \sigma(x_2, x_3).$$

Lemma 4.17 (Bridges into PC congruences)**.** *Let δ be a bridge from a PC congruence σ_1 on $\mathbf{A}_1$ to an irreducible congruence σ_2 on $\mathbf{A}_2$, with $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$ and $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$. Then*

(a) *σ_2 is a PC congruence;*
(b) *$\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$;*
(c) *$\{(a/\sigma_1, b/\sigma_2) : (a, b) \in \widetilde{\delta}\}$ is bijective;*
(d) *$\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_3) \wedge \widetilde{\delta}(x_2, x_4)$ or $\delta(x_1, x_2, x_3, x_4) = \widetilde{\delta}(x_1, x_4) \wedge \widetilde{\delta}(x_2, x_3)$.*

Two results connect the new vocabulary to the linear and PC subuniverses of the original proof.

Lemma 4.18. *If σ is a linear congruence on $\mathbf{A} \in \mathcal{V}_n$ with $\sigma^* = A^2$, then $\mathbf{A}/\sigma \cong \mathbf{Z}_p$ for a prime p.*

Lemma 4.19. *If σ is a PC congruence on $\mathbf{A}$ with $\sigma^* = A^2$ and $\mathbf{A}/\sigma$ is BA and centre free, then $\mathbf{A}/\sigma$ is polynomially complete.*

Caveat 4.20 (What the two statements on top rest on)**.** Lemma 4.18 is proved in §7.7 from Lemma 7.47(c), and the route matters: with $\sigma^* = A^2$ that lemma has a single block, so $\mathbf{A}/\sigma \cong \mathbb{Z}_p^m$ as a group, and irreducibility then forces $m = 1$. That the resulting bijection is an isomorphism of *algebras* rather than of groups is Proposition 2.7, without which the statement does not typecheck against $\mathbf{Z}_p \in \mathcal{V}_n$.

Lemma 4.19 is *stated only.* Its proof needs one thing this document does not establish — that an idempotent algebra is polynomially complete exactly when every *reflexive* subuniverse of a power is a conjunction of equalities, those being precisely the invariants of the clone generated by the basic operations together with all constants — and one thing that is now a hypothesis: that $\mathbf{A}/\sigma$ is BA and centre free, without which the step $\{a_i\} <_{\mathrm{PC}} \mathbf{A}/\sigma$ is not licensed by Definition 4.22. Nothing below consumes the lemma; it is retained only to connect the vocabulary with [13], and the extra hypothesis therefore costs nothing.

Caveat 4.21 (Polynomial completeness)**.** An algebra is *polynomially complete* if the clone generated by its basic operations together with all constants is the clone of *all* operations on its universe. Lemma 4.19 is where the name "PC" comes from, and it is the only place below where polynomial completeness means more than a label; proving it needs the Istinger–Kaiser characterization [9]. Everywhere else "PC congruence" may be read as "irreducible and not linear".

4.3 The types

Definition 4.22 (The six types)**.** Let $\varnothing \neq C \lneq B \leq A$. We write:

- $C <^A_{\text{BA}} B$ if C is a binary absorbing subuniverse of $\mathbf{B}$;
- $C <^A_{\text{C}} B$ if C is a central subuniverse of $\mathbf{B}$;
- $C <^A_{\text{D}} B$ if there is an irreducible congruence σ on $\mathbf{A}$ with
  (i) $B^2 \subseteq \sigma^*$,
  (ii) $C = B \cap E$ for some block E of σ,
  (iii) $\mathbf{B}/\sigma$ is BA and centre free;
- $C <^A_{\text{L}} B$ if $C <^A_{\text{D}} B$ with the witnessing σ linear;
- $C <^A_{\text{PC}} B$ if $C <^A_{\text{D}} B$ with the witnessing σ a PC congruence;
- $C <^A_{\text{S}} B$ as in Definition 4.6.

When the witnessing congruence matters we write $C <^A_{T(\sigma)} B$; for $T \in \{\text{BA}, \text{C}, \text{S}\}$, where no congruence is involved, σ is taken to be A^2.

Caveat 4.23 (The superscript A is not decoration)**.** For $T \in \{\text{D}, \text{L}, \text{PC}\}$ the witnessing congruence lives on $\mathbf{A}$, not on $\mathbf{B}$, and condition (i) relates the two: $B^2 \subseteq \sigma^*$ says B sits inside a single block of σ^*. This is why the relation carries the ambient algebra as a superscript. For $T \in \{\text{BA}, \text{C}, \text{S}\}$ the superscript is genuinely irrelevant, and it is retained only so that one relation symbol serves all six types.

Note also that $\mathbf{B}/\sigma$ in (iii) means B/σ with the induced algebra structure, which requires B to be a subuniverse and σ restricted to B; since $B^2 \subseteq \sigma^*$ and not necessarily $B^2 \subseteq \sigma$, the quotient B/σ is in general *not* a single point, and its being BA and centre free is a real condition.

Definition 4.24 (Multi-types)**.** For $T \in \{\text{L}, \text{PC}, \text{D}\}$ write $C <^A_{\mathcal{M}T} B$ if $C \neq \varnothing$ and $C = C_1 \cap \cdots \cap C_t$ with $t \geq 1$ and $C_i <^A_T B$ for every $i \in [t]$.

Definition 4.25 (The strong-reduction relation)**.** Write $C \lhd\!\lhd\!\lhd^A B$ if there are $B_0, \ldots, B_m \subseteq B$ and $T_1, \ldots, T_m \in \{\text{BA}, \text{C}, \text{S}, \text{D}\}$ with

$$C = B_m <^A_{T_m} B_{m-1} <^A_{T_{m-1}} \cdots <^A_{T_1} B_0 = B.$$

Here $m = 0$ is allowed, so $\lhd\!\lhd\!\lhd^A$ is reflexive. A congruence *comes from* $C \lhd\!\lhd\!\lhd^A B$ if it is one of the witnessing congruences in such a chain. We write $B \lhd\!\lhd\!\lhd A$ for $B \lhd\!\lhd\!\lhd^A A$, and $C \leq^A_{T(\sigma)} B$ for "$C = B$ or $C <^A_{T(\sigma)} B$".

Caveat 4.26 ($\lhd\!\lhd\!\lhd$ is data, not a proposition)**.** It is tempting to read $C \lhd\!\lhd\!\lhd^A B$ as the reflexive transitive closure of $\bigcup_T <^A_T$ and to forget the chain. That is wrong, and the error is not cosmetic: Definition 4.25 also defines "the congruences coming from $C \lhd\!\lhd\!\lhd^A B$", which is a function of the chain and not of the pair. Two places consume it, and in both the *choice* of chain is what makes the argument work.

(a) In the "moreover" clause of Lemma 7.39(d) the conclusion is $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$ where the δ_i are the congruences of the chains for $B \lhd\!\lhd\!\lhd A$ and $C \lhd\!\lhd\!\lhd A$ *fixed at the top of that proof*; the proof establishes exactly $\sigma \cap \omega \subseteq \delta$ for those two intersections.
(b) In the type-D case of Lemma 7.54 the clause is applied to C_2 rather than to B_2, and its output is the intersection of ω with σ_2 — which is correct only because the chain witnessing $C_2 \lhd\!\lhd\!\lhd A$ is chosen to be the chain witnessing $B_2 \lhd\!\lhd\!\lhd A$ extended by the single step $C_2 <_{\text{D}(\sigma_2)} B_2$. A different chain gives a different family and the step fails.

So $\lhd\!\lhd\!\lhd^A$ carries, at each step, the intermediate set, its type, and its witnessing congruence, and extension by one step is an operation on that data. The bare existential statement may be used only where no congruence is named, which is most but not all of what follows. What *is* true, and is the central simplification over [13], is that the intermediate *types* may be forgotten: L and PC are absent from the list in Definition 4.25 because D subsumes them.

Convention 4.27 (Dotted variants)**.** $C \dot{<}^A_T B$, $C \dot{\leq}^A_T B$ and $C \mathrel{\dot{\lhd}\dot{\lhd}\dot{\lhd}}^A B$ mean that the undotted relation holds *or* $C = \varnothing$. See Caveat 2.5.

5 Instances

This section is entirely combinatorial: no algebra is used beyond the fact that each domain carries a w and each constraint relation is a subuniverse of a product of domains. It is also where the most representation decisions are forced, because instances are syntactic objects manipulated by operations — weakening, covering, renaming — that are usually described in words.

5.1 Instances and solutions

Definition 5.1 (Instance)**.** Fix a finite set V of variables and, for each $x \in V$, an algebra $\mathbf{D}_x \in \mathcal{V}_n$. A *constraint* is a pair $C = (\bar{x}, R)$ where $\bar{x} = (x_1, \ldots, x_m)$ is a tuple of variables and $R \leq \mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$. An *instance* $\mathcal{I}$ is a finite list of constraints; we write $C \in \mathcal{I}$, and $\mathrm{Var}(C)$, $\mathrm{Var}(\mathcal{I})$ for the variables occurring. A *subinstance* is a sublist.

A *solution* is a map s with $s(x) \in D_x$ for all x such that $(s(x_1), \ldots, s(x_m)) \in R$ for every constraint. The solution set of $\mathcal{I}$ is *subdirect* if for every x and every $a \in D_x$ there is a solution with $s(x) = a$.

Caveat 5.2 (Lists, not sets; tuples, not sets of variables)**.** Two representation choices are forced by the way the arguments below go, and both differ from the naive one.

Constraints form a list, not a set. Proofs delete "a constraint $C \in \mathcal{I}$" and reason about $\mathcal{I} \setminus \{C\}$, and the same relation may occur twice on different variable tuples — and, in the expanded coverings of §5.5, the same relation on the *same* tuple can legitimately occur twice. Multiset or list semantics is required.

A constraint's scope is a tuple, not a set. Repeated variables in a scope are allowed and occur — $\zeta(x, x', z)$ in §8 arises by splitting one variable into two, and $R(x, x, \ldots, x)$ appears explicitly in Definition 5.19. So $\mathrm{Var}(C)$ is the image of the scope, and $|\mathrm{Var}(C)|$ may be smaller than the arity.

Once repeated variables are allowed, every semantic operation that names a *variable* rather than a *position* must be pulled back along the equal-variable diagonal first. Definition 5.3 does that once, for all of them.

Definition 5.3 (The relation of a constraint)**.** Let $C = (\bar{x}, R)$ be a constraint with $\bar{x} = (x_1, \ldots, x_m)$ and $\mathrm{Var}(C) = \{x_1, \ldots, x_m\}$ as a *set*. The *relation of* C is

$$R^C = \Big\{ \bar{a} \in \prod_{z \in \mathrm{Var}(C)} D_z \ : \ (a_{x_1}, \ldots, a_{x_m}) \in R \Big\} \ \leq \ \prod_{z \in \mathrm{Var}(C)} \mathbf{D}_z,$$

the pullback of R along the map $\prod_{z \in \mathrm{Var}(C)} D_z \to \prod_{i \in [m]} D_{x_i}$ that repeats entries. It is indexed by *variables*, not by positions (Convention 2.6), and it is a subuniverse because the repeating map is a homomorphism.

Every semantic operation below that names a variable is taken on R^C, never on R: the projections $\mathrm{pr}_z(C)$ and $\mathrm{pr}_{z,z'}(C)$, the induced congruences $\mathrm{Con}_1(C, z)$, adjacency at a common variable, weakening by a set of variable names, and clause (ii) of Definition 5.11. If the entries of $\bar{x}$ are pairwise distinct then $R^C = R$ up to reindexing, and the distinction is invisible.

Caveat 5.4 (Why the pullback, and not the raw projection)**.** Repeated variables in a scope are allowed and they occur (Caveat 5.2), and once they do, a projection of R at a *position* is not a statement about the *variable*. Take $C = R(x, x)$ with $R = \{(0, 1), (1, 0)\}$. Then $\mathrm{pr}_1(R) = \{0, 1\}$ is full, while $R^C = \varnothing$: the constraint admits no value for x at all. Read positionally, C is 1-consistent; read correctly, it is unsatisfiable. The same collapse affects paths, adjacency and instance irreducibility, each of which would otherwise quantify over positions where it means variables.

One notion genuinely depends on the position and cannot be repaired this way: $\mathrm{Con}_1(R, i)$ of Definition 5.22 is defined for a relation and an index. For constraints we therefore use $\mathrm{Con}_1(C, z) = \mathrm{Con}_1(R^C, z)$, which changes *all* occurrences of z at once — the right notion for "the congruence C induces on the variable z" — and which is defined for every $z \in \mathrm{Var}(C)$ whether or not it repeats.

Definition 5.5 (Relations defined by an instance)**.** For $x_1, \ldots, x_m \in \mathrm{Var}(\mathcal{I})$, $\mathcal{I}(x_1, \ldots, x_m)$ is the set of $(a_1, \ldots, a_m)$ such that $\mathcal{I}$ has a solution with $s(x_i) = a_i$ for all i. It is a subuniverse of $\mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_m}$, being defined by a primitive positive formula over the constraint relations.

5.2 Reductions

Definition 5.6 (Reduction)**.** A *reduction* for $\mathcal{I}$ is a map $D^{(\top)}$ assigning to each $x \in \mathrm{Var}(\mathcal{I})$ a subuniverse $D_x^{(\top)} \subseteq D_x$, possibly empty. It is *nonempty* if every $D_x^{(\top)} \neq \varnothing$. The instance $\mathcal{I}^{(\top)}$ is $\mathcal{I}$ with each domain replaced by $D_x^{(\top)}$ and each constraint relation intersected with the corresponding box. For reductions we write $D^{(\perp)} \lhd\!\lhd\!\lhd\, D^{(\top)}$ and $D^{(\perp)} \leq_T D^{(\top)}$ when the corresponding relation holds at every variable. For $T = \textsc{ba}$ we write $D^{(\perp)} \leq_{\textsc{ba}(t)} D^{(\top)}$ for the stronger relation in which *one* binary term t witnesses the absorption at every variable; see Caveat 5.7.

Caveat 5.7 (A reduction of type BA carries a term, or it is useless)**.** $D^{(\perp)} \leq_{\textsc{ba}} D^{(\top)}$, read variablewise, gives a binary term t_x for each x and nothing that relates them. Its consumer, Lemma 6.12, needs a term common to all the coordinates it is applied to, and by Lemma 6.14 that is strictly more than the separate absorptions supply. So the uniform relation $\leq_{\textsc{ba}(t)}$ is the one that propagates, and the variablewise one is recorded here only to say that it is not enough.

Caveat 5.8 (Empty reductions)**.** Reductions are allowed to be empty at some or all variables — the maximal 1-consistent reduction below $D^{(B,\top)}$ in the proof of Theorem 8.1 is constructed exactly so that it may come out empty, and the case split on whether it does is the branch point of the whole argument. But $D^{(\perp)} \lhd\!\lhd\!\lhd\, D^{(\top)}$ requires nonemptiness at every variable (Convention 2.4). Both notions are needed and they must be different objects.

5.3 Consistency

Definition 5.9 (Paths)**.** A *path* in $\mathcal{I}$ is an alternating sequence $z_1 - C_1 - z_2 - \cdots - C_{l-1} - z_l$ with $z_i, z_{i+1} \in \mathrm{Var}(C_i)$. It *connects b and c* if there are $a_i \in D_{z_i}$ with $a_1 = b$, $a_l = c$, and $(a_i, a_{i+1}) \in \mathrm{pr}_{z_i, z_{i+1}}(R^{C_i})$ for every i; if $z_i = z_{i+1}$ this reads $a_i = a_{i+1}$. $\mathcal{I}$ is a *tree-instance* if it has no path with $l \geq 3$, $z_1 = z_l$ and $C_1, \ldots, C_{l-1}$ pairwise distinct.

Definition 5.10 (Consistency)**.** $\mathcal{I}$ is 1-*consistent* if $\mathrm{pr}_z(R^C) = D_z$ for every constraint C and every $z \in \mathrm{Var}(C)$. A reduction $D^{(\top)}$ is 1-consistent for $\mathcal{I}$ if $\mathcal{I}^{(\top)}$ is 1-consistent. $\mathcal{I}$ is *cycle-consistent* if it is 1-consistent and every path from z to z connects a to a, for every variable z and every $a \in D_z$.

Definition 5.11 (Linked, fragmented, irreducible)**.** $\mathcal{I}$ is *linked* if for every $z \in \mathrm{Var}(\mathcal{I})$ and every $a, b \in D_z$ some path from z to z connects a and b. $\mathcal{I}$ is *fragmented* if $\mathrm{Var}(\mathcal{I})$ splits into two disjoint nonempty sets X_1, X_2 with $\mathrm{Var}(C) \subseteq X_1$ or $\mathrm{Var}(C) \subseteq X_2$ for every C. $\mathcal{I}$ is *irreducible* if there is no instance $\mathcal{I}'$ with

(i) $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$;
(ii) every constraint of $\mathcal{I}'$ is $\mathrm{pr}_Y(R^C)$ on the scope Y, for some constraint C of $\mathcal{I}$ and some $\varnothing \neq Y \subseteq \mathrm{Var}(C)$;
(iii) $\mathcal{I}'$ is not fragmented;

(iv) $\mathcal{I}'$ is not linked;
(v) the solution set of $\mathcal{I}'$ is not subdirect.

Caveat 5.12 (Irreducibility is a Π_1 condition over a large family)**.** Conditions (i)–(v) quantify over *all* instances built from projections of constraints of $\mathcal{I}$ — a finite but exponentially large family (any multiset of projections of any constraints onto any subsets of their variables). It is finite, so the predicate is decidable, but it is not something to unfold. Every use in §8 goes through Lemma 8.14 and *never* through the definition.

5.4 Weakening and cruciality

Definition 5.13 (Weakening)**.** A constraint C_1 is *weaker or equivalent to* a constraint C_2 if $\mathrm{Var}(C_1) \subseteq \mathrm{Var}(C_2)$ and $\mathrm{pr}_{\mathrm{Var}(C_1)}(R^{C_2}) \subseteq R^{C_1}$ — that is, every assignment satisfying C_2 restricts to one satisfying C_1. It is *weaker* if the containment is proper or the variable set is. An instance $\mathcal{I}'$ is a *weakening* of $\mathcal{I}$ if $\mathrm{Var}(\mathcal{I}') \subseteq \mathrm{Var}(\mathcal{I})$ and every constraint of $\mathcal{I}'$ is weaker or equivalent to a constraint of $\mathcal{I}$. A *weakening step* on $\mathcal{I}$ replaces one constraint C by a finite nonempty family of constraints each weaker than C. Two cases are named: $\mathcal{I}[C \mapsto C']$ replaces C by the single C', and $\mathcal{I}[C \Uparrow]$ replaces C by *all* constraints weaker than C. Constraint relations are subuniverses (Definition 5.3), so the family of constraints weaker than C is finite and $\mathcal{I}[C \Uparrow]$ is well defined.

Definition 5.14 (The violation set)**.** For a constraint C of $\mathcal{I}$ put $\mathrm{Viol}(C) = \{f \in \prod_{x\in\mathrm{Var}(\mathcal{I})} D_x : f|_{\mathrm{Var}(C)} \notin R^C\}$, the assignments it rules out.

Lemma 5.15 (Weakening terminates)**.** *For a constraint C put $\nu(C) = \big(|\,\mathrm{Viol}(C)|, |\,\mathrm{Var}(C)|\big)$, and let V be the (finite) set of values ν takes on constraints weaker than or equal to a constraint of $\mathcal{I}$, ordered lexicographically. Then*

(a) *$\nu(C') < \nu(C)$ whenever C' is weaker than C;*
(b) *the set W of weakenings of $\mathcal{I}$ is finite;*
(c) *writing $v(\mathcal{J}) \in \mathbb{N}^V$ for the vector counting the constraints of $\mathcal{J}$ by their ν-value, every weakening step strictly decreases v in the lexicographic order on $\mathbb{N}^V$ that compares the coordinates from the largest element of V downwards. That order is well founded, so there is no infinite sequence of weakening steps.*

Proof. (a) If f satisfies C then $f|_{\mathrm{Var}(C')} = \mathrm{pr}_{\mathrm{Var}(C')}(f|_{\mathrm{Var}(C)}) \in \mathrm{pr}_{\mathrm{Var}(C')}(R^C) \subseteq R^{C'}$, so f satisfies C'; contrapositive, $\mathrm{Viol}(C') \subseteq \mathrm{Viol}(C)$, and $\mathrm{Var}(C') \subseteq \mathrm{Var}(C)$. If $\mathrm{Viol}(C') \subsetneq \mathrm{Viol}(C)$ the first coordinate drops. Otherwise $\mathrm{Viol}(C') = \mathrm{Viol}(C)$, and the variable sets cannot be equal: they would force $R^{C'} = R^C$, since every domain is nonempty, so a tuple on $\mathrm{Var}(C)$ extends to some f and membership in R^C is exactly non-membership of f in $\mathrm{Viol}(C)$ — and then C' would not be *weaker*. So $\mathrm{Var}(C') \subsetneq \mathrm{Var}(C)$ and the second coordinate drops.

(b) A constraint of a weakening has a scope inside $\mathrm{Var}(\mathcal{I})$ and a relation that is a subuniverse of the corresponding box, so it is drawn from a finite pool; a weakening is a set of such constraints.

(c) A weakening step removes one constraint of value ν and inserts finitely many of values strictly below ν, by (a). So the coordinate at ν drops by one, the coordinates above ν are untouched, and only coordinates below ν grow. A lexicographic order on a product of copies of $\mathbb{N}$ indexed by a *finite* ordered set is well founded. ☐

Caveat 5.16 (The two things a reading of "weaken C" has to do)**.** It has to make the process terminate, and it has to be strong enough for the consumers of cruciality. Two readings each fail one of these, and the failures pull in opposite directions.

Counting tuples fails on termination. A properly weaker constraint admits *more* tuples, so the total size of the constraint relations *increases* along the process and any measure built from it points the wrong way. What decreases is the number of assignments *excluded*, which is what Definition 5.14 counts; the second component of ν is there for the dummy-variable steps, which leave the violation set unchanged.

Quantifying over single *weaker constraints fails on strength.* It is tempting, because a step that replaces C by one C' visibly makes progress. But the resulting notion of cruciality is too weak for its consumers: Lemma 8.14 contradicts cruciality by exhibiting *two* weaker constraints R_1, R_2 with $R_1 \cap R_2 = R$, and replacing R by either one alone does gain a solution — it is replacing it by both at once that does not. So Definition 5.17 quantifies over $\mathcal{I}[C \Uparrow]$, and a weakening step is allowed to replace one constraint by a whole family.

And the family reading does not cost termination. The objection to it — that the conjunction of all constraints weaker than R can be R itself, as intersecting all the one-tuple extensions of R recovers R, so that $\mathcal{I}[C \Uparrow]$ has the same solutions as $\mathcal{I}$ — is true and does not matter. Progress is measured on the *constraints*, not on the solution set: each member of the family is strictly weaker than C, so Lemma 5.15(c) applies whether or not the solution set moves.

No algorithm performs weakening — it appears only inside proofs, as the device for reaching a crucial instance — which is why Definition 5.13 makes it a relation between instances and names the steps separately.

Definition 5.17 (Crucial)**.** Let $D^{(\top)}$ be a reduction for $\mathcal{I}$. A variable y_i of a constraint $R(y_1, \dots, y_t)$ is *dummy* if R does not depend on its i-th coordinate. A constraint $C \in \mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has no dummy variables, $\mathcal{I}^{(\top)}$ has no solution, and $\mathcal{I}[C \Uparrow]^{(\top)}$ has a solution. The instance $\mathcal{I}$ is *crucial in* $D^{(\top)}$ if it has at least one constraint and all of its constraints are crucial in $D^{(\top)}$.

Proposition 5.18 (Reaching a crucial instance)**.** *Let $D^{(\top)}$ be a reduction with $\mathcal{I}^{(\top)}$ unsolvable. Then finitely many weakening steps, each preserving unsolvability in $D^{(\top)}$, lead from $\mathcal{I}$ to an instance crucial in $D^{(\top)}$.*

Proof. Repeat the following while some constraint C of the current instance $\mathcal{J}$ fails to be crucial in $D^{(\top)}$; unsolvability of $\mathcal{J}^{(\top)}$ is an invariant, so the failure is of one of the other two clauses. If C has a dummy variable y, replace C by its projection off y: that projection is weaker, and since R^C does not depend on the y-coordinate the two have the same violation set, so unsolvability is preserved. Otherwise $\mathcal{J}[C \Uparrow]^{(\top)}$ has no solution, and we pass to $\mathcal{J}[C \Uparrow]$.

Both are weakening steps, so Lemma 5.15(c) makes the process halt, and it can halt only at an instance all of whose constraints are crucial. Every relation introduced is a projection of a subuniverse or a constraint relation, hence a subuniverse of the corresponding product of domains. □

5.5 Coverings

Definition 5.19 (Expanded covering)**.** $\mathcal{I}'$ is an *expanded covering* of $\mathcal{I}$, written $\mathcal{I}' \in \mathrm{Exp}(\mathcal{I})$, if there is $S\colon \mathrm{Var}(\mathcal{I}') \to \mathrm{Var}(\mathcal{I})$ with

(i) $S(x) = x$ for $x \in \mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}')$;
(ii) $D_x = D_{S(x)}$ for every $x \in \mathrm{Var}(\mathcal{I}')$;
(iii) for every constraint $R(x_1, \dots, x_m)$ of $\mathcal{I}'$, either $S(x_1), \dots, S(x_m)$ are pairwise distinct and $R(S(x_1), \dots, S(x_m))$ is weaker or equivalent to a constraint of $\mathcal{I}$, or $S(x_1) = \cdots = S(x_m)$ and $\{(a, \dots, a) : a \in D_{x_1}\} \subseteq R$.

It is a *covering* if moreover $R(S(x_1), \dots, S(x_m)) \in \mathcal{I}$ for every constraint; a *tree-covering* if it is a covering and a tree-instance. $S(x)$ is the *parent* of x, and x a *child*.

Proposition 5.20 (Basic properties)**.** *Let $\mathcal{I}'$ be an expanded covering of $\mathcal{I}$, with parent map S.*

(a) *Replacing every x by $S(x)$ and deleting the constraints $R(x,\dots,x)$ yields a weakening of $\mathcal{I}$.*
(b) *A weakening is an expanded covering with $S = \mathrm{id}$.*
(c) *Every solution of $\mathcal{I}$ lifts to a solution of $\mathcal{I}'$: $\alpha \circ S$ solves $\mathcal{I}'$ whenever α solves $\mathcal{I}$.*
(d) *If $\mathcal{I}$ is 1-consistent and $\mathcal{I}'$ is a tree-covering, the solution set of $\mathcal{I}'$ is subdirect.*
(e) *The union of two expanded coverings whose only common variables lie in $\mathrm{Var}(\mathcal{I})$ is an expanded covering (renaming arranges the proviso, Caveat 5.21).*
(f) *An expanded covering $\mathcal{I}''$ of an expanded covering $\mathcal{I}'$ of $\mathcal{I}$, with $\mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}'') \subseteq \mathrm{Var}(\mathcal{I}')$, is an expanded covering of $\mathcal{I}$.*
(g) *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*
(h) *Every reduction of $\mathcal{I}$ extends to $\mathcal{I}'$ by $D_x^{(\top)} := D_{S(x)}^{(\top)}$, and a 1-consistent one extends to a 1-consistent one.*

Proof. Two remarks on Definition 5.13 are used repeatedly. *Monotonicity*: an assignment satisfying a constraint satisfies every constraint weaker than or equivalent to it (the containment argument of Lemma 5.15(a)). *Renaming*: weaker-or-equivalent is transitive, and is preserved by an injective substitution applied to both scopes — both immediate from the definition.

(a) By Definition 5.19(iii), each surviving constraint $R(x_1,\dots,x_m)$ has $S(x_1),\dots,S(x_m)$ pairwise distinct and $R(S(x_1),\dots,S(x_m))$ weaker than or equivalent to a constraint of $\mathcal{I}$; the deleted ones are exactly those with S constant on the scope. The resulting variable set lies in $\mathrm{Var}(\mathcal{I})$, so Definition 5.13 is met.

(b) Take $S = \mathrm{id}$ on $\mathrm{Var}(\mathcal{I}_1) \subseteq \mathrm{Var}(\mathcal{I})$. Conditions (i) and (ii) of Definition 5.19 are trivial. For (iii): under the repeated-variable policy (Definition 5.3) a constraint's scope is a set, so the identity is pairwise distinct on it, and the constraint itself is the required weaker-or-equivalent constraint of $\mathcal{I}$.

(c) Let α solve $\mathcal{I}$ and put $\alpha' = \alpha \circ S$, legitimate since $D_x = D_{S(x)}$. A diagonal constraint $R(x,\dots,x)$ is satisfied because α' is constant on its scope and $\{(a,\dots,a)\} \subseteq R$. Any other constraint $R(x_1,\dots,x_m)$ has $R(S(x_1),\dots,S(x_m))$ weaker than or equivalent to a constraint of $\mathcal{I}$, which α satisfies; by monotonicity α satisfies $R(S(x_1),\dots,S(x_m))$, and that assertion *is* α' satisfying $R(x_1,\dots,x_m)$.

(d) A diagonal constraint is vacuous under the scope policy: its scope is the single variable x, and $\{(a,\dots,a) : a \in D_x\} \subseteq R$ gives $R^C = D_x$. Discard them, and induct on the number of remaining constraints; the claim is that for every $x_0 \in \mathrm{Var}(\mathcal{I}')$ and $a \in D_{x_0}$ some solution takes x_0 to a. With no constraints, assign a at x_0 and anything elsewhere. Otherwise first suppose some constraint $C_0 = R(x_1,\dots,x_m)$ of $\mathcal{I}'$ has x_0 in its scope. Since $\mathcal{I}'$ is a *covering*, $R(S(x_1),\dots,S(x_m))$ is a constraint of $\mathcal{I}$, and 1-consistency of $\mathcal{I}$ gives a tuple of R whose $S(x_0)$-entry is a; reading it along the scope $x_1,\dots,x_m$ — legitimate since $D_{x_i} = D_{S(x_i)}$ — gives $\bar\beta$ with $\beta_{x_0} = a$; assign C_0's variables by $\bar\beta$. Deleting C_0 splits the remaining constraints into groups $G_1,\dots,G_r$: two constraints lie in one group when a path avoiding C_0 joins their scopes. Each G_j shares *at most one* variable with C_0: two shared variables $z \neq w$ would give a path from z to w through G_j avoiding C_0, which closed up through C_0 is a path forbidden by Definition 5.9's tree condition. Each G_j is a tree-covering of $\mathcal{I}$ with fewer constraints, so by the inductive hypothesis its solution set is subdirect; take a solution of G_j agreeing with $\bar\beta$ at the shared variable if there is one, and any solution otherwise. The assignments agree wherever scopes meet, and every constraint of $\mathcal{I}'$ is satisfied. If no constraint contains x_0, run the same argument from any constraint (its groups likewise), and set $x_0 = a$ separately.

(e) The two parent maps agree on the common variables — both are the identity there, by the proviso and Definition 5.19(i) — so their union S is a function, and each clause of

Definition 5.19 holds because it holds on each half.

(f) Let $S_2\colon \mathrm{Var}(\mathcal{I}'') \to \mathrm{Var}(\mathcal{I}')$ and $S_1\colon \mathrm{Var}(\mathcal{I}') \to \mathrm{Var}(\mathcal{I})$ be the parent maps; take $S = S_1 \circ S_2$. (i): for $x \in \mathrm{Var}(\mathcal{I}) \cap \mathrm{Var}(\mathcal{I}'')$ the proviso puts $x \in \mathrm{Var}(\mathcal{I}')$, so $S_2(x) = x$ and $S_1(x) = x$. (ii) composes. (iii): let $R(x_1, \ldots, x_m)$ be a constraint of $\mathcal{I}''$. If S_2 is constant on the scope then so is S, and the diagonal clause carries over. Otherwise S_2 is pairwise distinct there and $C' = R(S_2(x_1), \ldots, S_2(x_m))$ is weaker than or equivalent to a constraint C_1 of $\mathcal{I}'$; note $\mathrm{Var}(C') \subseteq \mathrm{Var}(C_1)$. Apply (iii) for $\mathcal{I}'$ to C_1. If S_1 is constant on $\mathrm{Var}(C_1)$ then S is constant on the scope of R, and the diagonal of the common domain lies in R^{C_1}, hence — projecting to $\mathrm{Var}(C')$, by weaker-or-equivalence — in R. If S_1 is pairwise distinct on $\mathrm{Var}(C_1)$, then by renaming $C'(S_1\,\cdot)$ is weaker than or equivalent to $C_1(S_1\,\cdot)$, which is weaker than or equivalent to a constraint of $\mathcal{I}$; transitivity finishes, and S is pairwise distinct on the scope because S_1 is injective on $\mathrm{Var}(C_1) \supseteq \mathrm{Var}(C')$.

(g) This is Lemma 8.11, imported.

(h) $D^{(\top)}_{S(x)}$ is a subuniverse of $\mathbf{D}_{S(x)} = \mathbf{D}_x$, so the extension is a reduction. For 1-consistency, let C be a constraint of $\mathcal{I}'^{(\top)}$ and z in its scope. If C is diagonal, $R \supseteq \{(a, \ldots, a)\}$ meets the box $(D^{(\top)}_x)^m$ in all of $D^{(\top)}_x$ at every coordinate. Otherwise let C_0 be the constraint of $\mathcal{I}$ that $R(S(x_1), \ldots, S(x_m))$ weakens or equals; for $a \in D^{(\top)}_z = D^{(\top)}_{S(z)}$, 1-consistency of $\mathcal{I}^{(\top)}$ at C_0 gives a tuple of R^{C_0} in C_0's box with $S(z)$-entry a. Its projection to $\mathrm{Var}(C')$ lies in $R^{C'}$ by weaker-or-equivalence and keeps the $S(z)$-entry a; reading it back along the scope $x_1, \ldots, x_m$ — whose boxes match through S, by the definition of the extension — gives a tuple of C's relation in C's box with z-entry a. So pr_z of each reduced constraint is all of $D^{(\top)}_z$. □

Caveat 5.21 (Variable renaming carries obligations)**.** The constructions of §8 say "rename the variables so that the only common variable of Υ_x and $\mathcal{J}$ is x", and form conjunctions $\bigwedge_{\mathcal{J}\in\Omega} \mathcal{J}$ of instances whose variable sets must first be made disjoint. Renaming is not free: each of 1-consistency, cycle-consistency, cruciality and being a covering has to be shown invariant under it. Those four facts are used throughout §8 and are nowhere stated.

5.6 Induced congruences and connectedness

Definition 5.22 (Con_1)**.** For R of arity m and $i \in [m]$, $\mathrm{Con}_1(R, i)$ is the binary relation

$$\{(y, y') \;:\; \exists\, \bar{x} \;\; R(\bar{x}[i \mapsto y]) \wedge R(\bar{x}[i \mapsto y'])\}.$$

For a constraint C and $z \in \mathrm{Var}(C)$ we write $\mathrm{Con}_1(C, z)$ for $\mathrm{Con}_1(R^C, z)$, the index set of R^C being $\mathrm{Var}(C)$ (Definition 5.3). For an instance, $\mathrm{Con}(\mathcal{I}, x) = \{\mathrm{Con}_1(C, x) : C \in \mathcal{I},\ x \in \mathrm{Var}(C)\}$ and $\mathrm{Con}(\mathcal{I}) = \bigcup_x \mathrm{Con}(\mathcal{I}, x)$.

Lemma 5.23. *Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $i \in [m]$.*

(a) $\mathrm{Con}_1(R, i)$ *is a symmetric subuniverse of* $\mathbf{A}_i^2$*, reflexive on* $\mathrm{pr}_i(R)$.

(b) *The i-th variable of R is rectangular if and only if it is stable under* $\mathrm{Con}_1(R, i)$.

(c) *If the i-th variable of R is rectangular then* $\mathrm{Con}_1(R, i)$ *is transitive; if moreover* $\mathrm{pr}_i(R) = A_i$ *then* $\mathrm{Con}_1(R, i)$ *is a congruence on* $\mathbf{A}_i$.

Proof. (a) Symmetry is the symmetry of the defining formula in y, y'. For $a \in \mathrm{pr}_i(R)$, a witnessing tuple $\bar{x}[i \mapsto a] \in R$ serves as its own partner, giving $(a, a) \in \mathrm{Con}_1(R, i)$. Compatibility: if $(y_j, y'_j) \in \mathrm{Con}_1(R, i)$ for $j \in [n]$, with witnesses $\bar{x}^j$, then applying w coordinatewise to the tuples $\bar{x}^j[i \mapsto y_j]$ gives $w(\bar{x}^1, \ldots, \bar{x}^n)[i \mapsto w(\bar{y})] \in R$, and likewise with y'_j; the common tuple $w(\bar{x}^1, \ldots, \bar{x}^n)$ witnesses $\big(w(\bar{y}), w(\bar{y}')\big) \in \mathrm{Con}_1(R, i)$.

(b) Suppose the variable is stable and let $\bar{a}[i \mapsto b_i]$, $\bar{b}[i \mapsto a_i]$, $\bar{b} \in R$. The pair $\bar{b}$, $\bar{b}[i \mapsto a_i]$ witnesses $(b_i, a_i) \in \mathrm{Con}_1(R, i)$, and stability applied to $\bar{a}[i \mapsto b_i] \in R$ replaces b_i by a_i, giving

$\bar{a} \in R$: the variable is rectangular. Conversely suppose it is rectangular, let $\bar{a} \in R$ and $(a_i, y) \in \mathrm{Con}_1(R, i)$ with witness $\bar{x}$, so $\bar{x}[i \mapsto a_i]$, $\bar{x}[i \mapsto y] \in R$. Apply rectangularity to the target $\bar{a}[i \mapsto y]$ against $\bar{b} = \bar{x}[i \mapsto a_i]$: the three hypotheses read $\bar{a}[i \mapsto y][i \mapsto a_i] = \bar{a} \in R$, $\bar{b}[i \mapsto y] = \bar{x}[i \mapsto y] \in R$, and $\bar{b} \in R$; the conclusion is $\bar{a}[i \mapsto y] \in R$.

(c) Let (y, y') and (y', y'') lie in $\mathrm{Con}_1(R, i)$, with witnesses $\bar{x}$ and $\bar{z}$. By (b) the variable is stable, so from $\bar{z}[i \mapsto y'] \in R$ and $(y', y) \in \mathrm{Con}_1(R, i)$ — symmetry — we get $\bar{z}[i \mapsto y] \in R$; together with $\bar{z}[i \mapsto y''] \in R$ this witnesses $(y, y'') \in \mathrm{Con}_1(R, i)$. With $\mathrm{pr}_i(R) = A_i$, (a) makes the relation reflexive on all of A_i, and (a)'s compatibility makes the equivalence a congruence. □

Caveat 5.24. $\mathrm{Con}_1(R, i)$ is always reflexive and symmetric but is *not* transitive in general, hence not always a congruence; Lemma 5.23 is what makes it one, and its hypotheses (subdirect *and* rectangular at i) are both needed. The notation invites the reader to assume congruence-hood, and every use below must check the hypotheses.

Definition 5.25 (Types of relations and instances)**.** R is *of PC type* (resp. *linear type*) if R is rectangular and every $\mathrm{Con}_1(R, i)$ is a PC (resp. linear) congruence. A constraint C is of PC or linear type if R^C is; an instance is if all its constraints are.

Definition 5.26 (Adjacency, connectedness)**.** Two congruences σ_1, σ_2 on $\mathbf{D}_x$ are *adjacent* if there is a reflexive bridge from σ_1 to σ_2. Two constraints C_1, C_2 with R^{C_1}, R^{C_2} rectangular are *adjacent in a common variable* $x \in \mathrm{Var}(C_1) \cap \mathrm{Var}(C_2)$ if $\mathrm{Con}_1(C_1, x)$ and $\mathrm{Con}_1(C_2, x)$ are adjacent. An instance $\mathcal{I}$ is *connected* if all its constraints are rectangular, all congruences in $\mathrm{Con}(\mathcal{I})$ are irreducible, and the graph on the constraints with edges the adjacent pairs is connected.

Note that every proper congruence is adjacent to itself, via $\delta(x_1, x_2, x_3, x_4) = \sigma(x_1, x_3) \wedge \sigma(x_2, x_4)$.

6 Properties of strong subuniverses

This section is the *interface* between the algebraic theory and the CSP argument: §8 uses the results below and nothing else from §§4. The proofs are §7.

Its shape is worth naming. Four of the five groups say that the relation $\lhd\lhd\lhd$ and the types T *propagate*: forward and backward along surjective homomorphisms, down to quotients, across products of a congruence, and through subdirect relations. The fifth — Theorem 6.18 — is the one genuinely hard statement, and it is what converts a failure of consistency into a bridge. Everything the CSP proof does with bridges traces back to it.

Convention 6.1. Where a type is not specified, T ranges over $\{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{pc}, \textsc{l}, \textsc{d}\}$. Where a multi-type $\mathcal{M}T$ appears, T ranges over $\{\textsc{pc}, \textsc{l}, \textsc{d}\}$.

6.1 Ubiquity

Lemma 6.2 (Ubiquity)**.** *If $B \lhd\lhd\lhd A$ and $|B| > 1$, then $C <_T^A B$ for some C and some $T \in \{\textsc{ba}, \textsc{c}, \textsc{l}, \textsc{pc}\}$.*

Lemma 6.2 is the formal content of Informal Claim 8.1: every domain of size at least 2 that has been reached by a chain of strong reductions admits another strong reduction. It is what guarantees the outer loop of the algorithm makes progress, and it is the reason the algorithm terminates with singleton domains or a linear instance.

Caveat 6.3. The hypothesis is $B \lhd\lhd\lhd A$, not $B \leq A$. A subalgebra reached by no chain of strong reductions need admit nothing. In the algorithm the hypothesis is maintained as an invariant of the reduction loop, and it is carried rather than re-derived.

6.2 Propagation

The backward direction is separated out, and the separation is not cosmetic; Caveat 6.6 says why.

Lemma 6.4 (Reverse homomorphism)**.** *Let $f\colon \mathbf{A} \to \mathbf{A}'$ be a surjective homomorphism. Then*

(b) $C' \lhd\lhd\lhd^{A'} B' \Rightarrow f^{-1}(C') \lhd\lhd\lhd f^{-1}(B')$;

(bt) $C' <_{T(\sigma)}^{A'} B' \Rightarrow f^{-1}(C') <_{T(f^{-1}(\sigma))}^{A} f^{-1}(B')$;

(bm) $C' \leq_{\mathcal{M}T}^{A'} B' \lhd\lhd\lhd A' \Rightarrow f^{-1}(C') \leq_{\mathcal{M}T}^{A} f^{-1}(B')$.

Lemma 6.5 (Propagation along a homomorphism)**.** Let $f\colon \mathbf{A} \to \mathbf{A}'$ be a surjective homomorphism. Then clauses (b), (bt), (bm) of Lemma 6.4 hold, and moreover

(f) $C \lhd\lhd\lhd^{A} B \Rightarrow f(C) \lhd\lhd\lhd f(B)$;

(ft) $C <_{T(\sigma)}^{A} B \lhd\lhd\lhd A \Rightarrow (f(C) = f(B)$ or $f(C) <_{\textsc{s}} f(B)$ or $f(C) <_T^{A'} f(B))$;

(fs) $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ and $C <_T B \Rightarrow f(C) \leq_T f(B)$;

(fm) $C \leq_{\mathcal{M}T}^{A} B \lhd\lhd\lhd A$ and $f(B)$ $\textsc{s}$-free $\Rightarrow f(C) \leq_{\mathcal{M}T}^{A'} f(B)$.

Caveat 6.6 (Why the backward clauses are a separate lemma)**.** The backward direction is much the easier half, and several consumers need only it. Bundling it with the forward clauses makes those consumers cite the forward clauses too — and the forward clauses rest on §7.6, which rests back on them.

Concretely, with the clauses bundled the following is a cycle: Corollary 7.41 → Lemma 6.5 → Lemma 7.60 → Lemma 7.57 → Lemma 7.56 → Lemma 7.55 → Corollary 7.41. Every edge is a real citation. The cycle is spurious only because the first edge uses (b), which needs none of the rest; splitting the lemma is what makes that visible.

Caveat 6.7 (The three-way disjunction in (ft))**.** Clause (ft) is the characteristic shape of this theory and the reason it is usable. Pushing a strong subset forward along a homomorphism can

- collapse it ($f(C) = f(B)$: the reduction becomes vacuous),
- degrade it ($f(C) <_{\textsc{s}} f(B)$: still strong, but the type is lost),
- or preserve it ($f(C) <_T^{A'} f(B)$: same type).

The type S exists to be the value of the middle case. Every downstream proof that pushes a reduction through a quotient does a three-way case split here, and the middle case is the one that is easy to forget; in [13] it is what forces the "go forward and backward from global to local" induction that the present route eliminates.

Note the asymmetry: the backward clauses (b), (bt), (bm) are clean implications with no disjunction. Preimages behave; images do not.

Corollary 6.8 (Propagation to a quotient)**.** *Let δ be a congruence on $\mathbf{A}$. Then* (f) $C \lhd\lhd\lhd^A B \Rightarrow C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta$*;* (t) $C <_{T(\sigma)}^A B \lhd\lhd\lhd A \Rightarrow (C/\delta = B/\delta$ *or* $C/\delta <_{\textsc{s}} B/\delta$ *or* $C/\delta <_T^{A/\delta} B/\delta)$*;* (s) *for* $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$*,* $C <_T B \Rightarrow C/\delta \leq_T B/\delta$*;* (m) $C \leq_{\mathcal{M}T}^A B \lhd\lhd\lhd A$ *and* B/δ S*-free* $\Rightarrow C/\delta \leq_{\mathcal{M}T}^{A/\delta} B/\delta$*.*

Corollary 6.9 (Reflection from a quotient)**.** *Let δ be a congruence on $\mathbf{A}$ and $B, C \leq \mathbf{A}$. Then $C/\delta \lhd\lhd\lhd^{A/\delta} B/\delta \iff C \circ \delta \lhd\lhd\lhd^A B \circ \delta$, and $C/\delta <_T^{A/\delta} B/\delta \iff C \circ \delta <_T^A B \circ \delta$.*

Corollary 6.10 (Multiplying by a congruence)**.** *Let δ be a congruence on $\mathbf{A}$. Then* (f) $C \lhd\lhd\lhd^A B \Rightarrow C \circ \delta \lhd\lhd\lhd^A B \circ \delta$*;* (t) $C <_{T(\sigma)}^A B \lhd\lhd\lhd^A A \Rightarrow (C \circ \delta = B \circ \delta$ *or* $C \circ \delta <_{\textsc{s}}^A B \circ \delta$ *or* $C \circ \delta <_T^A B \circ \delta)$*;* (e) *if* $\delta \subseteq \sigma$ *and* $C <_{T(\sigma)}^A B \lhd\lhd\lhd^A A$ *then* $C \circ \delta <_T^A B \circ \delta$*.*

Corollary 6.11 (Propagation to relations)**.** Let $R \leq_{\text{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $B_i \lhd\lhd\lhd A_i$. Then

(r) $R \cap (B_1 \times \cdots \times B_m) \dot{\lhd}\dot{\lhd}\dot{\lhd} R$;
(r1) $\mathrm{pr}_1(R \cap (B_1 \times \cdots \times B_m)) \dot{\lhd}\dot{\lhd}\dot{\lhd} A_1$;
(b) if $C_i \lhd\lhd\lhd^{A_i} B_i$ for all i then $R \cap (C_1 \times \cdots \times C_m) \dot{\lhd}\dot{\lhd}\dot{\lhd}^R R \cap (B_1 \times \cdots \times B_m)$;
(b1) under the same hypothesis, $\mathrm{pr}_1(R \cap \prod C_i) \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A_1} \mathrm{pr}_1(R \cap \prod B_i)$;
(m) if $C_i \leq_{\mathcal{M}T}^{A_i} B_i$ for all i then $R \cap \prod C_i \dot{\leq}_{\mathcal{M}T}^R R \cap \prod B_i$;
(m1) if in addition $\mathrm{pr}_1(R \cap \prod B_i)$ is S-free, then $\mathrm{pr}_1(R \cap \prod C_i) \dot{\leq}_{\mathcal{M}T}^{A_1} \mathrm{pr}_1(R \cap \prod B_i)$.

For the two absorption types the conclusion is stronger — no chain, no dot on the type, and no subdirectness hypothesis:

Lemma 6.12 ([15, Cor. 6.1.2, 6.9.2])**.** *Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ with* $\mathrm{pr}_1(R) = A_1$*, and let $C_i \leq_T A_i$ for every i, where $T \in \{\textsc{ba}, \textsc{c}\}$ — and, when $T = \textsc{ba}$, with a* single *binary term t witnessing all m absorptions. Then* $\mathrm{pr}_1(R \cap (C_1 \times \cdots \times C_m)) \dot{\leq}_T A_1$*, again with the same t when $T = \textsc{ba}$.*

Caveat 6.13 (Both hypotheses of Lemma 6.12 are needed)**.** Lemma 6.12 is the workhorse of case (2) of Theorem 8.1, and neither of its hypotheses may be dropped.

Subdirectness. Without $\mathrm{pr}_1(R) = A_1$ the statement is false. Take $R = \{(0, 0)\} \leq \mathbf{Z}_p \times \mathbf{Z}_p$ — a subuniverse, by idempotence — and $C_i = A_i$, which is allowed because $\leq_T$ admits equality. Then $R \cap (C_1 \times C_2) = R$ and $\mathrm{pr}_1(R) = \{0\}$, which is neither empty nor all of $\mathbf{Z}_p$, so the conclusion would assert $\{0\} <_{\textsc{ba}} \mathbf{Z}_p$, contradicting Lemma 6.36.

The common term. In the BA case the m absorptions must be witnessed by a *single* binary t; that is why $\leq_{\textsc{ba}(t)}$ carries the term in the notation, and the conclusion returns the same t.

At the call site inside Theorem 8.1(2) the second hypothesis was, until now, not discharged: the theorem recorded that every coordinate reduction has type BA, which by Lemma 6.14 does not by itself produce a term common to all of them. The statement of Theorem 8.1(2) now carries the witness instead (Definition 5.6, Caveat 5.7); what that buys and what it costs is Caveat 6.15.

Lemma 6.14 (When a common witness exists)**.** *Let* $C_i \leq_{\mathrm{BA}} \mathbf{A}_i$ *for* $i \in [m]$*, with each* C_i *nonempty. The following are equivalent:*

(a) *some one binary term* t *witnesses all* m *absorptions;*
(b) $C_1 \times \cdots \times C_m \leq_{\mathrm{BA}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$.

Proof. If t witnesses all m, then for $\bar{c}$ in the product of the C_i and $\bar{a}$ in the product of the A_i, the tuple $t(\bar{c}, \bar{a})$ has i-th entry $t(c_i, a_i) \in C_i$, and symmetrically; so t witnesses (b). Conversely let t witness (b), fix i, and let $c \in C_i$, $a \in A_i$. Choose $\bar{c}$ in the product of the C_j with $\bar{c}_i = c$, possible since every C_j is nonempty, and $\bar{a}$ in the product of the A_j with $\bar{a}_i = a$. Then $t(\bar{c}, \bar{a})$ lies in the product of the C_j, so its i-th entry $t(c, a)$ lies in C_i; symmetrically on the other side. So t witnesses the i-th absorption. $\square$

Caveat 6.15 (What the equivalence does and does not settle)**.** Lemma 6.14 is a translation, not a construction: it says the missing hypothesis of Lemma 6.12 is exactly *productivity* of binary absorption over the family in question, and nothing here supplies that. The two ways out are to prove such a productivity statement, or to carry the witness in the data; this document takes the second, so Theorem 8.1(2) hypothesises $D^{(2)} \leq_{\mathrm{BA}(t)} D^{(1)}$ and Lemma 6.12 applies at its call site with that t.

The cost is that the debt moves rather than vanishing: every producer of a type-BA reduction must now be checked to produce a uniform one. The producers are the type-BA alternatives of Theorem 8.1(1) and of the reductions found by the algorithm, and **that check has not been made** — it lives in the part of [14] this document has not read adversarially. See §11.

6.3 Intersections

Lemma 6.16 (Intersecting with a strong subset)**.** *If* $B \lhd\!\lhd\!\lhd A$ *and* $D \lhd\!\lhd\!\lhd A$ *then* (i) $B \cap D\ \dot{\lhd}\!\dot{\lhd}\!\dot{\lhd}^{A}\ D$*, and* (t) $C <^{A}_{T(\sigma)} B \Rightarrow C \cap D \dot{\leq}^{A}_{T(\sigma)} B \cap D$.

Corollary 6.17. *Under the same hypotheses,* $B \cap D\ \dot{\lhd}\!\dot{\lhd}\!\dot{\lhd}\ A$.

The next theorem is the heart of §6. Read it as follows: if a family of strong reductions is *jointly* inconsistent but *minimally* so — dropping any one of them restores consistency — then the types must agree, and in the linear case one gets bridges between the witnessing congruences. Bridges are produced nowhere else.

Theorem 6.18 (Stable intersection)**.** Suppose

(i) $C_i <^{A}_{T_i(\sigma_i)} B_i \lhd\!\lhd\!\lhd A$ with $T_i \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}, \mathrm{L}, \mathrm{PC}\}$, for $i \in [m]$, $m \geq 2$;
(ii) $\bigcap_{i \in [m]} C_i = \varnothing$;
(iii) $B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing$ for every $j \in [m]$.

Then one of:

(ba) $T_1 = \cdots = T_m = \mathrm{BA}$;
(l) $T_1 = \cdots = T_m = \mathrm{L}$, and for all $k, \ell \in [m]$ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$;
(c) $m = 2$ and $T_1 = T_2 = \mathrm{C}$;
(pc) $m = 2$, $T_1 = T_2 = \mathrm{PC}$ and $\sigma_1 = \sigma_2$.

Corollary 6.19 (Stable intersection, relational form)**.** Suppose $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$, $C_i <^{A_i}_{T_i(\sigma_i)} B_i \lhd\!\lhd\!\lhd A_i$ with $T_i \in \{\mathrm{BA}, \mathrm{C}, \mathrm{S}, \mathrm{L}, \mathrm{PC}\}$ and $m \geq 2$, $R \cap (C_1 \times \cdots \times C_m) = \varnothing$, and $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_m) \neq \varnothing$ for every j. Then one of: (ba) all $T_i = \mathrm{BA}$; (l) all $T_i = \mathrm{L}$ and for all k, ℓ a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$; (c) $m = 2$ and $T_1 = T_2 = \mathrm{C}$; (pc) $m = 2$, $T_1 = T_2 = \mathrm{PC}$, $\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$, and $\{(a/\sigma_1, b/\sigma_2) : (a, b) \in R\}$ is bijective.

Remark 6.20. Several applications need more than one restriction on a single coordinate of R. The statement is kept simple; to apply it, duplicate the coordinate (form $R' = \{(\bar{a}, a_k) : \bar{a} \in R\}$) and restrict the copies separately.

Caveat 6.21 (Minimality is a hypothesis, not a construction)**.** Hypotheses (ii)–(iii) of Theorem 6.18 say the family is *critically* inconsistent. In applications one starts with an inconsistent family and shrinks it to a critical subfamily. The shrinking step is elementary — *if P is a monotone predicate on subsets of a finite set X and $P(X)$ holds, there is $M \subseteq X$ minimal with $P(M)$* — but it is performed at least four times below, each time with a different P, so it is worth naming rather than repeating.

A related step is used with the multi-types: an inclusion-minimal $\mathcal{M}T$ subuniverse of B *containing a given point b* is automatically minimal outright. That is not immediate, and it is what Lemma 8.12 consumes. It holds because $M(b) = B \cap \bigcap_{\sigma \in \Sigma} [b]_\sigma$, over the T-dividing congruences Σ for B, is the *least* $\mathcal{M}T$ subuniverse containing b (each $B \cap [b]_\sigma$ is $<^A_T B$, and any $\mathcal{M}T$ set containing b is an intersection of blocks each containing b), and because $c \in M(b)$ forces $[c]_\sigma = [b]_\sigma$ for every $\sigma \in \Sigma$ and hence $M(c) = M(b)$. So the image of $b \mapsto M(b)$ consists of pairwise disjoint minimal sets, and a minimal set containing b is $M(b)$, which has no proper $\mathcal{M}T$ subset.

6.4 Multi-types

Lemma 6.22. *If $C \leq^A_{\mathcal{M}T} B$ then there are $s \geq 0$ and $C = E_s <^A_T \ldots <^A_T E_0 = B$, and $C \lhd\!\lhd\!\lhd^A B$. The chain may be* empty*: $s = 0$ is the case $C = B$, which $\leq_{\mathcal{M}T}$ admits and which no strict chain covers.*

That is: an intersection of type-T subsets, though not itself obtained by a single reduction of type T, is reachable by a *chain* of type-T reductions. This is what makes $\mathcal{M}T$ compatible with $\lhd\!\lhd\!\lhd$, and it is used whenever the algorithm reduces several variables at once.

Lemma 6.23 (Preserving linkedness)**.** *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $C_i \leq^{A_i}_{\mathcal{M}\mathrm{D}} B_i \lhd\!\lhd\!\lhd A_i$ for $i = 1, 2$, let S be the rectangular closure of R, and suppose $R \cap (B_1 \times B_2) \neq \varnothing$ and $S \cap (C_1 \times C_2) \neq \varnothing$. Then $R \cap (C_1 \times C_2) \neq \varnothing$.*

Lemma 6.24 (Preserving linkedness, anchored form)**.** *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, let $C_1 \lhd\!\lhd\!\lhd A_1$, let $C_2 <^{A_2}_{\mathrm{D}(\sigma)} B_2 \lhd\!\lhd\!\lhd A_2$, and let S be the rectangular closure of R. Suppose*

(i) *every element of C_1 has an R-neighbour in B_2; and*
(ii) *$S \cap (C_1 \times C_2) \neq \varnothing$.*

Then $R \cap (C_1 \times C_2) \neq \varnothing$.

Caveat 6.25 (What the anchor buys)**.** Membership in the rectangular closure places a pair in the same linked component of R; both lemmas then ask for an actual edge of that component in a box. The difficulty in Lemma 6.23 is not the types. At a step of the right-hand chain where $(\mathrm{RightLinked}(R) \cap B^2)/\sigma$ is full, the intersection cannot die at all — Lemma 7.56, applied to the transpose with the left-hand set as its second factor, closes it outright. At a step where it is not full, Lemma 7.57(1) pins every linked component of R to a single σ-block of B; but then the known mixed nonemptiness $R\cap(C_1\times B_2)\neq\varnothing$ may lie in a *different* linked component from the witness of $S\cap(C_1\times C_2)\neq\varnothing$, and nothing in this section moves an edge between components while keeping one end in C_1. Hypothesis (i) of Lemma 6.24 dissolves exactly that obstruction: the witness's own neighbour is an edge of the right component, and the full/cut dichotomy then closes both ways.

The anchor is not free at the call site of Lemma 6.23 in Lemma 8.21: there it would demand that every section of the crucial constraint over one block of coordinates whose entries lie in $D^{(2)}$ extend to a full tuple with the remaining entries in $D^{(1)}$, and 1-consistency of $D^{(1)}$ is a per-variable statement, not a per-tuple one. So Lemma 6.23 stays as stated, with Lemma 6.24 as its proved core.

Lemma 6.26 (Anchored stable intersection)**.** *Let $W\leq_{\rm sd}\mathbf{A}_1\times\mathbf{A}_2$, let $C_1<^{A_1}_{T_1(\sigma_1)}B_1\lhd\!\lhd\!\lhd A_1$ with $T_1\in\{\textsc{ba},\textsc{c},\textsc{s},\textsc{l},\textsc{pc}\}$, and let $C_2\leq^{A_2}_{\mathcal{M}T}B_2\lhd\!\lhd\!\lhd A_2$ with $T\in\{\textsc{l},\textsc{pc},\textsc{d}\}$. Suppose*

(i) *$W\cap(B_1\times C_2)\neq\varnothing$ and $W\cap(C_1\times B_2)\neq\varnothing$;*
(ii) *$T_1\in\{\textsc{ba},\textsc{c},\textsc{s}\}$; or $T_1=\textsc{l}$ and $T=\textsc{pc}$; or $T_1=\textsc{pc}$ and $T=\textsc{l}$.*

Then $W\cap(C_1\times C_2)\neq\varnothing$.

Caveat 6.27 (What "anchored" means, and what the lemma does not cover)**.** Corollary 6.19 requires its relation subdirect over the *ambients* of the typed steps, and its leave-one-out condition frees a coordinate to the top of its own step. In applications the known nonemptiness lives lower: the relation is subdirect only over a standing reduction — 1-consistency of a reduction $D^{(1)}$ supplies exactly subdirectness over the $D^{(1)}_x$ — and the known emptiness has the reduction, not the full domain, at every other coordinate. Lemma 6.26 is the form that consumes such data: hypothesis (i) is the anchor, and the leave-one-out configurations that Corollary 6.19 needs are manufactured inside the proof, by walking down the chain of Lemma 6.22 to the first step at which the intersection dies. It is the analogue, in this development, of the fit-in lemmas of [13], which walk chains of subuniverses with an anchoring tuple.

Hypothesis (ii) is exactly the condition that *no* alternative of Corollary 6.19 admits the pair of types $\{T_1,T'\}$ with T' the class of a step of the chain: $\textsc{s}$ appears in no alternative, $\textsc{ba}$ and $\textsc{c}$ only alongside themselves, and the two dividing classes only alongside themselves. What the lemma therefore does *not* cover is a family of steps of one class — two $\textsc{pc}$ blocks, or a critically inconsistent family of $\textsc{ba}$ steps — where an alternative of the corollary is consistent with the data; those configurations retain their own routes (Lemma 7.24 for $\textsc{s}$ at set level, Claim 8.23 for what remains open).

Lemma 6.28 (Irreducibles above a typed family)**.** *Let $B\lhd\!\lhd\!\lhd A$ with $\mathbf{B}$ $\textsc{s}$-free, let $T\in\{\textsc{l},\textsc{pc},\textsc{d}\}$, and for $i\in[t]$, $t\geq 1$, let τ_i be an irreducible congruence on $\mathbf{A}$ — linear if $T=\textsc{l}$, PC if $T=\textsc{pc}$ — with $\tau_i^*\supseteq B^2$ and $\tau_i\not\supseteq B^2$. If δ is an irreducible congruence on $\mathbf{A}$ with $\tau_1\cap\cdots\cap\tau_t\subseteq\delta$ and $B^2\not\subseteq\delta$, then δ is linear if $T=\textsc{l}$ and PC if $T=\textsc{pc}$, and $\delta^*\supseteq B^2$.*

Caveat 6.29 (Stated only; the $t=1$ case is free)**.** Lemma 6.28 is *stated without proof.* For $t=1$ it is immediate: $\delta\supseteq\tau_1$ and $\delta\neq\tau_1$ would make δ a subuniverse of $\mathbf{A}^2$, stable under τ_1 and properly containing it, hence $\delta\supseteq\tau_1^*\supseteq B^2$ (Proposition 3.25(b)), which is excluded; so $\delta=\tau_1$. For $t\geq 2$ the content is that the interval above an intersection of typed congruences with large covers contains no irreducible of the other class and none with a small cover, and that is not a lattice fact: it needs the structure of the quotient $\mathbf{A}/(\tau_1\cap\cdots\cap\tau_t)$. Its one consumer is the good branch of Case A in the proof of Lemma 8.21.

Lemma 6.30 (Maximal multi-type extension)**.** Let $C_1<^{A}_{\mathcal{M}T}B_1\lhd\!\lhd\!\lhd A$, $B_2\lhd\!\lhd\!\lhd A$, $C_1\cap B_2=\varnothing$, $B_1\cap B_2\neq\varnothing$, and let σ be a maximal congruence on $\mathbf{A}$ with $(C_1\circ\sigma)\cap B_2=\varnothing$.

Write $C_1 = \bigcap_{j \le t} C_1^j$ with $C_1^j <_{T(\sigma_j)}^{A} B_1$ and put

$$F = \{\, j \le t \;:\; C_1^j/\sigma <_T^{A/\sigma} B_1/\sigma \,\}, \qquad P = \bigcap_{j \in F} C_1^j,$$

the factors that survive as type T modulo σ. Suppose in addition

(i) $(P \circ \sigma) \cap B_2 = \varnothing$ — *the type-T factors alone already cut B_2 out.*

Then $\sigma = \omega_1 \cap \cdots \cap \omega_s$ where each ω_i is a congruence of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.

Caveat 6.31 (Hypothesis i is an obligation, not a convenience)**.** Hypothesis i does not appear in [14], and it is not free: what the other hypotheses give is $(C_1 \circ \sigma) \cap B_2 = \varnothing$ with $C_1 \subseteq P$, and that is the *weaker* statement. Every attempt to remove it recorded so far has failed; the companion audit sets out three of them and why each fails. Lemma 6.30 is not used anywhere in this document, so adding the hypothesis costs nothing here — but it is a genuine debt at the one place the lemma is consumed in the CSP argument, and it is recorded as such in §11.

6.5 Auxiliary imports

The following are used but not proved in [14].

Lemma 6.32 ([10, Lem. 4.7])**.** *If w is an idempotent WNU operation of arity n on a finite set A, then* $\mathrm{Clo}(w)$ *contains a special idempotent WNU operation of arity $n^{n!}$.*

Lemma 6.33 ([13, Cor. 8.17.1])**.** *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta} = A^2$, then σ is a perfect linear congruence.*

Lemma 6.34. *If σ is an irreducible congruence on $\mathbf{A} \in \mathcal{V}_n$ and δ is a bridge from σ to σ with $\widetilde{\delta}$ linked, then σ is a perfect linear congruence.*

Proof. Let $\delta^{-1}(y_1, y_2, x_1, x_2) = \delta(x_1, x_2, y_1, y_2)$, which is again a bridge from σ to σ. By Caveat 6.35 the relation $\rho = \widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is reflexive and symmetric, and it is connected because $\widetilde{\delta}$ is linked; so $\rho^N = A^2$ for $N \ge |A|$. Composing $2N$ bridges $\delta, \delta^{-1}, \ldots$ by Lemma 3.32 — legitimate, all the congruences involved being σ, which is irreducible — gives a bridge δ' from σ to σ with $\widetilde{\delta'} = \rho^N = A^2$; apply Lemma 6.33. □

Caveat 6.35 (Where the reflexivity comes from)**.** The step $(\widetilde{\delta} \circ \widetilde{\delta}^{-1})^N = A^2$ is the elementary fact that a reflexive symmetric relation ρ on a finite set A with connected graph satisfies $\rho^N = A^2$ for $N \ge |A|$; reflexivity is what makes the powers increase monotonically.

The reflexivity of $\widetilde{\delta}$ itself does *not* follow from $\mathrm{pr}_{1,2}(\delta) \supseteq \sigma$, which gives only some tuple $\delta(a, a, c, d)$ and not $\delta(a, a, a, a)$. It follows instead from clause (d) of Definition 3.29 together with stability: from $\delta(a, a, c, d)$, clause (d) gives $(c, d) \in \sigma$, and stability of the last two variables under σ replaces d by c, giving $\delta(a, a, c, c)$. So $\widetilde{\delta}$ is left-total, and $\widetilde{\delta} \circ \widetilde{\delta}^{-1}$ is reflexive and symmetric — which is the relation the power is taken of.

Lemma 6.36. *If $B \le \mathbf{Z}_p \times \cdots \times \mathbf{Z}_p$ then there is no $C <_T B$ with $T \in \{\textsc{ba}, \textsc{c}\}$.*

Proof. $\mathbf{B}$ is affine over the product of the $\mathbf{Z}_p$, so every term operation acts as $\tau(\bar{x}) = \sum_i c_i x_i$ with $\sum_i c_i = 1$, and every nonempty subuniverse is an affine coset. Suppose $C <_T B$ with $T \in \{\textsc{ba}, \textsc{c}\}$; a central subuniverse is absorbing (Definition 4.3), so in either case some k-ary τ absorbs at every coordinate. Translating so that C is a subgroup, absorption at coordinate i says τ carries $C^{i-1} \times B \times C^{k-i}$ into C; taking all entries but the i-th to be 0 gives $c_i b \in C$ for every $b \in B$, so c_i annihilates B/C. Doing this at every i makes $\sum_i c_i = 1$ annihilate B/C as well, i.e. $B = C$, contradicting properness. □

7 Proofs of the strong-subalgebra theory

This section proves the statements of §6. Throughout, Convention 2.1 is in force — every algebra is finite, idempotent, and has a WNU term operation — and §2.1 fixes the readings.

7.1 What is imported

The development is self-contained except for the following, which are proved in the literature and used here as black boxes. Two are substantial; the rest are routine. §11.4 lists them again with their exact hypotheses.

Theorem 7.1 (Absorption Theorem)**.** Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ be a subdirect relation between finite idempotent Taylor algebras. If R is linked then one of the following holds: $R = A \times B$; or $\mathbf{A}$ has a proper nonempty binary absorbing or centrally absorbing subalgebra; or $\mathbf{B}$ has a proper nonempty subalgebra that is both binary absorbing and centrally absorbing.

Corollary 7.2 (a corollary of Theorem 7.1)**.** *Let $R \lneq_{sd} \mathbf{A} \times \mathbf{B}$ be a* proper *subdirect relation between finite idempotent Taylor algebras. If R is linked then $\mathbf{A}$ or $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse.*

Proof. Properness excludes the first alternative of Theorem 7.1. The second is the conclusion on $\mathbf{A}$; the third is stronger than the conclusion on $\mathbf{B}$. ☐

Caveat 7.3 (Both words in "proper nonempty" are load-bearing)**.** Without them Corollary 7.2 is vacuous, since every algebra is a binary absorbing subuniverse of itself and ∅ is vacuously one (§2.1, item iii); every use below is the proper nonempty one. Theorem 7.1 is the single heaviest import of this section, and it is stated here in the form its source gives rather than in the weaker symmetric form actually used, so that the strengthening is visible.

Lemma 7.4 ([4, Lem. 2.9], [15, Lem. 6.1, Thm. 6.9])**.** *Let $R \leq A^n$ be defined by a pp-formula Φ in which a relation S occurs, and let R' be defined by the formula obtained from Φ by replacing* every *occurrence of S by some $S' <_T S$, where $T \in \{\textsc{ba}, \textsc{c}\}$. Then $R' \dot{\leq}_T R$.*

Caveat 7.5 (The conclusion is dotted)**.** Replacing a relation by a proper strong subrelation can make the pp-defined result *empty*, so the undotted $R' \leq_T R$ — which under Convention 2.4 entails $R' \neq \varnothing$ — is not what the lemma gives.

Every use below has been checked against this. Lemma 7.15 states its conclusion dotted and passes it on. Lemmas 7.17, 7.46 and 7.49 exhibit a point of the smaller relation and so upgrade to the undotted form. Lemma 7.22 needs both nonemptiness *and* properness, and gets each from a different clause of Lemma 7.6 (Caveat 7.23). The remaining use is inside Lemma 7.55, which is an outline (§11).

Lemma 7.6 ([4, Prop. 2.14], [15, Lem. 3.2])**.** *For $B \leq \mathbf{A}$ and $n \geq 2$: B absorbs $\mathbf{A}$ with an operation of arity n if and only if there is no $S \leq A^n$ with $S \cap B^n = \varnothing$ and $S \cap (B^{i-1} \times A \times B^{n-i}) \neq \varnothing$ for every i.*

Lemma 7.7 ([15, Lem. 6.8])**.** *If $B \leq_T \mathbf{A}$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ and σ is a congruence on $\mathbf{A}$, then $B/\sigma \leq_T \mathbf{A}/\sigma$.*

Caveat 7.8 (Only the $\textsc{c}$ case is imported)**.** For $T = \textsc{ba}$ the statement is one line: if $t(B, A) \subseteq B$ and $t(A, B) \subseteq B$ then the same t works on images. The case $T = \textsc{s}$ follows from the other two by Definition 4.6. Only $\textsc{c}$ is genuinely imported.

Lemma 7.9 ([15, Lem. 6.24])**.** *If R is a nontrivial strong subuniverse of $\mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ of a type $T \neq \textsc{pc}$, then some $\mathbf{A}_i$ has a nontrivial subuniverse of type T. In particular $B <_T \mathbf{A}^n$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ yields some $C <_T \mathbf{A}$.*

Caveat 7.10 (One run of the proof serves all three types)**.** The cited lemma is stated for every type $T \neq$ PC, and its proof is *type-uniform*: it takes $\mathrm{pr}_1(R)$ if that is proper and otherwise a fibre $R' = \{(a_2, \ldots, a_n) : (a_1, \ldots, a_n) \in R\}$, and neither construction mentions the type — the type enters only through the facts that projections and fibres preserve BA and preserve C. So a single run produces one witness carrying *every* strong type that R has; in particular a subuniverse that is simultaneously binary absorbing and central yields one that is too, which is the case $T =$ S.

Theorem 7.11 ([15, Thm. 6.15])**.** Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ and $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then C is a central subuniverse of $\mathbf{A}$, or $\mathbf{B}$ has a nontrivial binary absorbing subuniverse, or $\mathbf{B}$ has a nontrivial *projective* subuniverse.

Caveat 7.12 (The heaviest import after Theorem 7.1)**.** Theorem 7.11 is taken on faith here. Its proof in [15] is two pages and uses only the Barto–Kazda criterion (Lemma 7.6) and the theory of projective subuniverses; none of that theory is needed anywhere else below, which is why absorbing it has been deferred. The same applies to [15, Lem. 3.4], used once, in the proof of Lemma 7.20.

Lemma 7.13 ([8])**.** *An algebra is Abelian if and only if some congruence of its square has the diagonal as a block; and a finite algebra with a WNU term operation is Abelian if and only if it is affine.*

Lemma 7.14. *Let $|A| > 1$. Then $\mathbf{A}$ is Abelian if and only if there is a bridge δ from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\widetilde{\delta} = \mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta) = A^2$.*

Proof. If $\mathbf{A}$ is Abelian, a congruence of $\mathbf{A}^2$ having the diagonal as a block is such a bridge; clause (c) needs $|A| > 1$, since for $|A| = 1$ no relation properly contains $0_{\mathbf{A}} = A^2$.

Conversely, regard δ as a binary relation $\hat{\delta}$ on A^2, $\hat{\delta}((x_1, x_2), (x_3, x_4)) \iff \delta(x_1, x_2, x_3, x_4)$, and put $\rho = \hat{\delta} \circ \hat{\delta}^{-1}$, a reflexive symmetric compatible relation on $\mathbf{A}^2$ — reflexive because $\mathrm{pr}_{1,2}(\delta) = A^2$ makes $\hat{\delta}$ left-total. Its powers increase, so $\theta = \rho^N$ for $N \geq |A|^2$ is reflexive, symmetric and transitive, hence a congruence of $\mathbf{A}^2$. Clause (d) of Definition 3.29 is preserved by each composition, so every pair of θ relates a diagonal element only to diagonal elements: the diagonal is a θ-block. Now apply Lemma 7.13. ☐

7.2 Absorption preliminaries

Lemma 7.15. *If $B \leq_T \mathbf{A}$ with $T \in \{$BA, C$\}$ and $C \leq \mathbf{A}$, then $B \cap C \dot{\leq}_T C$.*

Proof. C is defined by the pp-formula $A(x) \wedge C(x)$; replacing the occurrence of the full unary relation A by B yields $B \cap C$, so Lemma 7.4 applies. ☐

Lemma 7.16 (No absorbing diagonal)**.** *Let $\mathbf{C}$ be a finite idempotent algebra with $|C| \geq 2$. Then $0_{\mathbf{C}}$ is not an absorbing subuniverse of $\mathbf{C}^2$, of any arity; in particular $0_{\mathbf{C}} \not\leq_{\mathrm{BA}} \mathbf{C}^2$ and $0_{\mathbf{C}} \not\leq_{\mathrm{C}} \mathbf{C}^2$.*

Proof. Suppose t is k-ary and absorbs at every position. Fix i, take $a_j \in C$ for $j \neq i$ and $b, c \in C$, and apply t coordinatewise to the tuples $(a_j, a_j) \in 0_{\mathbf{C}}$ for $j \neq i$ and $(b, c) \in C^2$ at position i. The result $(t(\bar{a}; b), t(\bar{a}; c))$ lies in $0_{\mathbf{C}}$, so $t(\bar{a}; b) = t(\bar{a}; c)$. As b, c were arbitrary, t does not depend on its i-th argument; and this holds for every i, so t is constant. Idempotence then forces $|C| = 1$. A central subuniverse is absorbing by Definition 4.3, so the central case is included. ☐

Lemma 7.17 (Absorption preserves linkedness, subrelation form)**.** *Let $W \leq_{\mathrm{sd}} \mathbf{P} \times \mathbf{A}$ be linked and let $\varnothing \neq W' \leq_T W$ with $T \in \{$BA, C$\}$. Put $P' = \mathrm{pr}_1(W')$ and $A' = \mathrm{pr}_2(W')$. Then $W' \leq_{\mathrm{sd}} \mathbf{P}' \times \mathbf{A}'$ is linked.*

Proof. Let $\Phi_k(x, y)$ be the pp-formula in one binary relation symbol obtained by iterating $x \sim y \iff \exists z\, (W(x, z) \wedge W(y, z))$ k times. Because W is subdirect, $W \circ W^{-1}$ is reflexive and symmetric on P, so for $k \geq |P|$ the formula Φ_k interpreted in W defines the transitive closure of $W \circ W^{-1}$, which is P^2 since W is linked. Interpreted in W' and for the same $k \geq |P| \geq |P'|$ it defines $\lambda := \mathrm{LeftLinked}(W')$. By Lemma 7.4, replacing every occurrence of W by W' gives $\lambda \dot{\leq}_T P^2$, and $\lambda \neq \varnothing$ because $W' \neq \varnothing$ makes P' nonempty and λ reflexive on it, so $\lambda \leq_T P^2$. Since $\lambda \subseteq (P')^2 \leq P^2$, Lemma 7.15 gives $\lambda \leq_T (P')^2$.

Now λ is a congruence on $\mathbf{P}'$: reflexive because W' is subdirect over P', symmetric by construction, transitive because the powers stabilise. So $\lambda \times \lambda$ is a congruence on the algebra $(P')^2$, and the image of λ under $(P')^2 \twoheadrightarrow (P'/\lambda)^2$ is exactly $0_{\mathbf{P}'/\lambda}$; by Lemma 7.7, $0_{\mathbf{P}'/\lambda} \leq_T (\mathbf{P}'/\lambda)^2$. Lemma 7.16 forces $|P'/\lambda| = 1$, i.e. $\lambda = (P')^2$, which says W' is linked. □

Caveat 7.18 (The box form is not enough)**.** The familiar form of this principle, [3, Prop. 2.15(i)], restricts by a *box*: if $W \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ is linked, B_i absorbs $\mathbf{A}_i$, and $W \cap (B_1 \times B_2)$ is subdirect in $B_1 \times B_2$, then $W \cap (B_1 \times B_2)$ is linked. That form cannot reach the restrictions used in Lemma 7.49, and the two really do differ: for the meet-semilattice on $\{0, 1, 2\}$ with $B = \{0, 1\}$, the subalgebra $K = \{(0, 0), (1, 2)\}$ of A^2 meets B^2 and has $\mathrm{pr}_1(K) \cap B = \{0, 1\}$ while $\mathrm{pr}_1(K \cap B^2) = \{0\}$. The subrelation form also discharges the box form's subdirectness hypothesis for free.

Lemma 7.19 (A WNU forbids essentially unary quotients)**.** *Let* $\mathbf{U}$ *be a finite algebra with a WNU term operation and* $|U| \geq 2$*. Then some term operation of* $\mathbf{U}$ *depends on more than one variable. Consequently, if* $\mathbf{A}$ *has a WNU term operation then no* $\mathbf{U} \in \mathrm{HS}(\mathbf{A})$ *with* $|U| \geq 2$ *is essentially unary.*

Proof. Suppose every term operation of $\mathbf{U}$ has at most one non-dummy variable, and let w be a WNU term operation of arity $k \geq 2$; it is idempotent. If w has no non-dummy coordinate it is constant, and idempotence gives $|U| = 1$. Otherwise $w(\bar{x}) = g(x_i)$ for a unary g, and $g(x) = w(x, \ldots, x) = x$, so w is the i-th projection. The WNU identities equate the k terms in which a single y sits at each position against a background of x's; two of them are w with y at position i, which evaluates to y, and w with y at some position $\neq i$, which evaluates to x. Hence $x = y$ for all x, y and $|U| = 1$. The consequence follows because idempotence and the WNU identities are identities, hence hold in every member of $\mathrm{HS}(\mathbf{A})$. □

Lemma 7.20 (Central relations)**.** Let $\mathbf{A}, \mathbf{B}$ satisfy Convention 2.1, let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$, and put $C = \{c \in A : \{c\} \times B \subseteq R\}$. Then

(a) $C = \varnothing$, or
(b) C is a central subuniverse of $\mathbf{A}$, or
(c) $\mathbf{B}$ has a proper nonempty binary absorbing subuniverse.

Proof. Apply Theorem 7.11. Its first alternative gives (a) or (b) according as C is empty or not, its second gives (c). In its third, $\mathbf{B}$ has a nontrivial projective subuniverse P; by [15, Lem. 3.4], either P is a binary absorbing subuniverse of $\mathbf{B}$, which is (c) since P is nontrivial, or there is an essentially unary $\mathbf{U} \in \mathrm{HS}(\mathbf{B})$ with $|U| \geq 2$, which Lemma 7.19 forbids. □

Caveat 7.21 (Alternative (a) is not optional)**.** Under Convention 2.4 the two-alternative form (b)–(c) is false: $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with $R = \{(x, x + 1)\}$ is subdirect, has $C = \varnothing$, and $\mathbf{Z}_p$ has no nontrivial binary absorbing subuniverse at all. It is true in [15] only because $\varnothing$ counts as central there. Every call site below discharges (a) by exhibiting a point of C first, and two further obligations go with each contrapositive use: the centre produced must be shown *proper*, and a strong subuniverse of a *power* must be moved down by Lemma 7.9.

Lemma 7.22. *Let $R \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{B}$ with $\mathbf{A}$ BA and centre free,* $\mathrm{LeftLinked}(R) = A^2$*, and $C = \{c \in B : A \times \{c\} \subseteq R\}$. Then $C \neq \varnothing$ and $C \leq_{\mathrm{BA,C}} \mathbf{B}$.*

Proof. If $|A| = 1$ then subdirectness gives $A \times \{b\} \subseteq R$ for every $b \in B$, so $C = B$ and the conclusion holds with equality. Assume $|A| \geq 2$.

Step 1: $C \neq \varnothing$. For $k \geq 1$ put $W_k = \{(a_1, \ldots, a_k) : \exists b \, \forall i \, R(a_i, b)\}$, a subuniverse of $\mathbf{A}^k$, with $W_1 = \mathrm{pr}_1(R) = A$. If $W_{|A|} = A^{|A|}$ then a witness b for the tuple listing all of A has $A \times \{b\} \subseteq R$, so $b \in C$ and we are done. Otherwise choose $k \geq 2$ minimal with $W_k \neq A^k$, so $W_{k-1} = A^{k-1}$.

View W_k as a binary relation in $\mathbf{A} \times \mathbf{A}^{k-1}$. It is *subdirect*: for $a \in A$, subdirectness of R gives b with $R(a, b)$ and hence $(a, a, \ldots, a) \in W_k$, so the left projection is A; and for $\bar{a} \in A^{k-1}$, $W_{k-1} = A^{k-1}$ gives b with $R(a_i, b)$ for all i, while $\mathrm{pr}_2(R) = B$ gives some a_0 with $R(a_0, b)$, so $(a_0, \bar{a}) \in W_k$ and the right projection is A^{k-1}.

It is *linked*. If $a, c \in A$ share an R-neighbour b — that is, $R(a, b)$ and $R(c, b)$ — then b also witnesses $(a, a, \ldots, a), (c, a, \ldots, a) \in W_k$, so $(a, c) \in \mathrm{LeftLinked}(W_k)$ by Definition 3.11. Since $\mathrm{LeftLinked}(R)$ is the least equivalence closed under that step and $\mathrm{LeftLinked}(W_k)$ is an equivalence containing each such pair, $\mathrm{LeftLinked}(R) \subseteq \mathrm{LeftLinked}(W_k)$; and $\mathrm{LeftLinked}(R) = A^2$ by hypothesis, so $\mathrm{LeftLinked}(W_k) = A^2$.

So W_k is a proper linked subdirect relation, and Corollary 7.2 gives a proper nonempty binary absorbing or central subuniverse of $\mathbf{A}$ or of $\mathbf{A}^{k-1}$; Lemma 7.9 moves the second case down to $\mathbf{A}$. Either way this contradicts the hypothesis on $\mathbf{A}$. Hence $W_{|A|} = A^{|A|}$ after all, and $C \neq \varnothing$.

Step 2: C is central in $\mathbf{B}$. Apply Lemma 7.20 to $R^{-1} \leq_{\mathrm{sd}} \mathbf{B} \times \mathbf{A}$, whose associated set is exactly C. Alternative (a) is excluded by Step 1 and alternative (c) — a proper nonempty binary absorbing subuniverse of $\mathbf{A}$ — by hypothesis, so (b) holds.

Step 3: C is binary absorbing in $\mathbf{B}$. Suppose not. By Lemma 7.6 at $n = 2$ there is $S \leq B \times B$ with

$$S \cap (C \times C) = \varnothing, \qquad S \cap (C \times B) \neq \varnothing, \qquad S \cap (B \times C) \neq \varnothing.$$

Write $U_X = \{b \in B : \exists c \, (S(b, c) \wedge X(c))\}$ for a unary X on B, and consider the pp-formula

$$\Phi_X(a_1, \ldots, a_{|A|}) = \exists b \, \Big(U_X(b) \wedge \bigwedge_i R(a_i, b) \Big),$$

in which the unary relation X occurs once. Put $W = \Phi_C$ and $W^\circ = \Phi_B$. Since $C <_{\mathrm{C}} \mathbf{B}$ by Step 2 — *proper*, because $C = B$ would put $S \cap (C \times C) = S \neq \varnothing$ — Lemma 7.4 gives $W \dot{\leq}_{\mathrm{C}} W^\circ$. Three observations turn that into what is wanted, and each spends one of the three displayed conditions.

$W^\circ = A^{|A|}$. $S \cap (C \times B) \neq \varnothing$ supplies $b_0 \in C$ with $b_0 \in U_B$. Since $b_0 \in C$ we have $A \times \{b_0\} \subseteq R$, so b_0 witnesses $\Phi_B(\bar{a})$ for *every* $\bar{a} \in A^{|A|}$.

$W \neq \varnothing$. $S \cap (B \times C) \neq \varnothing$ supplies $(b_1, c_1) \in S$ with $c_1 \in C$, so $b_1 \in U_C$; and $\mathrm{pr}_2(R) = B$ supplies a with $R(a, b_1)$, so $(a, \ldots, a) \in W$.

$W \neq A^{|A|}$. If it were, the tuple enumerating A would give $b \in U_C$ with $A \times \{b\} \subseteq R$, i.e. $b \in C$; and $b \in U_C$ means $S(b, c)$ for some $c \in C$, so $(b, c) \in S \cap (C \times C) = \varnothing$.

So $W <_{\mathrm{C}} A^{|A|}$ is proper and nonempty, and Lemma 7.9 produces a proper nonempty central subuniverse of $\mathbf{A}$ — again a contradiction. ☐

Caveat 7.23 (All three Barto–Kazda conditions are spent)**.** Step 3 is the only place in this section where the three side conditions of Lemma 7.6 are used separately, one apiece: S meeting $C \times B$ is what makes the *ambient* of the pp-propagation step the full power $A^{|A|}$ rather than some subrelation of it; S meeting $B \times C$ is what makes W nonempty; and $S \cap (C \times C) = \varnothing$ is what makes it proper. Dropping any one leaves a conclusion that Lemma 7.9 cannot consume, since $<_{\mathrm{C}}$ requires a proper nonempty subuniverse of a product.

Lemma 7.24. *If $C \lhd\lhd\lhd^A B \lhd\lhd\lhd A$ and $D <_{\text{BA,C}} B$ then $C \cap D \neq \varnothing$.*

Proof. Suppose not, and choose C'' minimal with $C \lhd\lhd\lhd^A C' <^A_{T(\sigma)} C'' \lhd\lhd\lhd^A B$ and $C'' \cap D \neq \varnothing$. By Lemma 7.15 $C'' \cap D <_{\text{BA,C}} C''$, so by Lemma 7.28 $T = \text{D}$; and then Lemma 7.7 gives $(C'' \cap D)/\sigma <_{\text{BA}} C''/\sigma$, contradicting the requirement in Definition 4.22 that C''/σ be BA and centre free. □

Lemma 7.25 (No critical central family of size ≥ 3)**.** *Let $n \geq 3$, let $B_1, \dots, B_n \leq \mathbf{A}$, and let $C_i <_{\text{C}} B_i$ for $i \in [n]$. Then $\bigcap_i C_i = \varnothing$ and $B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing$ for every j cannot both hold.*

Proof. Suppose they do. Put $D = \bigcap_i B_i$ and $E_i = C_i \cap D$. Then D is a nonempty subuniverse, containing $B_1 \cap C_2 \cap \cdots \cap C_n$; each $E_i \leq_{\text{C}} D$ by Lemma 7.15, and properly, since $E_j = D$ would give $\bigcap_i E_i = \bigcap_{i \neq j} E_i \neq \varnothing$. Moreover $\bigcap_i E_i = \varnothing$ and, because $C_i \subseteq B_i$,

$$\textstyle\bigcap_{i \neq j} E_i = D \cap \bigcap_{i \neq j} C_i = B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing.$$

Now the diagonal $R = \{(a, \dots, a) : a \in D\} \leq D^n$ is a subuniverse by idempotence, and the two displays say exactly that R is $(E_1, \dots, E_n)$-essential. By [15, Lem. 6.11] no such relation exists for $n \geq 3$ when the E_i are proper nonempty central subuniverses. □

Corollary 7.26 (Relational form)**.** *Let $R \leq_{\text{sd}} \mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ with $n \geq 3$, $B_i \leq \mathbf{A}_i$, and $C_i <_{\text{C}} B_i$ for $i \in [n]$. If $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_n) \neq \varnothing$ for every j, then $R \cap (C_1 \times \cdots \times C_n) \neq \varnothing$.*

Proof. Write $f_i \colon \mathbf{R} \to \mathbf{A}_i$ for the projections and $P_i = f_i^{-1}(B_i)$, $Q_i = f_i^{-1}(C_i)$, subuniverses of $\mathbf{R}$. The formula $R(x_1, \dots, x_n) \wedge X(x_i)$ is positive primitive in X and defines P_i at $X = B_i$ and Q_i at $X = C_i$, so Lemma 7.4 gives $Q_i \dot{\leq}_{\text{C}} P_i$; and Q_i is nonempty, containing $R \cap (C_1 \times \cdots \times B_j \times \cdots \times C_n)$ for any $j \neq i$, which exists since $n \geq 3$. So $Q_i \leq_{\text{C}} P_i$.

If $Q_i = P_i$ for some i, the conclusion is immediate: $R \cap (C_1 \times \cdots \times C_n) = Q_i \cap \bigcap_{j \neq i} Q_j = P_i \cap \bigcap_{j \neq i} Q_j = R \cap (C_1 \times \cdots \times B_i \times \cdots \times C_n) \neq \varnothing$. Otherwise every $Q_i <_{\text{C}} P_i$ is proper, and Lemma 7.25 applies inside $\mathbf{R}$ with $B_i := P_i$ and $C_i := Q_i$: its two forbidden conditions are exactly $R \cap (C_1 \times \cdots \times C_n) = \varnothing$ and the displayed nonemptiness, read through the f_i. So the first fails. □

Caveat 7.27 (Why this is a separate statement)**.** Lemma 7.25 is the type-C half of case (c) of Theorem 6.18, and it is stated on its own because Lemma 7.46 needs *only* this half. Citing the whole theorem there would make the bridge theory rest on the bridge classification, which rests back on it:

$$6.19 \to 6.18 \to 4.17 \to 4.15 \to 4.13 \to 7.47 \to 7.46 \to 6.19.$$

Every edge of that is a real citation, and the loop is broken only by observing that the appeal at the last step uses no bridge at all: it is $n = 3$ with all types C, and the argument for that case runs through [15, Lem. 6.11] and nothing else in this section. Splitting the statement is what makes the observation checkable rather than a matter of trust.

A second observation, found by the layer check rather than by reading: the $\lhd\lhd\lhd$ hypotheses these two statements used to carry are consumed by *nothing* in either proof — nonemptiness of $\bigcap_i B_i$ comes from the displayed conditions, and the essential-relations import needs only proper nonempty central subuniverses. They are now stated for plain subalgebras, and the corollary pulls back through Lemma 7.4 instead of the $\lhd\lhd\lhd$-pullback of Lemma 6.4, with the equality case of the pullback giving the conclusion outright. That is what places the entire bridge classification below the strong-subuniverse theory rather than beside it.

Lemma 7.28 ([15, Lem. 6.25])**.** *If $B <_{T_1} \mathbf{A}$, $C <_{T_2} \mathbf{A}$, $B \cap C = \varnothing$ and $T_1, T_2 \in \{\text{BA}, \text{C}, \text{S}\}$, then $T_1 = T_2 \in \{\text{BA}, \text{C}\}$.*

Lemma 7.29 (A BA-and-central subuniverse meets every strong one)**.** *If $D <_{\text{BA,C}} \mathbf{A}$ and $E <_T \mathbf{A}$ with $T \in \{\text{BA}, \text{C}, \text{S}\}$, then $D \cap E \neq \varnothing$.*

Proof. Suppose $D \cap E = \varnothing$. Lemma 7.28 applied to the pair (D, E) with $T_1 = \text{BA}$ gives $T = \text{BA}$; applied to the same pair with $T_1 = \text{C}$ it gives $T = \text{C}$. The two conclusions are incompatible. □

Caveat 7.30 (Where the conjunction reading of $<_{\text{BA,C}}$ pays)**.** Lemma 7.29 is false with $<_{\text{BA,C}}$ read as a disjunction: two disjoint binary absorbing subuniverses are exactly what Lemma 7.28 permits. What makes the argument run is that D carries *both* types at once, so whichever of BA and C the subuniverse E has, the other is available for D and the two types differ. This is the reading fixed in Caveat 4.9; it is the mechanism already used inside the proof of Lemma 7.24, and the two appeals in Lemma 7.39(s) are what make it worth naming.

7.3 The intersection property

The goal is Lemma 7.39: if $B \lhd\lhd\lhd A$, $C \lhd\lhd\lhd A$ meet and δ is a dividing congruence for B, then $(B \cap C)/\delta$ is a single point or all of B/δ. Everything in §7 that produces a bridge goes through it.

Lemma 7.31 (The shift manoeuvre)**.** *Let $k \geq 2$ and let $R \leq A^k$ be totally symmetric and closed under $(a_1, \dots, a_k) \mapsto (a_1, a_1, a_2, \dots, a_{k-1})$. Identify a tuple of R with its entry multiset, a k-element multiset over A. If $\bar{a} \in R$ has entry multiset M, then for any $u, v \in M$ —* occupying distinct positions of $\bar{a}$, though possibly with $u = v$ as elements of A — *the multiset $M + \{u\} - \{v\}$ is again the entry multiset of a tuple of R. Consequently every k-element multiset over the set S of distinct entries of $\bar{a}$ is reached, provided it is reached by such steps; in particular, for any $u \in S$, the constant multiset $k \cdot \{u\}$ is reached, in at most $k - 1$ steps.*

Proof. Total symmetry lets us place u first and v last; the shift then duplicates u and deletes v, giving $M + \{u\} - \{v\}$. For the last clause, apply this $k - 1$ times with the same u, deleting a different position each time. □

Caveat 7.32 (The shift deletes the last entry)**.** A single application of the shift to $(a, b_2, \dots, b_{k-1}, c)$ destroys c, so $(a, \dots, a, c) \in R$ does not follow from one step; it needs $k - 2$ steps interleaved with permutations, which is what Lemma 7.31 packages. The lemma is used twice below and the one-step reading fails both times.

Lemma 7.33. *Let $k \geq 2$, let $\mathbf{A}$ be BA and centre free, and let $R \leq A^k$ be totally symmetric, closed under the shift, with $\mathrm{pr}_{1,2}(R) = A^2$. Then $R = A^k$.*

Proof. Induct on k; for $k = 2$ the hypothesis is the conclusion. Let $k > 2$.

Step 1: $(a, \dots, a, c) \in R$ for all a, c. By total symmetry $\mathrm{pr}_{1,k}(R) = A^2$, so there is a tuple of R with first entry a and last entry c. Apply Lemma 7.31 $k - 2$ times with $u = a$, deleting on each step one position other than the last: the entry multiset becomes $(k - 1) \cdot \{a\} + \{c\}$, which by total symmetry gives $(a, \dots, a, c) \in R$. Deleting c as well gives $(a, \dots, a) \in R$.

Step 2: the left projection is full. $R' = \mathrm{pr}_{1,\dots,k-1}(R)$ inherits total symmetry, has $\mathrm{pr}_{1,2}(R') = A^2$, and is shift-closed: from $\bar{a} \in R'$ extend to R, shift there, and project. By the inductive hypothesis $R' = A^{k-1}$.

Step 3. Only now may R be viewed as a subdirect relation $R \leq_{\text{sd}} A^{k-1} \times A$: Step 2 gives the left projection and Step 1 gives the right. It is linked, since by Step 1 every c is adjacent to the single left vertex $(a, \dots, a)$. If $R \neq A^k$ then Corollary 7.2 gives a proper nonempty binary absorbing or central subuniverse on $\mathbf{A}^{k-1}$ or on $\mathbf{A}$, and Lemma 7.9 moves the first case to $\mathbf{A}$ — contradicting the hypothesis either way. □

Lemma 7.34. *Let σ be a dividing congruence for $B \leq A$ and let $R \leq (\mathbf{A}/\sigma)^k$ be reflexive, totally symmetric and shift-closed. Then $(B/\sigma)^k \subseteq R$, or R is the set of constant tuples.*

Proof. If $\mathrm{pr}_{1,2}(R)$ is the equality relation then, by total symmetry, all coordinates of every tuple of R agree, and reflexivity supplies all constant tuples.

Otherwise $\mathrm{pr}_{1,2}(R)$ is a subuniverse of $(\mathbf{A}/\sigma)^2$ containing $0_{\mathbf{A}/\sigma}$ properly and stable under it, so by minimality of the cover (Proposition 3.25, transported by Lemma 3.27) it contains σ^*/σ, hence $(B/\sigma)^2$ — the last step by Definition 4.22(i). Now apply Lemma 7.33 to $R \cap (B/\sigma)^k$ over the algebra B/σ, which is BA and centre free by Definition 4.22(iii). Its hypothesis $\mathrm{pr}_{1,2}(R \cap (B/\sigma)^k) = (B/\sigma)^2$ is not immediate from $\mathrm{pr}_{1,2}(R) \supseteq (B/\sigma)^2$, since a witnessing tuple may leave B/σ; Lemma 7.31 repairs it, pulling $(x_1, x_2, y_3, \ldots, y_k)$ back to $(x_1, x_2, x_1, \ldots, x_1)$. ☐

Lemma 7.35 (Full fibre)**.** *Let F be a finite set with $|F| \leq N$ and let $W \subseteq F^N \times E$ satisfy $\mathrm{pr}_{1..N}(W) = F^N$ and be closed under coordinate substitution: if $(x_1, \ldots, x_N, e) \in W$ and $g : [N] \to [N]$ then $(x_{g(1)}, \ldots, x_{g(N)}, e) \in W$. Then $F^N \times \{d\} \subseteq W$ for some $d \in E$.*

Proof. Write $F = \{F_1, \ldots, F_t\}$ with $t \leq N$ and put $\bar{x} = (F_1, \ldots, F_t, F_t, \ldots, F_t) \in F^N$. By surjectivity pick d with $(\bar{x}, d) \in W$. Every $\bar{y} \in F^N$ is $\bar{x} \circ g$ for some $g : [N] \to [N]$, since every entry of $\bar{y}$ occurs among the entries of $\bar{x}$; closure gives $(\bar{y}, d) \in W$. ☐

Caveat 7.36 (Why the arity is $|A|$)**.** Lemma 7.35 is used three times below, always in the shape "since $\mathrm{pr}_{1,\ldots,|A|}(R') = (B/\delta)^{|A|}$, there is d with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$". Surjectivity of a projection does not by itself produce a full fibre; what makes the step valid is that each such R' is cut out by a conjunction of unary and binary conditions on its first $|A|$ entries, hence closed under substituting entries for one another, and that $|B/\delta| \leq |A|$. **The exponent $|A|$ is load-bearing**: it must be at least the number of blocks, so that one tuple can enumerate them. Normalising the arity away breaks all three proofs.

Lemma 7.37. Let $B_k <^A_{T_k(\sigma_k)} \cdots <^A_{T_1(\sigma_1)} B_0 = A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, let δ be a congruence on $\mathbf{A}$, and let $m \in [k]$ with $T_m = \textsc{d}$. Then $((B_k \circ \delta) \cap B_{m-1})/\sigma_m$ is either B_{m-1}/σ_m or B_m/σ_m; and in the second case $\sigma_m \supseteq \delta \cap \sigma_1 \cap \cdots \cap \sigma_{m-1}$.

Proof. The induction is on k, and δ, m and the chain are all quantified *inside* the inductive statement: subsubcase 1B3 below applies the hypothesis with a different congruence in the role of δ, so δ must not be held fixed.

Two conventions are in force throughout. For every j with $T_j = \textsc{d}$ we have $B_j = B_{j-1} \cap E_j$ for a block E_j of σ_j, so $B_j^2 \subseteq \sigma_j$, $|B_j/\sigma_j| = 1$, and $\mathbf{B}_{j-1}/\sigma_j$ is BA and centre free with $|B_{j-1}/\sigma_j| \geq 2$ (Definition 4.22(iii), and properness of B_j in B_{j-1}); in particular $|A| \geq 2$. For every other j, $\sigma_j = A^2$.

Case 1: $k = m$**,** so $T_k = \textsc{d}$. Put $\tau = \delta \cap \sigma_1 \cap \cdots \cap \sigma_{k-1}$ and, for $0 \leq n \leq k$,

$$Q_n = \{(a_1, \ldots, a_{|A|}) \in B_n^{|A|} : (a_i, a_j) \in \tau \text{ for all } i, j\}, \qquad S_n = Q_n/\sigma_k,$$

the quotient again an image. Each Q_n is cut out of $\mathbf{A}^{|A|}$ by unary and binary conditions, so each S_n is a subuniverse of $(\mathbf{A}/\sigma_k)^{|A|}$ closed under substitution of coordinates and hence totally symmetric and shift-closed; S_0, whose Q_0 ranges over $B_0 = A$, is in addition reflexive; and $S_n \supseteq S_{n'}$ for $n \leq n'$. Since σ_k is a dividing congruence for $B_{k-1} \leq A$, Lemma 7.34 applies to S_0.

Subcase 1A: $S_0 = \{(a/\sigma_k, \ldots, a/\sigma_k) : a \in A\}$. For $(a, b) \in \tau$ the tuple $(a, b, b, \ldots, b)$ lies in Q_0, so its image is constant and $a/\sigma_k = b/\sigma_k$: that is $\sigma_k \supseteq \tau$, which is the additional clause. It also settles the main one. Every σ_ℓ with $\ell < k$ contains B_{k-1}^2 — for $T_\ell = \textsc{d}$ because $B_{k-1} \subseteq B_\ell$ and $B_\ell^2 \subseteq \sigma_\ell$, otherwise because $\sigma_\ell = A^2$ — so

$$\delta \cap B_{k-1}^2 \subseteq \tau \cap B_{k-1}^2 \subseteq \sigma_k.$$

Hence if $a \in B_{k-1}$ and $(a,b) \in \delta$ with $b \in B_k \subseteq B_{k-1}$ then $(a,b) \in \sigma_k$, so $a \in E_k$ and therefore $a \in B_{k-1} \cap E_k = B_k$. Thus $(B_k \circ \delta) \cap B_{k-1} = B_k$ and the projection is $B_k/\sigma_k = B_m/\sigma_m$.

Subcase 1B: $(B_{k-1}/\sigma_k)^{|A|} \subseteq S_0$. Suppose first that $(B_{k-1}/\sigma_k)^{|A|} \subseteq S_{k-1}$. Since $|B_{k-1}/\sigma_k| \leq |A|$, one tuple $x \in (B_{k-1}/\sigma_k)^{|A|}$ enumerates every class of B_{k-1}/σ_k; a witness for x in Q_{k-1} is a family $a_1, \dots, a_{|A|} \in B_{k-1}$, pairwise τ-related and so in particular pairwise δ-related, whose σ_k-classes exhaust B_{k-1}/σ_k. Let E be the δ-class containing all of them, so that

$$(E \cap B_{k-1})/\sigma_k = B_{k-1}/\sigma_k.$$

One of those classes is the class of B_k, so some a_i lies in $B_{k-1} \cap E_k = B_k$; hence E meets B_k, so $E \subseteq B_k \circ \delta$, and $((B_k \circ \delta) \cap B_{k-1})/\sigma_k \supseteq (E \cap B_{k-1})/\sigma_k = B_{k-1}/\sigma_k$. That is the first alternative, and this branch is done.

So assume $(B_{k-1}/\sigma_k)^{|A|} \not\subseteq S_{k-1}$ and choose n minimal with $(B_{k-1}/\sigma_k)^{|A|} \not\subseteq S_n$; then $1 \leq n \leq k-1$ and $(B_{k-1}/\sigma_k)^{|A|} \subseteq S_{n-1}$. For every $b \in B_{k-1} \subseteq B_n$ the constant tuple $(b, \dots, b)$ lies in Q_n, so

$$\varnothing \neq S_n \cap (B_{k-1}/\sigma_k)^{|A|} \subsetneq (B_{k-1}/\sigma_k)^{|A|}. \tag{7.1}$$

We show each of the three types of T_n impossible, so that this branch does not occur.

Subsubcase 1B1: $T_n \in \{\textsc{ba}, \textsc{c}\}$. The formula $\bigwedge_{i,j} \tau(x_i, x_j) \wedge \bigwedge_i X(x_i)$ defines Q_{n-1} at $X = B_{n-1}$ and Q_n at $X = B_n$, so Lemma 7.4 and (7.1) give $Q_n \leq_{T_n} Q_{n-1}$; Lemma 7.7 pushes this through σ_k and Lemma 7.15 restricts it along $(B_{k-1}/\sigma_k)^{|A|} \leq \mathbf{S}_{n-1}$, giving by (7.1) a proper nonempty $<_{T_n} (B_{k-1}/\sigma_k)^{|A|}$. Lemma 7.9 moves it to B_{k-1}/σ_k, which is BA and centre free — a contradiction.

Subsubcase 1B2: $T_n = \textsc{s}$. Choose a nonempty $G \subseteq B_n$ with $G <_{\textsc{ba,c}} B_{n-1}$. Lemma 7.24, applied along $B_k \lhd\lhd\lhd^A B_{n-1} \lhd\lhd\lhd A$, gives $G \cap B_k \neq \varnothing$, and $B_k \subseteq B_{k-1}$, so replacing B_n by G leaves the intersection with $(B_{k-1}/\sigma_k)^{|A|}$ nonempty; it stays proper because $G \subseteq B_n$. Now 1B1 applies verbatim with G for B_n and contradicts BA-freeness of B_{k-1}/σ_k.

Subsubcase 1B3: $T_n = \textsc{d}$. Let R' be the image in $(\mathbf{A}/\sigma_k)^{|A|} \times \mathbf{A}/\sigma_n$ of

$$\{(a_1, \dots, a_{|A|}, b) : b \in B_{n-1},\ \forall i\ a_i \in B_{n-1} \cap (B_{k-1} \circ \sigma_k),\ \forall i,j\ (a_i, a_j) \in \tau,\ \forall i\ (a_i, b) \in \sigma_n\}.$$

Because $n \leq k-1$, the congruence σ_n is one of those intersected in τ, so $\tau \subseteq \sigma_n$. Four facts.

(i) $\mathrm{pr}_{1,\dots,|A|}(R') = (B_{k-1}/\sigma_k)^{|A|}$. A tuple x on the right lies in S_{n-1} by minimality of n, so it has witnesses $a_i \in B_{n-1}$, pairwise τ-related, with $a_i/\sigma_k = x_i$ and hence $a_i \in B_{k-1} \circ \sigma_k$; and $\tau \subseteq \sigma_n$ makes $b = a_1 \in B_{n-1}$ a legitimate last entry.

(ii) $\mathrm{pr}_{|A|+1}(R') = ((B_{k-1} \circ \sigma_k) \cap B_{n-1})/\sigma_n$. A witnessing tuple has $(a_1, b) \in \sigma_n$ with $a_1 \in B_{n-1} \cap (B_{k-1} \circ \sigma_k)$; conversely such a c is witnessed by $(c, \dots, c, c)$.

(iii) $(B_{k-1}/\sigma_k)^{|A|} \times B_n/\sigma_n \not\subseteq R'$. Otherwise, fixing $b_0 \in B_{k-1} \subseteq B_n$, every $x \in (B_{k-1}/\sigma_k)^{|A|}$ has a witness with $(a_i, b_0) \in \sigma_n$; then $b_0 \in B_n \subseteq E_n$ puts each a_i in $B_{n-1} \cap E_n = B_n$, so $x \in S_n$, contradicting the choice of n.

(iv) $\mathrm{pr}_{|A|+1}(R') = B_{n-1}/\sigma_n$. The inductive hypothesis at $k-1$ — applied to the chain $B_{k-1} <^A_{T_{k-1}(\sigma_{k-1})} \cdots <^A_{T_1(\sigma_1)} B_0$, the congruence σ_k in the role of δ, and the index n — leaves B_{n-1}/σ_n or B_n/σ_n for the set computed in (ii). The latter is a single point, which with (i) would give the containment excluded by (iii).

Now R' is subdirect by (i) and (iv), it is closed under substitution in its first $|A|$ coordinates, and $|B_{k-1}/\sigma_k| \leq |A|$, so Lemma 7.35 supplies d with $(B_{k-1}/\sigma_k)^{|A|} \times \{d\} \subseteq R'$. The associated set $C^\star \subseteq B_{n-1}/\sigma_n$ is therefore nonempty, and proper by (iii); so Lemma 7.20 applied to R'^{-1} yields either a proper nonempty central subuniverse of B_{n-1}/σ_n or, through Lemma 7.9, a proper nonempty binary absorbing subuniverse of B_{k-1}/σ_k — both excluded. So subcase 1B ends at its first branch.

Case 2: $k > m$**.** Then $B_k \subseteq B_{k-1} \subseteq B_m \subseteq B_{m-1}$. The inductive hypothesis at $k-1$, with the same δ and m, makes $((B_{k-1} \circ \delta) \cap B_{m-1})/\sigma_m$ either B_m/σ_m or B_{m-1}/σ_m.

If it is B_m/σ_m, that set is a single point, and $\varnothing \neq B_k \subseteq (B_k \circ \delta) \cap B_{m-1} \subseteq (B_{k-1} \circ \delta) \cap B_{m-1}$ forces $((B_k \circ \delta) \cap B_{m-1})/\sigma_m = B_m/\sigma_m$ as well; the additional clause is carried over unchanged from the inductive hypothesis, which had the same δ and the same m. So assume

$$((B_{k-1} \circ \delta) \cap B_{m-1})/\sigma_m = B_{m-1}/\sigma_m, \tag{7.2}$$

and split on T_k. In each subcase the answer is B_{m-1}/σ_m, so the additional clause is not triggered.

Subcase 2A: $T_k \in \{\text{BA}, \text{C}\}$. The formula $\exists y\, (X(y) \wedge \delta(x, y)) \wedge B_{m-1}(x)$ defines $(B_{k-1} \circ \delta) \cap B_{m-1}$ at $X = B_{k-1}$ and $(B_k \circ \delta) \cap B_{m-1}$ at $X = B_k$, and the latter is nonempty as it contains B_k; so Lemmas 7.4 and 7.7 give $((B_k \circ \delta) \cap B_{m-1})/\sigma_m \leq_{T_k} ((B_{k-1} \circ \delta) \cap B_{m-1})/\sigma_m = B_{m-1}/\sigma_m$. A *proper* such containment is a proper nonempty binary absorbing or central subuniverse of B_{m-1}/σ_m, which is excluded, so it is an equality.

Subcase 2B: $T_k = \text{S}$. Choose a nonempty $G \subseteq B_k$ with $G <_{\text{BA,C}} B_{k-1}$ and run 2A with G for B_k: it gives $((G \circ \delta) \cap B_{m-1})/\sigma_m = B_{m-1}/\sigma_m$, and $G \subseteq B_k$ makes the left side a subset of $((B_k \circ \delta) \cap B_{m-1})/\sigma_m$.

Subcase 2C: $T_k = \text{D}$. Put

$$R = \{(a/\sigma_k, b/\sigma_m) : a \in B_{k-1},\ b \in B_{m-1},\ (a, b) \in \delta\} \leq \mathbf{B}_{k-1}/\sigma_k \times \mathbf{B}_{m-1}/\sigma_m.$$

It is subdirect: $B_{k-1} \subseteq B_{m-1}$ lets $b = a$ serve for the left projection, and the right projection is B_{m-1}/σ_m by (7.2). Moreover $B_{k-1}/\sigma_k \times B_m/\sigma_m \subseteq R$: here $k > m$ gives $B_{k-1} \subseteq B_m$, so for $a \in B_{k-1}$ the pair (a, a) witnesses $(a/\sigma_k, a/\sigma_m) \in R$ with a/σ_m the single class of B_m/σ_m. **This is the only use of $k > m$ in the subcase, and without it R need not have a centre at all.**

So the set $C^\star = \{c \in B_{m-1}/\sigma_m : \{c\} \times B_{k-1}/\sigma_k \subseteq R^{-1}\}$ is nonempty. Lemma 7.20 applied to $R^{-1} \leq_{\text{sd}} \mathbf{B}_{m-1}/\sigma_m \times \mathbf{B}_{k-1}/\sigma_k$ leaves only its alternative (b), alternative (c) being excluded by BA-freeness of B_{k-1}/σ_k; and since B_{m-1}/σ_m is centre free the central subuniverse $C^\star$ cannot be proper. Hence $C^\star = B_{m-1}/\sigma_m$ and R is the full product.

Finally, fullness transfers from B_{k-1} to B_k. Let $a \in B_k$ and $b \in B_{m-1}$. Fullness gives $a' \in B_{k-1}$ and $b' \in B_{m-1}$ with $(a', b') \in \delta$, $a'/\sigma_k = a/\sigma_k$ and $b'/\sigma_m = b/\sigma_m$. From $a \in B_k \subseteq E_k$ and $a' \,\sigma_k\, a$ we get $a' \in E_k$, so $a' \in B_{k-1} \cap E_k = B_k$; hence $b' \in (B_k \circ \delta) \cap B_{m-1}$ and $b'/\sigma_m = b/\sigma_m$. As b was arbitrary, $((B_k \circ \delta) \cap B_{m-1})/\sigma_m = B_{m-1}/\sigma_m$. ☐

Caveat 7.38 (Three steps this proof supplies)**.** Subcase 1B's first branch is where the projection is actually computed, and it needs two things beyond the displayed identity $(E \cap B_{k-1})/\sigma_k = B_{k-1}/\sigma_k$: that a *single* δ-class E works, which comes from the witnesses for one tuple enumerating all of B_{k-1}/σ_k and so from $|B_{k-1}/\sigma_k| \leq |A|$ again; and that E *meets* B_k, without which E has nothing to do with $B_k \circ \delta$. The second follows from the first, since the classes exhausted include the class of B_k.

Subcase 2C ends with a transfer step: R full is a statement about B_{k-1}, and the conclusion is about B_k. It goes through only because B_k is cut out of B_{k-1} by a σ_k-block, so a σ_k-move inside B_{k-1} cannot leave B_k — the same observation that drives subcase 1A.

The proof does not need δ to be irreducible, or comparable with any σ_i; it is an arbitrary congruence, and the additional clause is the only place any relation between δ and σ_m is asserted.

Lemma 7.39 (The intersection property)**.** Let $B \lhd\!\lhd\!\lhd A$, $C \lhd\!\lhd\!\lhd A$ with $B \cap C \neq \varnothing$. Then

(d) if δ is a dividing congruence for $B \leq A$ then $|(B \cap C)/\delta| = 1$ or $(B \cap C)/\delta = B/\delta$; and if $|(B \cap C)/\delta| = 1$ then $\delta \supseteq \delta_1 \cap \cdots \cap \delta_s$, where $\delta_1, \ldots, \delta_s$ are the congruences of the *chosen* chains witnessing $B \lhd\!\lhd\!\lhd A$ and $C \lhd\!\lhd\!\lhd A$;

(s) if $G <_{\text{BA,C}} B$ then $G \cap C \neq \varnothing$.

Proof. Fix chains

$$B = B_k <^A_{T_k(\sigma_k)} \cdots <^A_{T_1(\sigma_1)} B_0 = A, \qquad C = C_\ell <^A_{\mathcal{T}_\ell(\omega_\ell)} \cdots <^A_{\mathcal{T}_1(\omega_1)} C_0 = A$$

witnessing $B \lhd\lhd\lhd A$ and $C \lhd\lhd\lhd A$, with all $T_i, \mathcal{T}_j \in \{\text{BA}, \text{C}, \text{S}, \text{D}\}$ and, by Definition 4.22, $\sigma_i = A^2$ whenever $T_i \neq \text{D}$ and $\omega_j = A^2$ whenever $\mathcal{T}_j \neq \text{D}$. Put $\sigma = \bigcap_{i=1}^k \sigma_i$ and $\omega = \bigcap_{j=1}^\ell \omega_j$, so that $\sigma \cap \omega = \delta_1 \cap \cdots \cap \delta_s$ is exactly the intersection named in the "moreover" clause. The induction is on $k + \ell$ over *pairs of chains* — every appeal below names the shorter pair it is made to — and proves (s) and (d) together.

Three facts are used throughout and are recorded once.

(F1) $B^2 \subseteq \sigma$ and $C^2 \subseteq \omega$. If $T_i = \text{D}$ then $B_i = B_{i-1} \cap E_i$ for a block E_i of σ_i, so $B_i^2 \subseteq \sigma_i$ and a fortiori $B^2 \subseteq \sigma_i$; otherwise $\sigma_i = A^2$. Likewise for C. In particular $(a, b) \in \sigma \cap \omega$ for all $a, b \in B \cap C$.

(F2) $\sigma \subseteq \sigma_i$ and $\omega \subseteq \omega_j$ for every i and j, by construction.

(F3) $|A| \geq 2$. That δ is a dividing congruence for $B \leq A$ means, by Definition 4.22, that $B' <^A_{\text{D}(\delta)} B$ for some B'; such a B' is a nonempty proper subset $B \cap E$ of B cut out by a δ-block, so $|B/\delta| \geq 2$, and hence $|A| \geq |B| \geq 2$. The same appeal supplies the two properties of δ used below: $\mathbf{B}/\delta$ is BA and centre free, and $B^2 \subseteq \delta^*$.

Part (s). Let $G <_{\text{BA,C}} B$. If $\ell = 0$ then $C = A \supseteq G \neq \varnothing$. So let $\ell \geq 1$. The chain for C truncated at $C_{\ell-1}$ witnesses $C_{\ell-1} \lhd\lhd\lhd A$ with $\ell - 1$ steps, and $B \cap C_{\ell-1} \supseteq B \cap C \neq \varnothing$; so the inductive hypothesis (s), at $k + (\ell - 1)$, gives

$$G \cap C_{\ell-1} \neq \varnothing. \tag{7.3}$$

Write $A' = B \cap C_{\ell-1}$, a nonempty subuniverse of $\mathbf{A}$. Since $G \subseteq B$ we have $G \cap X = G \cap A' \cap X$ for every $X \subseteq C_{\ell-1}$, and $B \cap C_\ell \supseteq B \cap C \neq \varnothing$. By Lemma 7.15 applied twice, once for BA and once for C,

$$G \cap C_{\ell-1} = G \cap A' \dot{\leq}_{\text{BA,C}} A', \tag{7.4}$$

and by (7.3) the dot may be dropped. Split on $\mathcal{T}_\ell$.

Case $\mathcal{T}_\ell = \text{D}$, with $C_\ell = C_{\ell-1} \cap E$ for a block E of ω_ℓ. Apply the inductive hypothesis (d) to the pair $(C_{\ell-1}, B)$ — that is, with $C_{\ell-1}$ in the role of the first subuniverse and B in the role of the second — and to the dividing congruence ω_ℓ for $C_{\ell-1} \leq A$, at $(\ell - 1) + k$. Either $|A'/\omega_\ell| = 1$ or $A'/\omega_\ell = C_{\ell-1}/\omega_\ell$.

In the first case the sole ω_ℓ-class meeting A' is E, because $\varnothing \neq B \cap C \subseteq A' \cap E$; hence $A' \subseteq E$ and $A' = A' \cap E = B \cap C_\ell$, so $G \cap C_\ell = G \cap A' \cap C_\ell = G \cap A' = G \cap C_{\ell-1} \neq \varnothing$.

In the second case Lemma 7.7 applied to (7.4) gives $(G \cap C_{\ell-1})/\omega_\ell \leq_{\text{BA,C}} A'/\omega_\ell = C_{\ell-1}/\omega_\ell$. If that containment is proper it exhibits a proper nonempty binary absorbing subuniverse of $C_{\ell-1}/\omega_\ell$, contradicting the requirement of Definition 4.22(iii) for the step $C_\ell <^A_{\text{D}(\omega_\ell)} C_{\ell-1}$ that $\mathbf{C}_{\ell-1}/\omega_\ell$ be BA and centre free. If it is an equality then every ω_ℓ-class meeting $C_{\ell-1}$ already meets $G \cap C_{\ell-1}$; the class E does meet $C_{\ell-1}$, since $C_\ell \neq \varnothing$, so there is $g \in G \cap C_{\ell-1} \cap E = G \cap C_\ell$.

Case $\mathcal{T}_\ell \in \{\text{BA}, \text{C}\}$. By Lemma 7.15, $B \cap C_\ell = A' \cap C_\ell \dot{\leq}_{\mathcal{T}_\ell} A'$, and it is nonempty. If $G \cap C_{\ell-1} = A'$ then $A' \subseteq G$, and $\varnothing \neq B \cap C_\ell \subseteq A'$ gives $G \cap C_\ell \neq \varnothing$. If $B \cap C_\ell = A'$ then $G \cap C_\ell = G \cap A' \cap C_\ell = G \cap A' \neq \varnothing$. Otherwise both containments are proper, so $G \cap C_{\ell-1} <_{\text{BA,C}} A'$ and $B \cap C_\ell <_{\mathcal{T}_\ell} A'$, and Lemma 7.29 inside $\mathbf{A}'$ makes their intersection nonempty; that intersection is contained in $G \cap C_\ell$.

Case $\mathcal{T}_\ell = \text{S}$. By Definition 4.6 there is a nonempty $G' \subseteq C_\ell$ with $G' <_{\text{BA,C}} C_{\ell-1}$ — proper, since $G' \subseteq C_\ell \subsetneq C_{\ell-1}$. The inductive hypothesis (s) applied to the pair $(C_{\ell-1}, B)$ with the subuniverse G', at $(\ell - 1) + k$, gives $B \cap G' \neq \varnothing$; so $B \cap G' = A' \cap G' \leq_{\text{BA,C}} A'$ by Lemma 7.15. If $G \cap C_{\ell-1} = A'$ then $\varnothing \neq B \cap G' \subseteq A' \subseteq G$ and $B \cap G' \subseteq C_\ell$, so $G \cap C_\ell \neq \varnothing$. If $B \cap G' = A'$

then $A' \subseteq G' \subseteq C_\ell$, so $G \cap C_\ell \supseteq G \cap A' \neq \varnothing$. Otherwise both are proper and Lemma 7.29 gives $\varnothing \neq (G \cap C_{\ell-1}) \cap (B \cap G') \subseteq G \cap C_\ell$.

Part (d). For $0 \leq m \leq k$ and $0 \leq n \leq \ell$ put

$$P_{m,n} = \{(a_1, \ldots, a_{|A|}) \in (B_m \cap C_n)^{|A|} : (a_i, a_j) \in \sigma \cap \omega \text{ for all } i, j\}, \qquad S_{m,n} = P_{m,n}/\delta,$$

the quotient being coordinatewise and, as always, an *image* (§2.1, item viii): a tuple $x \in (A/\delta)^{|A|}$ lies in $S_{m,n}$ exactly when *one* tuple of representatives of its entries lies in $P_{m,n}$. Each $P_{m,n}$ is a subuniverse of $\mathbf{A}^{|A|}$, being cut out by unary and binary conditions, so each $S_{m,n}$ is a subuniverse of $(\mathbf{A}/\delta)^{|A|}$; each is closed under substitution $x \mapsto x \circ g$ along any $g : [|A|] \to [|A|]$, because every defining condition is unary in a_i or binary in (a_i, a_j), and in particular each is totally symmetric and shift-closed; and $S_{m,n} \supseteq S_{m',n'}$ whenever $m \leq m'$ and $n \leq n'$. Write $S = S_{0,0}$; since $B_0 \cap C_0 = A$ and $(a, a) \in \sigma \cap \omega$ for every a, this one — and only this one — is also reflexive.

By (F3) the arity $|A|$ is at least 2, so Lemma 7.34 applies to S and the dividing congruence δ for $B \leq A$, leaving two cases.

Case 1: $S = \{(a/\delta, \ldots, a/\delta) : a \in A\}$. Let $(a, b) \in \sigma \cap \omega$. Since $|A| \geq 2$ the tuple $(a, b, b, \ldots, b)$ lies in $P_{0,0}$, its entries being pairwise $\sigma \cap \omega$-related; so its image lies in S and is therefore constant, giving $a/\delta = b/\delta$. Hence $\sigma \cap \omega \subseteq \delta$, which is the "moreover" clause. By (F1) any $a, b \in B \cap C$ satisfy $(a, b) \in \sigma \cap \omega \subseteq \delta$, and $B \cap C \neq \varnothing$, so $|(B \cap C)/\delta| = 1$.

Case 2: $(B/\delta)^{|A|} \subseteq S$. We show that the only surviving subcase is 2C, in which $(B \cap C)/\delta = B/\delta$; by (F3) that alternative has $|(B \cap C)/\delta| = |B/\delta| \geq 2$, so the "moreover" clause is not triggered in Case 2 and nothing further is owed.

Subcase 2A: $(B/\delta)^{|A|} \not\subseteq S_{k,0}$. Choose m minimal with $(B/\delta)^{|A|} \not\subseteq S_{m,0}$; since $S_{0,0} = S$ we have $m \geq 1$ and $(B/\delta)^{|A|} \subseteq S_{m-1,0}$, so $(B/\delta)^{|A|}$ is a subalgebra of $\mathbf{S}_{m-1,0}$. Note that for every $b \in B$ the constant tuple $(b, \ldots, b)$ lies in $P_{m,0}$, because $B \subseteq B_m$ and $(b, b) \in \sigma \cap \omega$; so

$$\varnothing \neq S_{m,0} \cap (B/\delta)^{|A|} \subsetneq (B/\delta)^{|A|}, \tag{7.5}$$

the properness being the choice of m. Split on T_m.

Subsubcase 2A1: $T_m \in \{\textsc{ba}, \textsc{c}\}$. The formula

$$\Phi_X(x_1, \ldots, x_{|A|}) = \bigwedge_{i,j} (\sigma \cap \omega)(x_i, x_j) \wedge \bigwedge_i X(x_i)$$

is positive primitive in the unary relation X, and defines $P_{m-1,0}$ at $X = B_{m-1}$ and $P_{m,0}$ at $X = B_m$. Since $B_m <_{T_m} B_{m-1}$, Lemma 7.4 gives $P_{m,0} \dot{\leq}_{T_m} P_{m-1,0}$, and $P_{m,0} \neq \varnothing$ by the constant tuples, so $P_{m,0} \leq_{T_m} P_{m-1,0}$. Lemma 7.7, applied to that containment inside $\mathbf{P}_{m-1,0}$ with the congruence induced by $\delta^{|A|}$, gives $S_{m,0} \leq_{T_m} S_{m-1,0}$, and then Lemma 7.15 restricts along the subalgebra $(B/\delta)^{|A|} \leq \mathbf{S}_{m-1,0}$:

$$S_{m,0} \cap (B/\delta)^{|A|} \dot{\leq}_{T_m} (B/\delta)^{|A|}.$$

By (7.5) the left side is nonempty and proper, so this is $<_{T_m}$, and Lemma 7.9 produces a proper nonempty subuniverse of $\mathbf{B}/\delta$ of type $T_m \in \{\textsc{ba}, \textsc{c}\}$. That contradicts (F3).

Subsubcase 2A2: $T_m = \textsc{s}$. By Definition 4.6 choose a nonempty $G \subseteq B_m$ with $G <_{\textsc{ba,c}} B_{m-1}$. Since $B \lhd\!\lhd\!\lhd^A B_{m-1} \lhd\!\lhd\!\lhd A$ along the two halves of the chain, Lemma 7.24 gives $G \cap B \neq \varnothing$. Run 2A1 with G in place of B_m: writing $P^G = \Phi_G$ and $S^G = P^G/\delta$, the same three lemmas give $S^G \cap (B/\delta)^{|A|} \dot{\leq}_{\textsc{ba}} (B/\delta)^{|A|}$; it is nonempty, since $b \in G \cap B$ contributes its constant tuple, and proper, since $G \subseteq B_m$ makes it a subset of the proper set in (7.5). Lemma 7.9 again contradicts (F3). (Only the $\textsc{ba}$ half of $G <_{\textsc{ba,c}} B_{m-1}$ is spent here.)

Subsubcase 2A3: $T_m = \textsc{d}$, with $B_m = B_{m-1} \cap E$ for a block E of σ_m. Let $B \circ \delta$ denote the δ-saturation of B, a subuniverse of $\mathbf{A}$, and note $a/\delta \in B/\delta \iff a \in B \circ \delta$. Put

$$P' = \{(a_1, \dots, a_{|A|}, b) : b \in B_{m-1},\ \forall i\ a_i \in B_{m-1} \cap (B \circ \delta),\ \forall i, j\ (a_i, a_j) \in \sigma \cap \omega,\ \forall i\ (a_i, b) \in \sigma_m\}$$

and let $R' \le (\mathbf{A}/\delta)^{|A|} \times \mathbf{A}/\sigma_m$ be its image under $(a_1, \dots, a_{|A|}, b) \mapsto (a_1/\delta, \dots, a_{|A|}/\delta, b/\sigma_m)$, so that $R' \subseteq (B/\delta)^{|A|} \times B_{m-1}/\sigma_m$. Four facts about R'.

(i) $\mathrm{pr}_{1,\dots,|A|}(R') = (B/\delta)^{|A|}$. Let $x \in (B/\delta)^{|A|}$. By minimality of m, $x \in S_{m-1,0}$, so there are $a_i \in B_{m-1}$, pairwise $\sigma \cap \omega$-related, with $a_i/\delta = x_i$; each a_i then lies in $B \circ \delta$. By (F2), $(a_i, a_j) \in \sigma \subseteq \sigma_m$, so $b = a_1 \in B_{m-1}$ satisfies $(a_i, b) \in \sigma_m$ for every i, and $(a_1, \dots, a_{|A|}, b) \in P'$.

(ii) $\mathrm{pr}_{|A|+1}(R') = ((B \circ \delta) \cap B_{m-1})/\sigma_m$. For $\subseteq$: a witnessing tuple has $a_1 \in B_{m-1} \cap (B \circ \delta)$ and $(a_1, b) \in \sigma_m$, so $b/\sigma_m = a_1/\sigma_m$. For $\supseteq$: given c in that set, the tuple $(c, \dots, c, c)$ lies in P'.

(iii) $(B/\delta)^{|A|} \times B_m/\sigma_m \not\subseteq R'$. Suppose otherwise and fix $b_0 \in B \subseteq B_m$. For any $x \in (B/\delta)^{|A|}$ the pair $(x, b_0/\sigma_m)$ lies in R', so it has a witness $(a_1, \dots, a_{|A|}, b) \in P'$ with $b/\sigma_m = b_0/\sigma_m$; then $(a_i, b_0) \in \sigma_m$, and $b_0 \in B_m \subseteq E$ forces $a_i \in E$, whence $a_i \in B_{m-1} \cap E = B_m$. So the witness lies in $P_{m,0}$ and $x \in S_{m,0}$ — for every x, contradicting the choice of m.

(iv) $\mathrm{pr}_{|A|+1}(R') = B_{m-1}/\sigma_m$. Apply Lemma 7.37 to the chain for B, the congruence δ and the index m: by (ii) the projection is either B_{m-1}/σ_m or B_m/σ_m. In the second case $B_m \subseteq E$ makes B_m/σ_m a single point, so by (i) every $x \in (B/\delta)^{|A|}$ is paired in R' with that point, i.e. $(B/\delta)^{|A|} \times B_m/\sigma_m \subseteq R'$, contradicting (iii).

By (i) and (iv), R' is subdirect in $(\mathbf{B}/\delta)^{|A|} \times \mathbf{B}_{m-1}/\sigma_m$; both factors are finite algebras with the single basic operation w, so Convention 2.1 holds for them. R' is closed under substitution in its first $|A|$ coordinates, since every defining condition of P' is unary in a_i or binary in (a_i, a_j) or (a_i, b); and $|B/\delta| \le |A|$. So Lemma 7.35, with $F = B/\delta$ and $N = |A|$, supplies $d \in B_{m-1}/\sigma_m$ with $(B/\delta)^{|A|} \times \{d\} \subseteq R'$.

Now apply Lemma 7.20 to the transpose $R'^{-1} = \{(y, x) : (x, y) \in R'\} \le_{\mathrm{sd}} \mathbf{B}_{m-1}/\sigma_m \times (\mathbf{B}/\delta)^{|A|}$, whose associated set is $C^\star = \{c \in B_{m-1}/\sigma_m : \{c\} \times (B/\delta)^{|A|} \subseteq R'^{-1}\}$. Alternative (a) fails, since $d \in C^\star$. And $C^\star$ is *proper*: were it all of B_{m-1}/σ_m we should have $R' = (B/\delta)^{|A|} \times B_{m-1}/\sigma_m$, which contains $(B/\delta)^{|A|} \times B_m/\sigma_m$ and contradicts (iii). So alternative (b) makes $C^\star$ a proper nonempty central subuniverse of B_{m-1}/σ_m, contradicting Definition 4.22(iii) for the step $B_m <^{A}_{\textsc{d}(\sigma_m)} B_{m-1}$; and alternative (c) gives a proper nonempty binary absorbing subuniverse of the *power* $(B/\delta)^{|A|}$, which Lemma 7.9 moves down to $\mathbf{B}/\delta$, contradicting (F3). Both obligations of Caveat 7.21 are thereby discharged.

Subcase 2B: $(B/\delta)^{|A|} \subseteq S_{k,0}$ but $(B/\delta)^{|A|} \not\subseteq S_{k,\ell}$. Choose n minimal with $(B/\delta)^{|A|} \not\subseteq S_{k,n}$; then $n \ge 1$ and $(B/\delta)^{|A|} \subseteq S_{k,n-1}$. Fixing once and for all $b_1 \in B \cap C \subseteq B_k \cap C_n$ — this is the one place where the hypothesis $B \cap C \neq \varnothing$ is spent inside Case 2 — its constant tuple lies in $P_{k,n}$, so

$$\varnothing \neq S_{k,n} \cap (B/\delta)^{|A|} \subsetneq (B/\delta)^{|A|}. \tag{7.6}$$

Split on $\mathcal{T}_n$.

Subsubcase 2B1: $\mathcal{T}_n \in \{\textsc{ba}, \textsc{c}\}$. Verbatim 2A1, with the pp-formula $\bigwedge_{i,j} (\sigma \cap \omega)(x_i, x_j) \wedge \bigwedge_i B_k(x_i) \wedge \bigwedge_i X(x_i)$, with X instantiated at C_{n-1} and at C_n, and with (7.6) in place of (7.5). It contradicts (F3).

Subsubcase 2B2: $\mathcal{T}_n = \textsc{s}$. Choose a nonempty $G \subseteq C_n$ with $G <_{\textsc{ba},\textsc{c}} C_{n-1}$. Here B and C_{n-1} lie on different chains, so Lemma 7.24 does not apply and the induction is used instead: the inductive hypothesis (s) for the pair (C_{n-1}, B), at $(n-1)+k$, with $C_{n-1} \cap B \supseteq C \cap B \neq \varnothing$, gives $G \cap B \neq \varnothing$. Now run 2A2 with G in place of C_n: the witness $b \in G \cap B$ lies in $B_k \cap C_{n-1}$ and makes the intersection nonempty, $G \subseteq C_n$ makes it proper by (7.6), and Lemma 7.9 contradicts (F3).

Subsubcase 2B3: $\mathcal{T}_n = \textsc{d}$, with $C_n = C_{n-1} \cap E$ for a block E of ω_n. Put

$$P' = \{(a_1, \dots, a_{|A|}, b) : b \in C_{n-1},\ \forall i\ a_i \in B_k \cap C_{n-1},\ \forall i, j\ (a_i, a_j) \in \sigma \cap \omega,\ \forall i\ (a_i, b) \in \omega_n\}$$

and let R' be its image in $(\mathbf{A}/\delta)^{|A|} \times \mathbf{A}/\omega_n$. No saturation appears in the first $|A|$ coordinates this time, because $a_i \in B_k = B$ already gives $a_i/\delta \in B/\delta$.

(i) $\mathrm{pr}_{1,\dots,|A|}(R') = (B/\delta)^{|A|}$: for $x \in (B/\delta)^{|A|} \subseteq S_{k,n-1}$ take witnesses $a_i \in B_k \cap C_{n-1}$; by (F2) $(a_i, a_j) \in \omega \subseteq \omega_n$, so $b = a_1 \in C_{n-1}$ serves. *(ii)* $\mathrm{pr}_{|A|+1}(R') = (B_k \cap C_{n-1})/\omega_n$, by the argument of 2A3(ii). *(iii)* $(B/\delta)^{|A|} \times C_n/\omega_n \not\subseteq R'$: with $b_1 \in B \cap C \subseteq B_k \cap C_n$ in place of b_0, the argument of 2A3(iii) puts every witness in $C_{n-1} \cap E = C_n$ and so every $x \in (B/\delta)^{|A|}$ in $S_{k,n}$, contradicting the choice of n. *(iv)* $\mathrm{pr}_{|A|+1}(R') = C_{n-1}/\omega_n$: the inductive hypothesis (d) for the pair (C_{n-1}, B_k) with the dividing congruence ω_n for $C_{n-1} \leq A$, at $(n-1)+k$, leaves $|(B_k \cap C_{n-1})/\omega_n| = 1$ or $(B_k \cap C_{n-1})/\omega_n = C_{n-1}/\omega_n$. In the first case the sole class is E, since $b_1 \in B_k \cap C_n \subseteq E$; so $B_k \cap C_{n-1} \subseteq E$ and $B_k \cap C_{n-1} = B_k \cap C_{n-1} \cap E = B_k \cap C_n$, whence $P_{k,n-1} = P_{k,n}$ and $S_{k,n-1} = S_{k,n}$, contradicting the choice of n.

From here 2A3 runs word for word with ω_n for σ_m, C_{n-1} for B_{m-1} and C_n for B_m: a full fibre by Lemma 7.35, a proper nonempty $C^\star$, and Lemma 7.20 contradicting either Definition 4.22(iii) for $C_n <^{A}_{\mathrm{D}(\omega_n)} C_{n-1}$ or (F3).

Subcase 2C: $(B/\delta)^{|A|} \subseteq S_{k,\ell}$. For $b \in B$ the constant tuple $(b/\delta, \dots, b/\delta)$ lies in $S_{k,\ell}$, so it has a witness in $(B_k \cap C_\ell)^{|A|} = (B \cap C)^{|A|}$, whose first entry a satisfies $a/\delta = b/\delta$ with $a \in B \cap C$. Hence $B/\delta \subseteq (B \cap C)/\delta$, and the reverse containment is trivial, so $(B \cap C)/\delta = B/\delta$. ☐

Caveat 7.40 (The four places this proof can be misread)**.** (a) *$S_{m,n}$ is an image, not an intersection.* A tuple lies in it when *one* tuple of representatives satisfies all the conditions at once, and those conditions couple the representatives to each other. So $S_{m,n}$ is not a power of $(B_m \cap C_n)/\delta$ cut down by S, and the membership tests in 2A3(iii), 2B3(iii) and 2C all work by producing or constraining a single witnessing tuple.

(b) *Lemmas 7.15 and 7.7 deliver $\leq$, never $<$.* Part (s) therefore splits five times on whether a containment is proper, and in each case the equality branch *proves the lemma* rather than contradicting anything: an absorbing subuniverse that is everything is no obstruction, but it does make the set on the left swallow the set one is trying to meet. Reading these lemmas as producing proper subuniverses collapses all five splits, and it is the most tempting error available here.

(c) *The second alternative of Lemma 7.37 has to be excluded by hand*, in 2A3(iv), and likewise its analogue in 2B3(iv). What kills it is that B_m lies inside a single σ_m-block: the alternative would make the last coordinate of R' constant and hence, with surjectivity of the first $|A|$, force exactly the containment that the minimal choice of m forbids. In 2B3(iv) the same manoeuvre also identifies *which* class the single one is — the one defining C_n, witnessed by a point of $B \cap C$.

(d) *Every appeal to Lemma 7.20 owes two discharges*, per Caveat 7.21: the centre must be shown proper — here by the same non-containment that the choice of m or n provides — and the binary absorbing subuniverse of alternative (c) lives on a power and must be moved down by Lemma 7.9.

Two dependencies remain outside the proof: it rests on Lemma 7.37, which is still an outline, and on the imported Lemma 7.9. Minimality of the chains for B and C is *not* among them, and could not be: the inductive appeals in part (s), 2B2 and 2B3 are to truncations of the chain for C, which need not be minimal even when the original is.

Corollary 7.41. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$, $B_1 \lhd\!\lhd\!\lhd A_1$, $B_2 \lhd\!\lhd\!\lhd A_2$, and let σ be a dividing congruence for $B_1 \lhd\!\lhd\!\lhd A_1$. Then $\mathrm{pr}_1(R \cap (B_1 \times B_2))/\sigma$ is empty, of size* 1*, or equal to B_1/σ.*

Proof. Let $f_i : \mathbf{R} \to \mathbf{A}_i$ be the projections. By Lemma 6.4(b), $f_i^{-1}(B_i) \lhd\!\lhd\!\lhd \mathbf{R}$; apply Lemma 7.39(d) inside $\mathbf{R}$ to these two, with the congruence $f_1^{-1}(\sigma)$, and read the conclusion back through f_1. ☐

Caveat 7.42. The alternative "empty" is present in Corollary 7.41 and absent from Lemma 7.39, which hypothesises $B \cap C \neq \varnothing$. It is a real alternative, and every call site below has to exclude it separately.

Lemma 7.43 (Parallelogram property for D)**.** *Let $B_1, B_2 \lhd\!\lhd\!\lhd A$, $C_1, C_1' <^A_{\mathrm{D}(\sigma_1)} B_1$ and $C_2, C_2' <^A_{\mathrm{D}(\sigma_2)} B_2$ with $C_1' \cap C_2$, $C_1 \cap C_2'$ and $C_1' \cap C_2'$ all nonempty. Then $C_1 \cap C_2 \neq \varnothing$.*

Proof. By Lemma 7.39(d), $(B_1 \cap B_2)/\sigma_1$ is of size 1 or all of B_1/σ_1. In the first case the single block is the one defining C_1, because $C_1 \cap C_2' \neq \varnothing$ puts a point of C_1 in $B_1 \cap B_2$; hence $B_1 \cap B_2 \subseteq C_1$ and $C_1 \cap C_2 = B_1 \cap C_2 \supseteq C_1' \cap C_2 \neq \varnothing$. The symmetric argument applies if $(B_1 \cap B_2)/\sigma_2$ has size 1.

So assume $(B_1 \cap B_2)/\sigma_1 = B_1/\sigma_1$ and $(B_1 \cap B_2)/\sigma_2 = B_2/\sigma_2$, and put $S = \{(a/\sigma_1, a/\sigma_2) : a \in B_1 \cap B_2\}$, which is then subdirect in $(B_1/\sigma_1) \times (B_2/\sigma_2)$. Apply Lemma 7.39(d) to each σ_1-class C of B_1 against B_2 with the congruence σ_2; the hypothesis $C \cap B_2 \neq \varnothing$ holds because S is subdirect. Each fibre of S is therefore a single point or the whole of B_2/σ_2.

If some fibre is full, then either S is full — and then the fibre of C_1/σ_1 contains C_2/σ_2, giving $C_1 \cap C_2 \neq \varnothing$ — or the centre of S is proper and nonempty, and Lemma 7.20 produces a proper nonempty binary absorbing or central subuniverse of B_1/σ_1 or B_2/σ_2, contradicting Definition 4.22(iii). If every fibre is a single point, then the fibre of C_1'/σ_1 is both C_2/σ_2 (witnessed by $C_1' \cap C_2$) and C_2'/σ_2 (witnessed by $C_1' \cap C_2'$), so $C_2 = C_2'$ and $C_1 \cap C_2 = C_1 \cap C_2' \neq \varnothing$. □

7.4 Bridges

Three of the statements below — Lemmas 7.44, 7.46 and 7.47 — require their bridge to be *pair-reflexive*, and the hypothesis is not cosmetic in any of them: Remark 7.45 exhibits a bridge satisfying every other hypothesis of the first and none of its conclusion. Lemma 3.38 is what makes the hypothesis free, since every bridge these statements are applied to below is a δ°.

Lemma 7.44. *Let σ be a congruence on $\mathbf{A}$ and δ a symmetric,* pair-reflexive *bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$. Then there is an abelian group $(G; +, -)$ with $(A/\sigma; \delta/\sigma) \cong (G; x_1 - x_2 = x_3 - x_4)$.*

Proof. Factor by σ (Lemma 3.27) and assume $\sigma = 0_{\mathbf{A}}$. By Lemma 7.14 and Lemma 7.13, $\mathbf{A}$ is affine, hence polynomially equivalent to a $\mathbb{Z}_p$-module $\mathbf{G}$ and Maltsev via $x - y + z$.

View δ as a binary relation on $\mathbf{A}^2$ as in the proof of Lemma 7.14. It is reflexive, by pair-reflexivity together with $\mathrm{pr}_{1,2}(\delta) = A^2$, and symmetric by hypothesis. In a Maltsev algebra a reflexive compatible binary relation is a congruence: if η is reflexive and compatible then for $(x, y), (y, z) \in \eta$ the Maltsev term applied to $(x, y), (y, y), (y, z)$ gives $(x, z) \in \eta$. So δ is a congruence of $\mathbf{G}^2$.

Clause (d) of Definition 3.29 says that a pair $((x_1, x_2), (x_3, x_4))$ of δ has $x_1 = x_2$ exactly when $x_3 = x_4$, so the diagonal of $\mathbf{G}^2$ is a δ-block. A congruence of $\mathbf{G}^2$ with the diagonal as a block is the kernel of $(x, y) \mapsto x - y$ composed with a quotient of $\mathbf{G}$; since the diagonal is exactly one block, no further collapsing occurs, and $\delta = \{x_1 - x_2 = x_3 - x_4\}$. Taking $G = \mathbf{G}$ finishes. □

Remark 7.45 (Pair-reflexivity is not removable)**.** Without pair-reflexivity the correct conclusion is $\delta = \{(x_1, x_2, x_3, x_4) : \varphi(x_1 - x_2) = x_3 - x_4\}$ for an automorphism φ of G: factoring out the diagonal exhibits δ as the preimage of a subgroup $K \leq G \times G$ that meets $G \times 0$ and $0 \times G$ trivially and is subdirect, hence a graph. Symmetry says $\varphi^2 = \mathrm{id}$; pair-reflexivity says $\varphi = \mathrm{id}$.

Both φ occur. On $\mathbf{Z}_3 \in \mathcal{V}_4$ — the operation $x_1 + x_2 + x_3 + x_4$ is idempotent since $4 \equiv 1$, and is a special WNU — take $\sigma = 0$ and

$$\delta = \{(x_1, x_2, x_3, x_4) \in \mathbb{Z}_3^4 : x_1 - x_2 + x_3 - x_4 = 0\},$$

which is $\varphi = -\mathrm{id}$. It is a subuniverse of $\mathbf{A}^4$, being a subgroup; clause (d) holds because $x_1 = x_2$ iff $x_3 = x_4$; $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$; and it is symmetric. Yet no bijection $f \colon \mathbb{Z}_3 \to \mathbb{Z}_3$ carries δ to $\{x_1 - x_2 = x_3 - x_4\}$: such an f would have to satisfy $f(x_1) - f(x_2) = f(x_3) - f(x_4)$ whenever $x_1 - x_2 + x_3 - x_4 = 0$, and putting $(x_1, x_2, x_3, x_4) = (0, b, b, 0)$ gives $2(f(0) - f(b)) = 0$, hence $f(0) = f(b)$ for every b since p is odd — so f is constant, not a bijection. Since every abelian group of order 3 is $\mathbb{Z}_3$, no G works, and Lemma 7.44 genuinely fails without its hypothesis.

Lemma 7.46. *Let δ be a bridge from $0_{\mathbf{A}}$ to $0_{\mathbf{A}}$ with $\mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$ and $\mathrm{pr}_{1,2}(\delta)$ linked. Then $\mathbf{A}$ is BA and centre free.*

Proof outline, with the delicate steps in full. Replace δ by $\sigma := \delta^\circ$ of Lemma 3.38 and put $\omega = \mathrm{pr}_{1,2}(\sigma) = \mathrm{pr}_{1,2}(\delta)$. Note first that $\widetilde{\delta}$ is *reflexive*: clause (c) of Definition 3.29 gives $0_A \subseteq \mathrm{pr}_{1,2}(\delta)$, and the hypothesis gives $\mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$. Hence δ is a reflexive bridge, so Lemma 3.38 applies and gives $\omega \subseteq \widetilde{\delta} \subseteq \widetilde{\sigma}$ with $\widetilde{\sigma}$ symmetric, which is what legitimises the "without loss of generality" below. Induct on $|A|$.

Suppose $B <_T \mathbf{A}$ with $T \in \{\textsc{ba}, \textsc{c}\}$. Since ω is linked, ω meets $(A \setminus B) \times B$ or $B \times (A \setminus B)$; say the first, and pick (a, b) there.

Case 1: some subalgebra $D \lneq A$ contains a and b. Put $\rho = (\omega \circ \omega^{-1}) \cap D^2$, which is a subalgebra of $\mathbf{D}^2$, reflexive because ω is reflexive and symmetric by construction. Its transitive closure is again a subalgebra, hence a congruence on $\mathbf{D}$; let E be the block containing a and b, a subuniverse. Because ω is reflexive, every undirected ω-edge is a ρ-edge and every ρ-edge is a two-edge undirected ω-path, so ρ and the undirected ω-graph have the same connected components; hence $\omega \cap E^2$ is linked, and a, b lie in one component. *Note that $(\omega \cup \omega^{-1}) \cap D^2$ would not serve: a union of two subuniverses need not be closed.* Moreover $\mathrm{pr}_{1,2}(\sigma \cap E^4) = \omega \cap E^2$: the inclusion $\subseteq$ is trivial and $\supseteq$ holds because pair-reflexivity puts (x_1, x_2, x_1, x_2) in σ for every $(x_1, x_2) \in \omega$. So the inductive hypothesis applies to $\sigma \cap E^4$ and makes E BA and centre free, while Lemma 7.15 makes $B \cap E$ a proper nonempty subuniverse of E of type T — proper because $a \in E \setminus B$, nonempty because $b \in B \cap E$. Contradiction.

Case 2: no such D. Then $\{a\} \circ \omega = A$, because $\{a\}$ is a subuniverse by idempotence, hence so is $\{a\} \circ \omega$, and it contains a and b. Put $\xi(x_1, x_2) = \sigma(a, x_1, x_2, b)$; from $(a, b, a, b), (a, a, b, b) \in \sigma$ both projections of ξ contain a and b, hence equal A. Put $C = B \circ \xi$, which is nonempty because B is and $\mathrm{pr}_1(\xi) = A$, so $C \leq_T \mathbf{A}$ by Lemma 7.4, with $a \in C$ and $b \notin C$ — the latter because $b \in C$ would give $(a, c, b, b) \in \sigma$ with $c \in B$, and clause (d) would force $a = c \in B$.

If $T = \textsc{ba}$, applying the absorbing term to (a, b, a, b) and (a, a, b, b) gives $(a, b_1, b_2, b) \in \sigma$ with $b_1, b_2 \in B$, so $C \cap B \neq \varnothing$. If $T = \textsc{c}$, the three tuples (a, b, a, b), (a, a, a, a) and (a, a, b, b) witness

$$\sigma \cap (\{a\} \times A \times C \times B), \quad \sigma \cap (\{a\} \times C \times C \times A), \quad \sigma \cap (\{a\} \times C \times A \times B) \neq \varnothing,$$

which is the leave-one-out hypothesis of Corollary 7.26 for the ternary relation $\{(x_2, x_3, x_4) : \sigma(a, x_2, x_3, x_4)\}$ with $n = 3$, all types $\textsc{c}$ and all ambients A — legitimate because $\lhd\lhd\lhd$ is reflexive. *The three tight boxes of these witnesses — $\{a\} \times B \times C \times B$, $\{a\} \times C \times C \times C$, $\{a\} \times C \times B \times B$ — are subsets of the displayed ones and do not match the corollary's shape; the widening is what makes the appeal legitimate.* The corollary's conclusion is therefore available: $\sigma \cap (\{a\} \times C \times C \times B) \neq \varnothing$. Put $C' = \mathrm{pr}_2$ of that set, nonempty because the set is. Then $C' \leq_{\textsc{c}} A$ by Lemma 7.4, using $\{a\} \circ \omega = A$, and hence $C' \leq_{\textsc{c}} C$ by Lemma 7.15; and by clause (d) either $a \notin C'$ or $C \cap B \neq \varnothing$.

So in every case we have $\varnothing \neq C' <_T C <_T A$ — taking $C' = B \cap C$ when $C \cap B \neq \varnothing$. Then $|C| > 1$, and applying the inductive hypothesis to $\sigma \cap C^4$ contradicts $C' <_T C$: its hypotheses hold because $\{a\} \circ \omega = A$ makes $\omega \cap C^2$ a star centred at a, hence linked. $\square$

Lemma 7.47. Let σ be an irreducible congruence on $\mathbf{A}$ and δ a *pair-reflexive* bridge from σ to σ with $\widetilde{\delta} \supsetneq \sigma$. Then

(a) σ^* is a congruence;
(b) B/σ is BA and centre free for every block B of σ^*;
(c) if δ is also symmetric then there is a prime p such that every block B of σ^* satisfies $(B/\sigma; (\delta \cap B^4)/\sigma) \cong (\mathbb{Z}_p^{n_B}; x_1 - x_2 = x_3 - x_4)$ for some $n_B \geq 0$.

Proof. By Lemma 3.27 we may assume $\sigma = 0_{\mathbf{A}}$. Since $\widetilde{\delta}$ is a subuniverse of $\mathbf{A}^2$ properly containing σ and stable under it, minimality gives $\sigma^* \subseteq \widetilde{\delta}$; the same argument applied to $\mathrm{pr}_{1,2}(\delta)$, which properly contains σ by clause (c), gives $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$.

Let B be a block of $\mathrm{LeftLinked}(\sigma^*)$ with $|B| > 1$ and put $\delta' = \delta \cap B^4 \cap (\sigma^* \times \sigma^*)$. Then $\mathrm{pr}_{1,2}(\delta') = \sigma^* \cap B^2$: for $(x_1, x_2) \in \sigma^* \cap B^2$ pair-reflexivity puts (x_1, x_2, x_1, x_2) in δ — legitimate because $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$ — and that tuple lies in B^4 and in $\sigma^* \times \sigma^*$ — *this is where pair-reflexivity is spent, and without it the projection can collapse to* 0_B. The resulting relation is linked precisely because B is a block, and is contained in $\widetilde{\delta'}$ because $\sigma^* \subseteq \widetilde{\delta}$; so Lemma 7.46 makes B BA and centre free. Then $\sigma^* \cap B^2$, being linked and subdirect on B, must be all of B^2 by Corollary 7.2. Blocks of size 1 satisfy $B^2 \subseteq \sigma^*$ trivially, so $\sigma^* = \mathrm{LeftLinked}(\sigma^*)$ is an equivalence relation and hence a congruence, giving (a); and (b) follows, one-element algebras being vacuously BA and centre free.

For (c), fix a block B of σ^* with $|B| \geq 2$ and apply Lemma 7.44 to $\delta \cap B^4$, whose hypotheses now hold: $\mathrm{pr}_{1,2}(\delta \cap B^4) = B^2$ by the same pair-reflexivity computation, and $\widetilde{\delta \cap B^4} = B^2$ because $B^2 \subseteq \sigma^* \subseteq \widetilde{\delta}$. This gives an abelian group G_B with $(B; \delta \cap B^4) \cong (G_B; x_1 - x_2 = x_3 - x_4)$.

It remains to make every G_B elementary abelian for one common prime. Put $\omega = \delta \cap (\sigma^* \times \sigma^*)$, which has the same properties. From $x_1 - x_2 = x_3 - x_4$ one pp-defines $(k+1)x_1 = k\,x_2 + x_3$ for every k, by

$$((k+1)x_1 = k\,x_2 + x_3) = \exists x_4\ (k\,x_1 = (k-1)x_2 + x_4) \wedge (x_1 - x_2 = x_3 - x_4),$$

and hence, identifying x_3 with x_1, the relation $q\,x_1 = q\,x_2$ for every q. Let S be what that pp-definition defines when $x_1 - x_2 = x_3 - x_4$ is replaced by ω. If some G_B had an element of order p_1^2, or elements of coprime orders, or if two blocks had elements of different prime orders, then for a suitable q the relation S would be a subuniverse of $\mathbf{A}^2$ with $S \supsetneq \sigma$ and $\sigma^* \not\subseteq S$, contradicting the minimality of σ^*. Hence every element of every G_B has one and the same prime order p, and $G_B \cong \mathbb{Z}_p^{n_B}$. $\square$

Caveat 7.48 (It is minimality, not irreducibility)**.** The contradiction at the end of the proof is with the *minimality of* σ^*, which is a consequence of irreducibility and not the same statement. The distinction matters wherever σ^* is treated as data attached to a proof of irreducibility (Caveat 3.24).

Proof of Lemma 4.13. *(a)* $\Rightarrow$ *(b)*: take $\delta = S$ of Definition 4.10, which by Lemma 4.12 is a bridge with $\widetilde{S} = \sigma^* \supsetneq \sigma$.

(b) $\Rightarrow$ *(a)*: given a bridge δ with $\widetilde{\delta} \supsetneq \sigma$ — reflexive by Lemma 3.36, since the hypothesis contains $\widetilde{\delta} \supseteq \sigma$ — form δ° of Lemma 3.38; it is symmetric and pair-reflexive, and $\widetilde{\delta^\circ} \supseteq \widetilde{\delta} \supsetneq \sigma$. Now Lemma 7.47 gives items (b) and (c) of Definition 4.10, with the witness $S = \delta^\circ \cap (\sigma^* \times \sigma^*)$. $\square$

Lemma 7.49. Let σ be a congruence on $\mathbf{A}$ and δ a reflexive bridge from σ to σ with

(i) $\delta(x_1, x_2, x_3, x_4) = \delta(x_3, x_4, x_1, x_2)$;
(ii) $(a, b, a, b), (b, a, b, a) \in \delta$ for every $(a, b) \in \mathrm{pr}_{1,2}(\delta)$;
(iii) $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\delta)) = A^2$.

Then $(a, a, b, b) \in \delta$ for some $a \neq b$.

Proof. Factor by σ and assume $\sigma = 0_{\mathbf{A}}$; clause (c) of Definition 3.29 then forces $|A| \geq 2$. Write $P = \mathrm{pr}_{1,2}(\delta)$, which is symmetric by (ii), and $W = \mathrm{pr}_{1,2,3}(\delta) \leq_{\mathrm{sd}} \mathbf{P} \times \mathbf{A}$. Induct on $|A|$.

Case 1: some $B <_T \mathbf{A}$ with $|B| > 1$, $T \in \{\mathrm{BA}, \mathrm{C}\}$. Put $W' = \mathrm{pr}_{1,2,3}(\delta \cap B^4)$ and write $W(x_1, x_2, x_3) = \exists x_4\, \delta(x_1, x_2, x_3, x_4) \wedge A(x_1) \wedge A(x_2) \wedge A(x_3) \wedge A(x_4)$ with A the full unary relation. Replacing A by B — a *single* application of Lemma 7.4, which replaces every occurrence, so no transitivity of $\leq_{\mathrm{C}}$ is needed — gives $W' \leq_T W$. It is nonempty, containing (b, b, b) for every $b \in B$, and its projections are $\mathrm{pr}_{1,2}(W') = P \cap B^2$, by (ii), and $\mathrm{pr}_3(W') = B$. So Lemma 7.17 makes W' linked, which is (iii) for $\delta \cap B^4$. Conditions (i) and (ii) are inherited and $\delta \cap B^4$ is reflexive. If it is a bridge, the inductive hypothesis applies and produces the required pair inside B. If it is not, then $\mathrm{pr}_{1,2}(\delta \cap B^4) = 0_B$, so by clause (d) every tuple of $\delta \cap B^4$ has the form (x, x, y, y); linkedness of W' on $|B| \geq 2$ points then forbids the perfect matching, so some $(a, a, b, b) \in \delta$ with $a \neq b$.

Case 2: some $\{a\} <_T \mathbf{A}$, $T \in \{\mathrm{BA}, \mathrm{C}\}$, and no such B of size > 1. By (i) the relation $\mathrm{pr}_{1,3}(\delta)$ is symmetric, and by (iii) it is linked, so $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq \{a\}$; pick $(a, b, c, d) \in \delta$ with $c \neq a$. If $a = b$ then $c = d$ by clause (d) and we are done. If $\{a\} \circ \mathrm{pr}_{1,3}(\delta) \neq A$ then that set is a strong subuniverse containing a and c, so of size > 1, which is Case 1. Otherwise choose $(a, e, b, f) \in \delta$. By (ii), $(b, a, b, a) \in \delta$. Let g be a ternary absorbing operation for $\{a\}$, available for $T = \mathrm{C}$ by Lemma 4.4. Applying g to (a, e, b, f), (b, a, b, a) and (a, a, a, a) gives $(a, a, g(b, b, a), a) \in \delta$, whence $g(b, b, a) = a$ by clause (d); applying g to (b, a, b, a), (b, a, b, a) and (a, e, b, f) then gives $(a, a, b, a) \in \delta$, whence $b = a$, a contradiction.

Case 3: $\mathbf{A}$ is BA and centre free. Put $C = \{(a, b) \in P : \{(a, b)\} \times A \subseteq W\}$. By Lemma 7.22, applied to the transpose of W, $C \leq_{\mathrm{BA}} \mathbf{P}$. If $C = P$ then in particular $(a, a) \in C$ for any a; otherwise $C <_{\mathrm{BA}} P$ and Lemma 7.50 gives $C \cap 0_{\mathbf{A}} \neq \varnothing$. Either way $(a, a) \in C$ for some a, so for every c there is d with $\delta(a, a, c, d)$, and clause (d) forces $d = c$. Taking $c \neq a$ finishes the proof. $\square$

Lemma 7.50 ([13, Lem. 7.2])**.** *If $0_{\mathbf{A}} \subseteq \sigma \leq \mathbf{A}^2$ and $\omega <_{\mathrm{BA}} \sigma$, then $\omega \cap 0_{\mathbf{A}} \neq \varnothing$.*

Proof of Lemma 4.16. Put $\xi = \delta * \delta^{-1}$, which is symmetric and pair-reflexive. Each of $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi))$ and $\mathrm{RightLinked}(\mathrm{pr}_{1,2,4}(\xi))$ is a congruence containing σ, so by minimality of σ^* each is σ or contains σ^*; this gives three cases.

If both equal σ, then for $(a, b, c, d) \in \xi$ the classes c/σ and d/σ are determined by $a/\sigma, b/\sigma$; since $(a, b, a, b) \in \xi$ we get $(a, c), (b, d) \in \sigma$, so ξ is trivial. *Transferring that determinacy from $\xi = \delta * \delta^{-1}$ to δ itself is a counting argument, and it is where the hypothesis* $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta)$ *is spent:* on σ^*/σ, which is finite, δ induces a bipartite relation whose left neighbourhoods are nonempty, pairwise disjoint — that is what the case gives — and cover the right side; equal finite cardinalities force every neighbourhood to be a singleton, so the induced relation is a bijection and determinacy holds in both directions.

With that, introduce

$$\begin{aligned} \zeta_1(x_1, x_2, x_3, x_4) &= \exists y \exists y' \exists z \exists z' \exists t_1 \exists t_2 \exists t_3 \exists t_4 \\ &\qquad \delta(x_1, y, z, t_1) \wedge \delta(x_2, y, z', t_2) \wedge \delta(x_3, y', z, t_3) \wedge \delta(x_4, y', z', t_4), \\ \zeta_2(x_1, x_2, x_3, x_4) &= \exists y \exists z \quad \delta(x_1, y, z, x_3) \wedge \delta(x_2, y, z, x_4), \end{aligned}$$

and pick $(a, b, c, d) \in \delta$ with $(a, b) \in \sigma^* \setminus \sigma$. From $(a, b, c, d), (b, b, b, b) \in \delta$ one gets $(a, b, a, b) \in \zeta_1$ and hence $\mathrm{pr}_{1,2}(\zeta_1) \supseteq \sigma^*$. Split.

If ζ_1 is a bridge, then $\widetilde{\zeta_1} = \sigma$ — forced, since σ is PC — and $(a, a, c, c) \in \zeta_1$ for every $(a, b, c, d) \in \delta$, so $\mathrm{pr}_{1,3}(\delta) = \sigma$.

If ζ_1 is not a bridge, there is $(a, a, b, c) \in \zeta_1$ with $(b, c) \notin \sigma$. Take witnesses $y = d$, $y' = d'$, $z = e$, $z' = e'$, $t_i = f_i$. Determinacy makes $(e, e') \in \sigma$, so $(b, d', e, f_3), (c, d', e, f_4) \in \delta$ and therefore $\mathrm{pr}_{1,2}(\zeta_2) \supsetneq \sigma$; determinacy in both directions then makes ζ_2 a bridge, whence $\widetilde{\zeta_2} = \sigma$ and $\mathrm{pr}_{1,4}(\delta) = \sigma$.

So $\mathrm{pr}_{1,3}(\delta) = \sigma$ or $\mathrm{pr}_{1,4}(\delta) = \sigma$, and by the same argument with x_1, x_2 interchanged, $\mathrm{pr}_{2,4}(\delta) = \sigma$ or $\mathrm{pr}_{2,3}(\delta) = \sigma$. *Two of the four combinations must now be excluded:* $\mathrm{pr}_{1,3} = \mathrm{pr}_{2,3} = \sigma$ would give $\mathrm{pr}_{1,2}(\delta) \subseteq \sigma$, against $\mathrm{pr}_{1,2}(\delta) = \sigma^* \supsetneq \sigma$, and likewise for $\mathrm{pr}_{1,4} = \mathrm{pr}_{2,4}$. The two surviving combinations are the conclusion, the reverse inclusion following from stability under σ.

If $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\xi)) \supseteq \sigma^*$, choose a block B of it that is not a σ-block. Then $a, b, d \in B$ whenever $c \in B$ and $(a, b, c, d) \in \xi$, so $\xi \cap (A^2 \times B \times A) \subseteq B^4$ and therefore $\mathrm{pr}_{1,2,3}(\xi \cap B^4) = \mathrm{pr}_{1,2,3}(\xi) \cap (A^2 \times B)$, which is linked. *If $\sigma^* \cap B^2 \supsetneq \sigma \cap B^2$ then $\xi \cap B^4$ is a bridge and Lemma 7.49 applies, contradicting PC-ness via Lemma 4.13. If not, the lemma cannot be cited — it is not a bridge — and the case must be inlined: clause (d) then puts both pairs of every tuple in σ, and linkedness on a B that is not a σ-block produces $(x_1, x_1, x_3, x_3) \in \xi$ with $(x_1, x_3) \notin \sigma$ directly.* The third case is the mirror image, using $\mathrm{pr}_{1,2,4}$. ☐

Proof of Lemma 4.15. Let σ_1, σ_2 be congruences on $\mathbf{A}_1, \mathbf{A}_2$; both are irreducible, σ_1 because it is linear (Definition 4.10(a)). Replace δ first by $\delta \cap (\sigma_1^* \times \sigma_2^*)$, which is again a bridge with the same collapse, and then by the ω of Lemma 3.33. By clause (d) of that lemma the containment in $\sigma_1^* \times \sigma_2^*$ survives, and by clause (c) neither linking equivalence changes, so the case analysis below is unaffected; by clause (b), $\widetilde{\delta}$ is now rectangular. Being a bridge inside $\sigma_1^* \times \sigma_2^*$, it has $\mathrm{pr}_{3,4}(\delta) \supsetneq \sigma_2$ and stable under σ_2, so minimality of the cover (Proposition 3.25) gives

$$\underset{3,4}{\mathrm{pr}}(\delta) = \sigma_2^*, \tag{7.7}$$

and symmetrically $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$.

The easy case. Suppose $\mathrm{RightLinked}(\widetilde{\delta}) \supsetneq \sigma_2$. The composite $\delta^{-1} * \delta$ is a bridge from σ_2 to σ_2 by Lemma 3.32 — both congruences are irreducible — with collapse $\widetilde{\delta}^{-1} \circ \widetilde{\delta}$. Iterating, the k-fold composite has collapse $(\widetilde{\delta}^{-1} \circ \widetilde{\delta})^k$, and for k large this is $\mathrm{RightLinked}(\widetilde{\delta}) \supsetneq \sigma_2$. So Lemma 4.13 makes σ_2 linear.

So assume $\mathrm{RightLinked}(\widetilde{\delta}) = \sigma_2$. Then $\mathrm{LeftLinked}(\widetilde{\delta}) \supseteq \sigma_1^*$: otherwise minimality of σ_1^* forces $\mathrm{LeftLinked}(\widetilde{\delta}) = \sigma_1$, and $\widetilde{\delta}$ then induces a bijection $\mathbf{A}_1/\sigma_1 \cong \mathbf{A}_2/\sigma_2$ carrying σ_1^*/σ_1 to σ_2^*/σ_2, so σ_2 is linear because σ_1 is.

Normalising the first and third coordinates. Put $\delta'(x_1, x_2, x_3, x_4) = \delta(x_1, x_2, x_3, x_4) \wedge \widetilde{\delta}(x_1, x_3)$.

If δ' is a bridge, it has $\mathrm{pr}_{1,3}(\delta') = \widetilde{\delta'}$ by construction, and we take $\omega = \delta'$.

If δ' is not a bridge, then — clauses (a), (b), (d) being inherited — clause (c) fails, so there is no $(a, b, c, d) \in \delta$ with $(a, b) \notin \sigma_1$ and $(a, c) \in \widetilde{\delta}$. Put

$$\delta''(x_1, x_2, x_3, x_4) = \exists z\, \widetilde{\delta}(x_1, x_3) \wedge \delta(x_2, z, x_3, x_4).$$

This is a bridge. For clause (d) in one direction: if $(x_1, x_2) \in \sigma_1$ then $(x_2, x_3) \in \widetilde{\delta}$, so the failure above applied to $(x_2, z, x_3, x_4) \in \delta$ forces $(x_2, z) \in \sigma_1$ and hence $(x_3, x_4) \in \sigma_2$. In the other direction, if $(x_3, x_4) \in \sigma_2$ then $(x_2, z) \in \sigma_1$ and $(x_2, x_3) \in \widetilde{\delta}$, whence $(x_1, x_2) \in \sigma_1$. For clause (c): since $\delta' \neq \delta$ there is $(a, b, c, d) \in \delta$ with $(a, c) \notin \widetilde{\delta}$. Some e has $(e, c) \in \widetilde{\delta}$: by (7.7) the reflexive pair $(c, c) \in \sigma_2 \subseteq \sigma_2^*$ is $\mathrm{pr}_{3,4}$ of some $(e, e', c, c) \in \delta$; clause (d) of Definition 3.29 puts $(e, e') \in \sigma_1$, and stability of the first two variables under σ_1 — clause (a) — replaces e' by e, giving $(e, e, c, c) \in \delta$. That e gives $(e, a, c, d) \in \delta''$ with $(e, a) \notin \sigma_1$. Take $\omega = \delta''$.

Either way we have a bridge ω from σ_1 to σ_2 with $\mathrm{pr}_{1,3}(\omega) = \widetilde{\omega} = \widetilde{\delta}$.

The contradiction. Suppose σ_2 is *not* linear, so it is PC. Put

$$\xi_1(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6\ \omega(x_5, x_6, x_1, x_2) \wedge \omega(x_5, x_6, x_3, x_4),$$

a bridge from σ_2 to σ_2; by Lemma 4.16 it is $\sigma_2(x_1, x_3) \wedge \sigma_2(x_2, x_4)$ — the other alternative is excluded by symmetry of ξ_1 in its two pairs. Put

$$\xi_2(x_1, x_2, x_3, x_4) = \exists x_5 \exists x_6\ \omega(x_5, x_6, x_1, x_2) \wedge \omega(x_6, x_5, x_3, x_4).$$

Here $(x_5, x_1), (x_6, x_3) \in \widetilde{\omega}$ and $(x_5, x_6) \in \sigma_1^* \subseteq \mathrm{LeftLinked}(\widetilde{\omega})$, which gives $(x_1, x_3) \in \sigma_2$; so ξ_2 too is $\sigma_2(x_1, x_3) \wedge \sigma_2(x_2, x_4)$ by Lemma 4.16. Comparing the two, $(b, a, c, d) \in \omega$ whenever $(a, b, c, d) \in \omega$.

Now put

$$\zeta(x_1, x_2, x_3, x_4) = \exists y_1 \exists y_2 \exists y_3\ \omega(y_1, y_2, x_1, x_2) \wedge \omega(y_1, y_3, x_1, x_3) \wedge \omega(y_2, y_3, x_1, x_4).$$

If $x_1 = x_2$ then $y_1 = y_2$ by the shape of ξ_1, and then $x_3 = x_4$. Take any $(c, d) \in \sigma_2^*$; since $\mathrm{pr}_{3,4}(\omega) = \sigma_2^*$ there is $(a, b) \in \sigma_1^*$ with $(a, b, c, d) \in \omega$. Now $(a, c) \in \widetilde{\omega}$ — which *is* $(a, a, c, c) \in \omega$, by the definition of the collapse — and $(a, b) \in \sigma_1^* \subseteq \mathrm{LeftLinked}(\widetilde{\omega})$, so rectangularity in the form of Lemma 3.33(b) gives $(b, c) \in \widetilde{\omega}$, that is $(b, b, c, c) \in \omega$. *This is the one step of the proof that consumes rectangularity.* Sending $(x_1, x_2, x_3, x_4) \mapsto (c, d, c, d)$ with $(y_1, y_2, y_3) = (a, b, a)$ witnesses $(c, d, c, d) \in \zeta$; sending it to (c, c, d, d) with $(y_1, y_2, y_3) = (a, a, b)$ witnesses $(c, c, d, d) \in \zeta$. So $\zeta(x_1, x_2, x_3, x_4) \wedge \zeta(x_3, x_4, x_1, x_2)$ is a bridge from σ_2 to σ_2 whose collapse contains $\sigma_2^* \supsetneq \sigma_2$, so σ_2 is linear by Lemma 4.13 — contradicting the supposition. □

Caveat 7.51 (Where rectangularity is spent)**.** The proof opens by normalising twice, and both steps were asserted in an earlier draft rather than argued.

Rectangularity of $\widetilde{\delta}$ is now Lemma 3.33: composing δ with itself enough times replaces $\widetilde{\delta}$ by $\mathrm{LeftLinked}(\widetilde{\delta}) \circ \widetilde{\delta}$, which is a union of full boxes, one per connected component, and leaves both linking equivalences and the containment in $\sigma_1^* \times \sigma_2^*$ alone. It is worth knowing that the normalisation is spent *once*, at $(b, b, c, c) \in \omega$ in the construction of ζ; nothing else in the proof uses it. Note also that this is not the situation of Caveat 3.37, where the property turned out to hold already: a genuine replacement of δ is being made here, and the lemma is what licenses it.

Choosing e with $(e, c) \in \widetilde{\delta}$ is the second, and it is three lines from $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$, clause (d) of Definition 3.29 and stability under σ_1. What made it look harder than it is was the earlier draft's appeal to $\mathrm{pr}_1(\widetilde{\delta})$, which is the wrong projection: what is needed is that c lies in $\mathrm{pr}_2(\widetilde{\delta})$, and the reflexive pair (c, c) supplies it.

Proof of Lemma 4.17. (a) is Lemma 4.15 contrapositively: if σ_2 were linear then so would σ_1 be, via δ^{-1}.

For (b) and (c), $\delta^{-1} * \delta$ is a bridge from σ_1 to σ_1, and σ_1 is PC, so by Lemma 4.16 it is trivial: $\sigma_1(x_1, x_3) \wedge \sigma_1(x_2, x_4)$ or $\sigma_1(x_1, x_4) \wedge \sigma_1(x_2, x_3)$. Its collapse is $\widetilde{\delta} \circ \widetilde{\delta}^{-1}$, which is therefore σ_1; so the relation $\{(a/\sigma_1, b/\sigma_2) : (a, b) \in \widetilde{\delta}\}$ has left-neighbourhoods that are pairwise disjoint. Running the same argument on $\delta * \delta^{-1}$ makes the right-neighbourhoods pairwise disjoint. Both sides are covered, since $\mathrm{pr}_{1,2}(\delta) = \sigma_1^*$ and $\mathrm{pr}_{3,4}(\delta) = \sigma_2^*$, so the induced relation is a bijection, which is (c), and it is an isomorphism of algebras because $\widetilde{\delta}$ is a subuniverse — that is (b).

For (d), put

$$\xi(x_1, x_2, x_5, x_6) = \exists x_3 \exists x_4\ \delta(x_1, x_2, x_3, x_4) \wedge \widetilde{\delta}(x_5, x_3) \wedge \widetilde{\delta}(x_6, x_4),$$

a bridge from σ_1 to σ_1 by (c). Lemma 4.16 gives $\xi = \sigma_1(x_1, x_5) \wedge \sigma_1(x_2, x_6)$ or $\sigma_1(x_1, x_6) \wedge \sigma_1(x_2, x_5)$, and transporting each alternative through the bijection of (c) gives the two alternatives of (d). □

Caveat 7.52 (The third case)**.** The third case is genuinely the mirror of the second: $(a,b,a,b) \in \xi$ places b in the block, and $\sigma^* \subseteq \mathrm{RightLinked}$ places a and c. Writing it out is mechanical, and it is not written out here.

7.5 Types interaction

Lemma 7.53 ([13, Lem. 8.19])**.** *Let $\omega, \sigma_1, \sigma_2$ be congruences on $\mathbf{A}$ with $\omega \cap \sigma_1 = \omega \cap \sigma_2$ and $\omega \setminus \sigma_1 \neq \varnothing$. Then there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.*

Proof. Put $\delta(x_1,x_2,y_1,y_2) = \exists z_1 \exists z_2\ \sigma_1(x_1,z_1) \wedge \sigma_1(x_2,z_2) \wedge \omega(z_1,z_2) \wedge \sigma_2(z_1,y_1) \wedge \sigma_2(z_2,y_2)$. Clauses (a) and (b) are visible. For (d): if $(x_1,x_2) \in \sigma_1$ then $(z_1,z_2) \in \sigma_1 \cap \omega = \sigma_2 \cap \omega$, so $(y_1,y_2) \in \sigma_2$; *and conversely,* if $(y_1,y_2) \in \sigma_2$ then $(z_1,z_2) \in \omega \cap \sigma_2 = \omega \cap \sigma_1$, so $(x_1,x_2) \in \sigma_1$ — the direction the source omits, and the one that spends the other half of the hypothesis. For (c): $\sigma_1 \subseteq \mathrm{pr}_{1,2}(\delta)$ by taking $z_i = y_i = x_1$, and choosing $(a,b) \in \omega \setminus \sigma_1$ gives $(a,b,a,b) \in \delta$ with $(a,b) \notin \sigma_1$ and, since $\omega \setminus \sigma_1 = \omega \setminus \sigma_2$, also $(a,b) \notin \sigma_2$. Finally $\widetilde{\delta} = \sigma_1 \circ \sigma_2$: the inclusion $\supseteq$ by taking $z_1 = z_2$, using reflexivity of ω, and $\subseteq$ by reading off $(x,z_1) \in \sigma_1$, $(z_1,y) \in \sigma_2$. ☐

Lemma 7.54. Let $C_1 <^{A}_{T_1(\sigma_1)} B_1 \lhd\lhd\lhd A$ and $C_2 <^{A}_{T_2(\sigma_2)} B_2 \lhd\lhd\lhd A$ with $T_1, T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$, $C_1 \cap B_2 \neq \varnothing$, $B_1 \cap C_2 \neq \varnothing$ and $C_1 \cap C_2 = \varnothing$. Then $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}, \textsc{d}\}$; and if $T_1 = T_2 = \textsc{d}$ there is a bridge δ from σ_1 to σ_2 with $\widetilde{\delta} = \sigma_1 \circ \sigma_2$.

Proof. No S. If $T_1 = \textsc{s}$, pick $S_1 \subseteq C_1$ with $S_1 <_{\textsc{ba},\textsc{c}} B_1$. Since $C_2 \lhd\lhd\lhd A$ — the chain for B_2 extended by one D step — and $B_1 \cap C_2 \neq \varnothing$, Lemma 7.39(s) gives $S_1 \cap C_2 \neq \varnothing$, hence $C_1 \cap C_2 \neq \varnothing$. Symmetrically for T_2.

Both absorbing. If $T_1, T_2 \in \{\textsc{ba}, \textsc{c}\}$ then by Lemma 7.15 $C_1 \cap B_2 \leq_{T_1} B_1 \cap B_2$ and $B_1 \cap C_2 \leq_{T_2} B_1 \cap B_2$; both are proper, since $C_1 \cap B_2 = B_1 \cap B_2$ would put $B_1 \cap C_2$ inside $C_1 \cap C_2 = \varnothing$. Their intersection is empty, so Lemma 7.28 gives $T_1 = T_2 \in \{\textsc{ba}, \textsc{c}\}$.

Mixed. Let $T_1 \in \{\textsc{ba}, \textsc{c}\}$ and $T_2 = \textsc{d}$; the mirror case $T_1 = \textsc{d}$, $T_2 \in \{\textsc{ba}, \textsc{c}\}$ is identical with the indices swapped, the hypotheses being symmetric. By Lemma 7.39(d), $(B_1 \cap B_2)/\sigma_2$ has size 1 or is all of B_2/σ_2. In the first case the single block is the one defining C_2, because $B_1 \cap C_2 \neq \varnothing$; then $B_1 \cap B_2 = B_1 \cap C_2$, so $C_1 \cap B_2 \subseteq C_1 \cap C_2 = \varnothing$, a contradiction. In the second, Lemmas 7.15 and 7.7 give $(C_1 \cap B_2)/\sigma_2 \leq_{T_1} B_2/\sigma_2$, and this is *proper*: equality would make the σ_2-block defining C_2 meet C_1, i.e. $C_1 \cap C_2 \neq \varnothing$. So B_2/σ_2 has a proper nonempty strong subuniverse, contradicting Definition 4.22(iii).

Both D. Let ω be the intersection of the dividing congruences of the chosen chains for $B_1 \lhd\lhd\lhd A$ and $B_2 \lhd\lhd\lhd A$. Apply Lemma 7.39(d) to B_1 and C_2 with the congruence σ_1 — *to C_2, not to B_2; against B_2 the step does not follow.* Its second alternative, $(B_1 \cap C_2)/\sigma_1 = B_1/\sigma_1$, would put a point of C_2 in the σ_1-block defining C_1; so $|(B_1 \cap C_2)/\sigma_1| = 1$ and the "moreover" clause gives $\sigma_1 \supseteq \omega \cap \sigma_2$, because the chain witnessing $C_2 \lhd\lhd\lhd A$ is the chain for B_2 extended by the step $C_2 <_{\textsc{d}(\sigma_2)} B_2$ (Caveat 4.26). Symmetrically $\sigma_2 \supseteq \omega \cap \sigma_1$, so $\omega \cap \sigma_1 = \omega \cap \sigma_2$. Finally, for $c_1 \in C_1 \cap B_2$ and $c_2 \in B_1 \cap C_2$ we have $(c_1,c_2) \in \omega$, since both lie in $B_1 \cap B_2$ and each B_i sits inside a single block of every dividing congruence of its own chain; and $(c_1,c_2) \notin \sigma_1$, since $c_2 \in B_1$ and $c_2\, \sigma_1\, c_1 \in C_1$ would give $c_2 \in C_1 \cap C_2$. Now Lemma 7.53 applies. ☐

7.6 Factorisation

Lemma 7.55. *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \mathbf{A}_2$ with $R \cap (B_1 \times B_2) \neq \varnothing$, $B_i \lhd\lhd\lhd A_i$, σ a congruence on $\mathbf{A}_1$ with B_1/σ BA and centre free. If some $c \in A_2$ has $(E \times \{c\}) \cap R \neq \varnothing$ for every $E \in B_1/\sigma$, then such a c may be chosen in B_2.*

Proof outline. If c may already be chosen in B_2 there is nothing to do; otherwise walk down a chain for $B_2 \lhd\lhd\lhd A_2$ and let $B_2'' <_{T(\delta)} B_2'$ be the first step at which the choice fails, so $B_2 \subseteq B_2''$. Put S' and S'' for the sets of σ-class tuples of length $|A_1|$ witnessed by some c in B_2', resp. B_2''. A single good $c \in B_2'$ makes $S' = (B_1/\sigma)^{|A_1|}$, and $S'' \neq \varnothing$ because $R \cap (B_1 \times B_2) \neq \varnothing$ and $B_2 \subseteq B_2''$ — *this is where that hypothesis is spent.* For $T \in \{\mathrm{BA}, \mathrm{C}\}$, Lemmas 7.4 and 7.9 then put a strong subuniverse on B_1/σ, a contradiction; for $T = \mathrm{S}$ the same follows once $R \cap (B_1 \times D_2) \neq \varnothing$, which Lemma 7.39(s) supplies inside $\mathbf{R}$.

For $T = \mathrm{D}$ one builds $S \leq (B_1/\sigma)^{|A_1|} \times B_2'/\delta$ and finds a full column over $d = c/\delta$, while $(B_1/\sigma)^{|A_1|} \times B_2''/\delta \not\subseteq S$ by minimality. Corollary 7.41 then computes $\mathrm{pr}_{|A_1|+1}(S)$. It offers three alternatives and only the third gives the contradiction. “Empty” is excluded by $R \cap (B_1 \times B_2) \neq \varnothing$. “Size one” is excluded thus: the single δ-class would have to be $[c]_\delta$, since c has an R-partner in B_1; and $B_2'' = B_2' \cap E$ for a δ-block E, so either $E = [c]_\delta$, putting c in B_2'' against the minimal choice of B_2', or $E \cap [c]_\delta = \varnothing$, making $R \cap (B_1 \times B_2'') = \varnothing$ against $B_2 \subseteq B_2''$. So S is a central relation, and Lemmas 7.20 and 7.9 contradict the hypotheses. ☐

Lemma 7.56. *Under the hypotheses of Lemma 7.55, if* $(\mathrm{LeftLinked}(R) \cap B_1^2)/\sigma = (B_1/\sigma)^2$ *then* $\mathrm{LeftLinked}(R \cap (B_1 \times B_2))/\sigma = (B_1/\sigma)^2$.

Proof outline. Write $R_n = (R \circ R^{-1})^n$, which is reflexive and symmetric because R is subdirect, so $R_{2n} = R_n \circ R_n^{-1}$ — *this is why the source restricts to* $n = 2^k$. If $(R_1 \cap B_1^2)/\sigma$ is already full, Lemma 7.22 applied to $S = \{(a/\sigma, b) : a \in B_1, (a,b) \in R\}$, subdirect over $(B_1/\sigma) \times \mathrm{pr}_2(S)$ rather than over $(B_1/\sigma) \times A_2$, produces the required c, which Lemma 7.55 moves into B_2; two classes are then linked through c. Otherwise take the largest $n = 2^k$ at which fullness fails, apply Lemma 7.22 to R_n, push c into B_1, and conclude by Lemma 7.20 that $(R_n \cap B_1^2)/\sigma$ is full after all. ☐

Lemma 7.57. Let σ be a dividing congruence for $B \lhd\lhd\lhd A$, let δ be a congruence on $\mathbf{A}$ with $(\delta \cap B^2)/\sigma \neq B^2/\sigma$, and let ω be the intersection of the dividing congruences of the chosen chain for $B \lhd\lhd\lhd A$. Then (1) $\delta \cap B^2 \subseteq \sigma \cap B^2$; (2) $\sigma \supseteq \delta \cap \omega$; (3) $(\delta \vee (\sigma \cap \omega)) \cap B^2 = \sigma \cap B^2$; (4) $(\delta \vee (\sigma \cap \omega)) \cap \omega = \sigma \cap \omega$.

Proof. Throughout we use $B^2 \subseteq \omega$, which holds because each B_i in the chain lies inside a single block of its own dividing congruence, and B is contained in every B_i. *It is used at least four times below.*

(2). The hypothesis gives σ-classes C_1, C_2 meeting B with $\big((C_1 \cap B) \times (C_2 \cap B)\big) \cap \delta = \varnothing$. Then $C_1 \cap B <_{\mathrm{D}(\sigma)} B$, and Lemma 7.37 with $m = k$ and $B_k = C_1 \cap B$ gives $\big((C_1 \cap B) \circ \delta \cap B\big)/\sigma = \{C_1\}$, since the value B/σ is excluded by the choice of C_2; its second clause is $\sigma \supseteq \delta \cap \omega$.

(1) follows, since $\delta \cap B^2 \subseteq \delta \cap \omega \subseteq \sigma$.

(4). Put $\delta' = \delta \vee (\sigma \cap \omega)$, which is $\mathrm{LeftLinked}(\delta \circ (\sigma \cap \omega))$. If $(\delta' \cap B^2)/\sigma$ were full then Lemma 7.56 would make LeftLinked of the restricted relation full, so some single linking step would cross σ-classes: there would be $a_1, a_2, c \in B$ and $b_1, b_2 \in A$ with $(a_i, b_i) \in \delta$, $(b_i, c) \in \sigma \cap \omega$ and $(a_1, a_2) \notin \sigma$. But $a_1, a_2, c \in B$ gives $(a_i, c) \in \omega$, hence $(a_i, b_i) \in \omega$, hence $(a_i, b_i) \in \sigma$ by (2), hence $(a_1, a_2) \in \sigma$ — a contradiction. So $(\delta' \cap B^2)/\sigma \neq (B/\sigma)^2$, and applying (2) to δ' gives $\sigma \supseteq \delta' \cap \omega$; with $\delta' \supseteq \sigma \cap \omega$ this is (4).

(3) follows from (4) by intersecting with $B^2 \subseteq \omega$. ☐

Lemma 7.58. *Let $B \lhd\lhd\lhd A$ and let σ be a congruence on $\mathbf{A}$ with $|B/\sigma| > 1$ and B/σ BA and centre free. Then there is a dividing congruence $\delta \supseteq \sigma$ for $B \leq A$.*

Proof. Choose $\delta \supseteq \sigma$ maximal with $|B/\delta| > 1$; it exists because σ qualifies. Then B/δ is BA and centre free, since the preimage under $B/\sigma \twoheadrightarrow B/\delta$ of a proper nonempty strong subuniverse would be one of B/σ.

It remains to see that δ is irreducible. Suppose $\delta = S_1 \cap \cdots \cap S_k$ with each $S_i \supsetneq \delta$ a subuniverse of $\mathbf{A}^2$ *stable under* δ. Some S_i has $B^2 \not\subseteq S_i$, else $B^2 \subseteq \delta$. Now $\mathrm{LeftLinked}(S_i)$

is a congruence containing $S_i \supsetneq \delta \supseteq \sigma$, so maximality of δ forces $B^2 \subseteq \mathrm{LeftLinked}(S_i)$, i.e. $(\mathrm{LeftLinked}(S_i) \cap B^2)/\delta = (B/\delta)^2$. By Lemma 7.56, applied *with the congruence* δ, $\mathrm{LeftLinked}(S_i \cap B^2)/\delta = (B/\delta)^2$, so $(S_i \cap B^2)/\delta$ is linked. It is also *proper*: stability under δ makes S_i a union of products of δ-blocks, so $(S_i \cap B^2)/\delta = (B/\delta)^2$ would give $B^2 \subseteq S_i$. Corollary 7.2 then produces a proper nonempty strong subuniverse of B/δ, a contradiction. $\square$

Caveat 7.59 (Where each hypothesis is spent)**.** Two things in that proof are easy to lose. The relation must be factored by δ and not by σ — $(S_i \cap B^2)/\sigma$ lives on B/σ and the conclusion is about B/δ, and it is B/δ that was shown BA and centre free. And properness needs the words *stable under* δ in the unfolding of irreducibility: that clause is what makes S_i a union of products of δ-blocks, and it is the only part of Definition 3.23 doing work here. Without it, $S_i \cap B^2 \subsetneq B^2$ survives factoring only by accident.

Lemma 7.60. *Let δ be a congruence on $\mathbf{A}$ and $C <^{A}_{T(\sigma)} B \lhd\lhd\lhd A$ with $T \in \{\textsc{pc}, \textsc{l}\}$. Then $C/\delta = B/\delta$, or $C/\delta <_{\textsc{s}} B/\delta$, or $C/\delta <^{A/\delta}_{T} B/\delta$; and in the last case $\delta \cap B^2 \subseteq \sigma \cap B^2$.*

Proof outline. Put $R = \{(a/\sigma, a/\delta) : a \in B\}$ and note $R \circ R^{-1} = (\delta \cap B^2)/\sigma$, so $\mathrm{LeftLinked}(R) = (B/\sigma)^2$ exactly when $(\delta \cap B^2)/\sigma = B^2/\sigma$. In that case Lemma 7.22 yields $U \leq_{\textsc{ba,c}} B/\delta$ with $(B/\sigma) \times U \subseteq R$, and $U \subseteq C/\delta$, giving the first two alternatives.

Otherwise Lemma 7.57(4) gives $\sigma \cap \omega = \delta' \cap \omega$ for $\delta' = \delta \vee (\sigma \cap \omega)$, and $B/\delta' \cong B/\sigma$ by (3), so B/δ' is BA and centre free and Lemma 7.58 produces a dividing congruence $\delta'' \supseteq \delta'$ for B. Applying Lemma 7.57(2) to δ'' — legitimate because $\sigma \cap B^2 \subseteq \delta''$ and $|B/\delta''| > 1$ give $(\delta'' \cap B^2)/\sigma \neq B^2/\sigma$ — yields $\delta'' \cap \omega = \sigma \cap \omega$, hence $\delta'' \cap B^2 = \sigma \cap B^2$, so the δ''-classes and σ-classes agree on B and $C/\delta <^{A/\delta}_{\mathcal{T}(\delta''/\delta)} B/\delta$. Finally $\mathcal{T} = T$: $\omega \setminus \delta'' \neq \varnothing$, since $\omega \supseteq B^2$ and $|B/\sigma| > 1$, so Lemma 7.53 gives a bridge from δ'' to σ, and Lemma 4.15 in both directions makes the two congruences of the same type. The last clause holds because $\delta \cap B^2 \subseteq \delta' \cap B^2 = \sigma \cap B^2$. $\square$

7.7 The headline properties

Proof of Lemma 6.2. If $\mathbf{B}$ has a proper nonempty binary absorbing or central subuniverse, take it. Otherwise $\mathbf{B}$ is BA and centre free and $|B| > 1$, so Lemma 7.58 with $\sigma = 0_{\mathbf{A}}$ gives a dividing congruence δ for B; take $C = B \cap E$ for any block E *of* δ with $B \cap E \neq B$, which exists because $|B/\delta| > 1$. Its type is $\textsc{l}$ or $\textsc{pc}$ according to δ. $\square$

Proof of Lemma 6.16 and Corollary 6.17. (t). For $T \in \{\textsc{ba}, \textsc{c}\}$ this is Lemma 7.15. For $T = \textsc{s}$, take $E \subseteq C$ with $E <_{\textsc{ba,c}} B$; if $C \cap D = \varnothing$ there is nothing to prove, and otherwise $B \cap D \neq \varnothing$, so Lemma 7.39(s) gives $E \cap D \neq \varnothing$ and Lemma 7.15 gives $E \cap D <_{\textsc{ba,c}} B \cap D$. For $T \in \{\textsc{pc}, \textsc{l}\}$, Lemma 7.39(d) makes $(B \cap D)/\sigma$ empty, a point, or all of B/σ; the three cases give $C \cap D = \varnothing$, $C \cap D \in \{\varnothing, B \cap D\}$, and $C \cap D <_{T(\sigma)} B \cap D$ respectively.

(i). Apply (t) to each step of a chain for $B \lhd\lhd\lhd^{A} A$, obtaining $B \cap D \ \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A} \ D$. The corollary follows by composing with $D \lhd\lhd\lhd A$. $\square$

Caveat 7.61 (The superscript in (i) is needed)**.** Corollary 6.17 is weaker than Lemma 6.16(i), and three later arguments need the stronger form: Corollary 6.11(b) and (m), which conclude $R'' \ \dot{\lhd}\dot{\lhd}\dot{\lhd}^{R} \ R'$, and Lemma 6.22. Stating only the corollary — as is natural, since it is the symmetric-looking form — makes all three unprovable.

Proof of Lemma 6.4. (bt) is proved directly, from the definitions; the reflection corollary of §6 is *derived* from this lemma and so may not be used here. For $T \in \{\textsc{ba}, \textsc{c}\}$ the witnessing term transfers: f is a homomorphism, so $f(t(\bar{a})) = t(f(\bar{a}))$, and a tuple with all but one entry in $f^{-1}(C')$ and the remaining entry in $f^{-1}(B')$ maps to one with all but one entry in C' and the remaining entry in B'; the centrality clause pulls back the same way, since $f(\mathrm{Sg}(X)) = \mathrm{Sg}(f(X))$ for surjective f. For $T = \textsc{s}$ the witness $D' \subseteq C'$ pulls back to $f^{-1}(D') \subseteq f^{-1}(C')$ and the previous case applies. For $T \in \{\textsc{d}, \textsc{l}, \textsc{pc}\}$, $C' = B' \cap E$ for a block E of σ, so $f^{-1}(C') = f^{-1}(B') \cap f^{-1}(E)$ with $f^{-1}(E)$ a block of $f^{-1}(\sigma)$; condition (i) of Definition 4.22 transfers because $f^{-1}(\sigma^*) = f^{-1}(\sigma)^*$ by Lemma 3.27, and condition (iii) because $f^{-1}(B')/f^{-1}(\sigma) \cong B'/\sigma$.

(b) follows by iterating (bt) along a chain, and (bm) from (bt) by intersecting. □

Proof of Lemma 6.5. The backward clauses are Lemma 6.4.

(fs) is Lemma 7.7. (ft) splits by type: for $T \in \{\textsc{ba}, \textsc{c}, \textsc{s}\}$ it follows from (fs); for $T \in \{\textsc{pc}, \textsc{l}\}$ it is Lemma 7.60; and for $T = \textsc{d}$, which that lemma does not cover, split the witnessing congruence into its linear and PC cases and apply the lemma to each, the two conclusions being the same three alternatives. Then (f) follows by iteration, all three outcomes of (ft) staying inside the type list of Definition 4.25.

(fm). Let $C = C_1 \cap \cdots \cap C_t$ with $C_i <^{A}_{T(\sigma_i)} B$ and induct on t. Write δ for the kernel of f, so that $f(X)$ may be read as X/δ throughout. For $t = 1$ this is (ft), and *the hypothesis that $f(B)$ is* S*-free is exactly what kills its middle alternative*, which $\mathcal{M}T$ cannot absorb. If $C_i/\delta = B/\delta$ for some i, say $i = t$, put $D_j = C_j \cap C_t$; by Lemma 6.16(t) $D_j \leq^{A}_{T(\sigma_j)} C_t$, and the inductive hypothesis applied to $C \leq_{\mathcal{M}T} C_t$ gives the claim. Otherwise every $C_i/\delta <^{A/\delta}_{T} B/\delta$, and $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ for every i by the last clause of Lemma 7.60. That is what makes the images meet correctly: $(C_i \circ \delta) \cap B = C_i$, so a δ-class E lying in every C_i/δ has $E \cap B \subseteq C_1 \cap \cdots \cap C_t$, whence

$$(C_1 \cap \cdots \cap C_t)/\delta = C_1/\delta \cap \cdots \cap C_t/\delta,$$

the inclusion $\subseteq$ being trivial. This is the one place where the failure of images to commute with intersections (§2.1, item viii) is repaired, and the saturation hypothesis $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ is precisely what buys it. □

Theorem 7.62 (Stable intersection, restated)**.** Let $C_i <^{A}_{T_i(\sigma_i)} B_i \lhd\!\lhd\!\lhd A$ with $T_i \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$ for $i \in [n]$, $n \geq 2$, and suppose $\bigcap_i C_i = \varnothing$ while $B_j \cap \bigcap_{i \neq j} C_i \neq \varnothing$ for every j. Then all T_i are BA; or all are L and for all k, ℓ there is a bridge δ from σ_k to σ_ℓ with $\widetilde{\delta} = \sigma_k \circ \sigma_\ell$; or $n = 2$ and $T_1 = T_2 = \textsc{c}$; or $n = 2$, $T_1 = T_2 = \textsc{pc}$ and $\sigma_1 = \sigma_2$.

Proof of Theorem 6.18. For $n = 2$ combine Lemmas 7.54 and 4.15. For general n put $B_2' = B_2 \cap C_3 \cap \cdots \cap C_n$ and $C_2' = C_2 \cap B_2'$; then $C_2' <_{T_2(\sigma_2)} B_2' \lhd\!\lhd\!\lhd A$ by Lemma 6.16, properly because $C_2' = B_2'$ would make $C_1 \cap B_2'$ empty. The three hypotheses of the pair case hold, so $T_1 = T_2$ and, for the dividing types, the bridge exists.

If $T_1 = \textsc{pc}$ then $\sigma_1 \circ \sigma_2 = \sigma_1 = \sigma_2$. Indeed, composing the bridge with its transpose gives a bridge from σ_1 to itself, and if $\sigma_1 \circ \sigma_2 \supsetneq \sigma_1$ its collapse properly contains σ_1, making σ_1 linear by Lemma 4.13. That leaves the possibility of proper nesting, $\sigma_1 \circ \sigma_2 = \sigma_1$ with $\sigma_1 \neq \sigma_2$, which is excluded by Lemma 4.17: a bridge between PC congruences makes $\mathbf{A}/\sigma_1 \cong \mathbf{A}/\sigma_2$ with the induced relation on blocks bijective, and two congruences on one finite algebra with equal quotient sizes and $\sigma_2 \subseteq \sigma_1$ are equal. With $\sigma_1 = \sigma_2 =: \sigma$, the blocks E_1, E_2 defining C_1, C_2 are *distinct* — equal blocks would put the point of $B_1 \cap C_2'$ in $C_1 \cap C_2'$ — so $C_1 \cap C_2 = \varnothing$; applying this to every pair and then the hypothesis at $j = 3$ forces $n = 2$.

For $T_1 = \textsc{c}$ the argument just given has no analogue — there is no congruence and no pair of blocks — and $n = 2$ is instead Lemma 7.25, whose two hypotheses are exactly (ii) and (iii) here. □

Caveat 7.63 (What the type-C case costs)**.** The type-C argument is the one place in this section that reaches outside the material developed here, to [15, Lem. 6.11]: *for $n \geq 3$ and*

E_i central in $\mathbf{D}$, there is no $(E_1,\dots,E_n)$-essential relation $R \le D^n$. Two notes for anyone checking it. Its printed proof cites *itself* three times where it means the preceding lemma, so a self-contained rendering must supply both. And its ingredients are exactly the Barto–Kazda criterion (Lemma 7.6) and "central implies ternary absorbing" (Lemma 4.4), both of which are already available above.

Nonemptiness and properness of each E_i are established before the essential-relation theorem is applied, and both are needed: essentiality is a statement about proper nonempty subsets.

Proof of Corollary 6.19. Pull back along the projections $f_i : \mathbf{R} \to \mathbf{A}_i$ using Lemma 6.4(b),(bt), apply Theorem 6.18 inside $\mathbf{R}$, and translate $\widetilde{\delta} = \sigma'_k \circ \sigma'_\ell$ back as $\sigma_k \circ \mathrm{pr}_{k,\ell}(R) \circ \sigma_\ell$. □

Proof of Lemma 6.22. With $C = \bigcap_{i\le n} C_i$ and $D_j = \bigcap_{i\le j} C_i$, Lemma 6.16(t) gives $D_{j+1} \le_T^A D_j$, using $D_j \lhd\lhd\lhd A$ from part (i); collapsing the equalities gives the chain. □

Proof of Lemma 6.26. By Lemma 6.22 fix a chain $C_2 = F_s <_T^{A_2} \dots <_T^{A_2} F_0 = B_2$. If $s = 0$ then $C_2 = B_2$ and the second part of (i) is the conclusion. Each truncation of the chain witnesses $F_j \lhd\lhd\lhd^{A_2} B_2$, and concatenating it with the chain for $B_2 \lhd\lhd\lhd A_2$ gives $F_j \lhd\lhd\lhd A_2$ for every j.

Suppose for contradiction that $W \cap (C_1 \times C_2) = \varnothing$. Along the chain, $W \cap (C_1 \times F_0) \ne \varnothing$ by (i) and $W \cap (C_1 \times F_s) = \varnothing$, so there is a least j with $W \cap (C_1 \times F_j) = \varnothing$; fix it, so that $W \cap (C_1 \times F_{j-1}) \ne \varnothing$. Write the step as $F_j <_{T(\xi)}^{A_2} F_{j-1}$ and let T' be the class of ξ — linear or PC, exhaustively, by Definition 4.10 — so that $F_j <_{T'(\xi)}^{A_2} F_{j-1}$; for $T \in \{\textsc{l}, \textsc{pc}\}$ the class is $T' = T$, and for $T = \textsc{d}$ it is whichever the step's congruence has.

Apply Corollary 6.19 to W with the two-member family

$$C_1 <_{T_1(\sigma_1)}^{A_1} B_1 \lhd\lhd\lhd A_1, \qquad F_j <_{T'(\xi)}^{A_2} F_{j-1} \lhd\lhd\lhd A_2.$$

Its hypotheses hold: W is subdirect; the joint intersection $W \cap (C_1 \times F_j)$ is empty by the choice of j; and both leave-one-out configurations are nonempty — $W \cap (B_1 \times F_j) \supseteq W \cap (B_1 \times C_2) \ne \varnothing$ by (i), since $C_2 = F_s \subseteq F_j$, and $W \cap (C_1 \times F_{j-1}) \ne \varnothing$ by the minimality of j. So one of the four alternatives holds of the pair $\{T_1, T'\}$. None can: $T' \in \{\textsc{l}, \textsc{pc}\}$, so (ba) and (c) fail; $\textsc{s}$ occurs in no alternative; and (l), (pc) require $T_1 = T' \in \{\textsc{l}, \textsc{pc}\}$, which (ii) excludes — for $T_1 = \textsc{l}$ it puts $T = \textsc{pc}$, making every step of the chain of class $\textsc{pc}$, and symmetrically for $T_1 = \textsc{pc}$. The contradiction refutes the assumption, so $W \cap (C_1 \times C_2) \ne \varnothing$. □

Proof of Lemma 6.24. Write $\Lambda_1 = \mathrm{LeftLinked}(R)$ and $\Lambda = \mathrm{RightLinked}(R)$ — congruences on $\mathbf{A}_1$ and $\mathbf{A}_2$ by Lemma 3.12(a), the projections being full by subdirectness. Fix $(c_1, c_2) \in S \cap (C_1 \times C_2)$ and, by (i), some $w \in B_2$ with $(c_1, w) \in R$. Let E be the σ-block with $C_2 = B_2 \cap E$ (Definition 4.22(ii)); since $c_2 \in C_2$, $E = [c_2]_\sigma$.

Step 1: $(w, c_2) \in \Lambda$. First a transport observation: *if $(x, x') \in \Lambda_1$ and $(x, y), (x', y') \in R$, then $(y, y') \in \Lambda$.* Indeed Λ_1 is the transitive closure of $R \circ R^{-1}$ (Definition 3.11), so there are $x = u_0, \dots, u_n = x'$ and $v_0, \dots, v_{n-1}$ with $(u_i, v_i), (u_{i+1}, v_i) \in R$; consecutive members of the list $y, v_0, \dots, v_{n-1}, y'$ share a left neighbour — u_0, then $u_1, \dots, u_{n-1}$, then u_n — so all are Λ-related. (For $n = 0$ the list is y, y', sharing $x = x'$.) Now put

$$T = \{(x, y) : \text{there is } (x', y') \in R \text{ with } (x, x') \in \Lambda_1 \text{ and } (y', y) \in \Lambda\}.$$

$T \supseteq R$, both linking equivalences being reflexive on the full domains. T is rectangular: let $(a, c), (a, d), (b, c) \in T$, with witnessing edges $(a_1, c^*), (a_2, d^*), (b_1, c^{**}) \in R$ respectively. Then $(a_1, a_2) \in \Lambda_1$, so the transport observation gives $(c^*, d^*) \in \Lambda$, whence $c^{**} \,\Lambda\, c \,\Lambda\, c^* \,\Lambda\, d^* \,\Lambda\, d$, and the edge (b_1, c^{**}) witnesses $(b, d) \in T$. Since the rectangular closure is the least rectangular relation containing R (Definition 3.18, Lemma 3.19(b)), $S \subseteq T$. In particular $(c_1, c_2) \in T$:

there is $(x', y') \in R$ with $(c_1, x') \in \Lambda_1$ and $(y', c_2) \in \Lambda$. The transport observation, applied to (c_1, x') with the edges (c_1, w) and (x', y'), gives $(w, y') \in \Lambda$, hence $(w, c_2) \in \Lambda$.

Step 2: the dichotomy. Suppose first $(\Lambda \cap B_2^2)/\sigma \neq (B_2/\sigma)^2$. The congruence σ is a dividing congruence for $B_2 \leq A_2$ — it witnesses the step $C_2 <_{\mathrm{D}(\sigma)} B_2$ — so Lemma 7.57(1), applied on $\mathbf{A}_2$ to B_2 with its chosen chain and the congruence Λ, gives $\Lambda \cap B_2^2 \subseteq \sigma$. By Step 1, $(w, c_2) \in \Lambda \cap B_2^2 \subseteq \sigma$, so $w \in B_2 \cap E = C_2$, and $(c_1, w) \in R \cap (C_1 \times C_2)$.

Suppose instead $(\Lambda \cap B_2^2)/\sigma = (B_2/\sigma)^2$. Apply Lemma 7.56 to $R^{-1} \leq_{\mathrm{sd}} \mathbf{A}_2 \times \mathbf{A}_1$ with first factor $B_2 \lhd\lhd\lhd A_2$, second factor $C_1 \lhd\lhd\lhd A_1$, and the congruence σ. The hypotheses of Lemma 7.55 hold: $\mathbf{B}_2/\sigma$ is BA and centre free by Definition 4.22(iii), and $R^{-1} \cap (B_2 \times C_1) \ni (w, c_1)$. Moreover $\mathrm{LeftLinked}(R^{-1}) = \Lambda$, so the fullness hypothesis is the present case. The conclusion, $\mathrm{LeftLinked}(R^{-1} \cap (B_2 \times C_1))/\sigma = (B_2/\sigma)^2$, attained at the pair (E, E), produces $u \in \mathrm{pr}_1(R^{-1} \cap (B_2 \times C_1))$ with $u \in E$; then $u \in B_2 \cap E = C_2$, and any R-partner of u in C_1 puts an edge in $R \cap (C_1 \times C_2)$. □

Proof of Lemma 4.18. By Lemma 4.13 there is a bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$, reflexive by Lemma 3.36; replace it by δ°, symmetric and pair-reflexive. Lemma 7.47(c) applies, and $\sigma^* = A^2$ is a congruence with the single block A, so $A/\sigma \cong \mathbb{Z}_p^m$ as a group. Now $m = 0$ gives $\sigma = A^2$, impossible for an irreducible congruence (§2.1, item iv). And $m \geq 2$ contradicts irreducibility: A/σ is affine, so the basic operation acts as $\sum_i c_i x_i$ with $\sum_i c_i = 1$, and the coordinate congruences $\theta_i = \{(x, y) : x_i = y_i\}$ are linear subspaces of $(\mathbb{Z}_p^m)^2$, hence subuniverses of $(\mathbf{A}/\sigma)^2$, each properly containing $0_{\mathbf{A}/\sigma}$, with $\bigcap_i \theta_i = 0_{\mathbf{A}/\sigma}$ — so $0_{\mathbf{A}/\sigma}$ is reducible, and by Lemma 3.27 so is σ. Hence $m = 1$, and the group isomorphism is an isomorphism of algebras by Proposition 2.7. □

Lemma 7.64. *Let $C_1 <^A_{\mathcal{M}T} B_1 \lhd\lhd\lhd A$ with $T \in \{\mathrm{PC}, \mathrm{L}, \mathrm{D}\}$, let $B_2 \lhd\lhd\lhd A$, $C_1 \cap B_2 = \varnothing$ and $B_1 \cap B_2 \neq \varnothing$. Then $(C_1 \circ (\omega_1 \cap \cdots \cap \omega_s)) \cap B_2 = \varnothing$, where $\omega_1, \ldots, \omega_s$ are all congruences of type T on $\mathbf{A}$ with $\omega_i^* \supseteq B_1^2$.*

Proof of Case 1, and of Case 2 from Claim 7.65. The induction is downward on $|B_1|$: the inductive hypothesis is the lemma for every larger first argument, and it starts at $B_1 = A$.

By Definition 4.24 write $C_1 = \bigcap_{i \in [t]} C_1^i$ with $t \geq 1$ and $C_1^i <^A_{T(\sigma_i)} B_1$ — of type T, not merely of type D, which is what makes each σ_i a congruence of type T with $\sigma_i^* \supseteq B_1^2$ and hence one of the ω_j. Put $K_i = C_1^i \circ \sigma_i$ and $\Omega = \omega_1 \cap \cdots \cap \omega_s$. Definition 4.22(ii) gives $C_1^i = K_i \cap B_1$, and Corollary 6.10(e),(f) gives $K_i <^A_{T(\sigma_i)} B_1 \circ \sigma_i \lhd\lhd\lhd A$, so in particular $K_i \lhd\lhd\lhd A$. Two inclusions are used, and both are immediate:

$$C_1 \circ \Omega \subseteq \bigcap_{i \in [t]} K_i, \qquad \text{and} \qquad X \subseteq Y \implies X \circ \Omega \subseteq Y \circ \Omega. \tag{7.8}$$

The first holds because $C_1 \subseteq C_1^i$ and $\Omega \subseteq \sigma_i$ for each i, the latter because σ_i is one of the ω_j.

Put $D = \bigcap_{i \in [t]} K_i \cap B_2$, and note $D \cap B_1 = \bigcap_i (K_i \cap B_1) \cap B_2 = C_1 \cap B_2 = \varnothing$.

Case 1: $D = \varnothing$. Then by (7.8) $(C_1 \circ \Omega) \cap B_2 \subseteq \bigcap_i K_i \cap B_2 = \varnothing$, which is the conclusion. This is the whole of the case $B_1 = A$, since there $D = D \cap B_1 = \varnothing$.

Case 2: $D \neq \varnothing$. Each $K_i \lhd\lhd\lhd A$ and $B_2 \lhd\lhd\lhd A$, so Corollary 6.17, applied t times, gives $D \mathrel{\dot{\lhd}\dot{\lhd}\dot{\lhd}} A$; as $D \neq \varnothing$, $D \lhd\lhd\lhd A$. Fix the chain $A = X_0 \supsetneq X_1 \supsetneq \cdots \supsetneq X_r = B_1$ witnessing $B_1 \lhd\lhd\lhd A$. Then $D \cap X_0 = D \neq \varnothing$ and $D \cap X_r = D \cap B_1 = \varnothing$, so there is a least $p \in [r]$ with $D \cap X_p = \varnothing$. Write $B_1' = X_p$ and $B_1'' = X_{p-1}$, so that

$$B_1 \subseteq B_1' <^A B_1'' \lhd\lhd\lhd A, \qquad D \cap B_1' = \varnothing, \qquad D \cap B_1'' \neq \varnothing. \tag{7.9}$$

Claim 7.65. *That step is of multi-type T: $B_1' <^A_{\mathcal{M}T} B_1''$.*

Granting the claim, the inductive hypothesis applies to $B_1' <^A_{\mathcal{M}T} B_1'' \lhd\lhd\lhd A$ and $D \lhd\lhd\lhd A$, whose hypotheses $B_1' \cap D = \varnothing$ and $B_1'' \cap D \neq \varnothing$ are (7.9), and which sits at a strictly larger first argument since $B_1 \subseteq B_1' \subsetneq B_1''$. It returns $(B_1' \circ \Omega'') \cap D = \varnothing$, where Ω'' is the intersection of all type-T congruences ω with $\omega^* \supseteq (B_1'')^2$. Every such ω has $\omega^* \supseteq B_1^2$, so that family is a *subfamily* of $\{\omega_1, \ldots, \omega_s\}$ and $\Omega'' \supseteq \Omega$; hence $(B_1' \circ \Omega) \cap D = \varnothing$ as well. Finally, by (7.8) and $C_1 \subseteq B_1 \subseteq B_1'$,

$$(C_1 \circ \Omega) \cap B_2 \;\subseteq\; (B_1' \circ \Omega) \cap \bigcap_{i \in [t]} K_i \cap B_2 \;=\; (B_1' \circ \Omega) \cap D \;=\; \varnothing. \qquad \square$$

Caveat 7.66 (Claim 7.65 is not proved, and it is the whole difficulty)**.** Everything above is a proof except the claim, and the claim carries two separate obligations. Neither is discharged here, so Lemma 7.64 is *not* available; see §11.

(a) The step is not of type BA, C *or* S. For S this is Lemma 7.39(s): B_1' then contains some $G <_{\text{BA,C}} B_1''$, and (s) applied to B_1'' and D — which meet, by (7.9) — gives $G \cap D \neq \varnothing$, contradicting $B_1' \cap D = \varnothing$. For BA and C separately the same route is *not* available, because $<_{\text{BA,C}}$ in (s) and in Lemma 7.24 is a conjunction (Caveat 7.30) and a single absorbing step supplies only one of the two types. Lemma 6.16(t) gives $B_1' \cap D \dot{\leq}_{\text{BA}} B_1'' \cap D$, whose dot is exactly the possibility to be excluded.

*(b) When $T \in \{\text{L}, \text{PC}\}$, a type-*D *step is not enough.* The inductive hypothesis needs $B_1' <_{\mathcal{M}T} B_1''$ for the *same* T. A D step carries an irreducible congruence, which is linear or PC by Caveat 4.14, but nothing so far forces it to match T. This obligation is invisible while the components are written as D, which is why the typing in the first paragraph of the proof matters.

Obligation (b) is the more serious: it is not a suppressed step but a missing hypothesis in the induction, and closing it plausibly means strengthening the inductive statement to carry the types of the intervening steps rather than patching the claim.

Proof of Lemma 6.30. Write $\bar{X}$ for X/σ and $\bar{\mathbf{A}} = \mathbf{A}/\sigma$. Note first that for any $X \subseteq A$, $(X \circ \sigma) \cap B_2 = \varnothing$ says exactly that no σ-class meets both X and B_2, i.e. that $\bar{X} \cap \bar{B}_2 = \varnothing$. So the hypotheses read $\bar{C}_1 \cap \bar{B}_2 = \varnothing$ and, by i, $\bar{P} \cap \bar{B}_2 = \varnothing$; and $\bar{B}_1 \cap \bar{B}_2 \neq \varnothing$ because $B_1 \cap B_2 \neq \varnothing$.

Step 1: images commute with intersection on F. For $j \in F$ the factor $\bar{C}_1^j$ lands in the type-T alternative of Lemma 6.5(ft), so the "moreover" clause of Lemma 7.60 gives $\sigma \cap B_1^2 \subseteq \sigma_j \cap B_1^2$. Hence

$$\bigcap_{j \in F} \bar{C}_1^j \;=\; \bar{P}.$$

Indeed $\subseteq$ is the only direction needing an argument. Let $[a] \in \bigcap_{j \in F} \bar{C}_1^j$; since each $\bar{C}_1^j \subseteq \bar{B}_1$ we may pick $a \in B_1$. For each $j \in F$ there is $a_j \in C_1^j$ with $(a, a_j) \in \sigma$; as $a, a_j \in B_1$ this puts $(a, a_j) \in \sigma \cap B_1^2 \subseteq \sigma_j$, so a lies in the same σ_j-block as a_j, and $C_1^j = B_1 \cap E_j$ for that block gives $a \in C_1^j$. So $a \in P$.

This is what §2.1, item viii warns about, and it is exactly where the warning does not bite: among the type-T factors, and only among them, the saturation identity holds and the image of the intersection is the intersection of the images.

Step 2: a proper multi-type reduction modulo σ. Each $\bar{C}_1^j <^{\bar{A}}_T \bar{B}_1$ for $j \in F$, and $\bar{P}$ is nonempty because $P \supseteq C_1 \neq \varnothing$. So by Step 1

$$\bar{P} \leq^{\bar{A}}_{\mathcal{M}T} \bar{B}_1,$$

and the containment is *proper*, since $\bar{P} \cap \bar{B}_2 = \varnothing$ while $\bar{B}_1 \cap \bar{B}_2 \neq \varnothing$. Also $\bar{B}_1 \lhd\lhd\lhd \bar{A}$ and $\bar{B}_2 \lhd\lhd\lhd \bar{A}$ by Corollary 6.8(f).

Step 3: multiply by all type-T congruences. Apply Lemma 7.64 inside $\bar{\mathbf{A}}$ to $\bar{P} <^{\bar{A}}_{\mathcal{M}T} \bar{B}_1 \lhd\lhd\lhd \bar{A}$ and $\bar{B}_2 \lhd\lhd\lhd \bar{A}$: it returns the congruences $\delta_1, \ldots, \delta_s$ of type T on $\bar{\mathbf{A}}$ with $\delta_i^* \supseteq \bar{B}_1^2$, and $\left(\bar{P} \circ \bigcap_i \delta_i\right) \cap \bar{B}_2 = \varnothing$. Note the ambient here is still $\bar{B}_1$, which is what the conclusion's condition $\omega_i^* \supseteq B_1^2$ requires.

Step 4: back to $\mathbf{A}$. Let ω_i be the preimage of δ_i under $\mathbf{A} \twoheadrightarrow \bar{\mathbf{A}}$, so $\omega_i \supseteq \sigma$. By Lemma 3.27 the correspondence $\omega \mapsto \omega/\sigma$ preserves irreducibility, linearity and PC-ness and carries covers to covers, so each ω_i is of type T with $\omega_i^* \supseteq B_1^2$. Pulling Step 3 back, $\left(P \circ \bigcap_i \omega_i\right) \cap B_2 = \varnothing$, and $C_1 \subseteq P$ gives $\left(C_1 \circ \bigcap_i \omega_i\right) \cap B_2 = \varnothing$.

Finally $\bigcap_i \omega_i \supseteq \sigma$, and it has the property that σ was chosen maximal for; so $\sigma = \bigcap_i \omega_i$. □

Caveat 7.67 (What Hypothesis i costs, and three failed routes)**.** What the other hypotheses give is $\bar{C}_1 \cap \bar{B}_2 = \varnothing$, and $C_1 \subseteq P$, so $\bar{C}_1 \subseteq \bar{P}$: Hypothesis i asks for emptiness of the *larger* set, and is therefore strictly stronger. Three routes to it have been tried and none works.

Drop the non-type-T factors one at a time. If $j_0 \notin F$ and $C_1' = \bigcap_{j \neq j_0} C_1^j$, one would like $\bar{C}_1' \cap \bar{B}_2 = \varnothing$ and then induct. But recovering that from $\bar{C}_1 \cap \bar{B}_2 = \varnothing$ needs $\bar{C}_1 = \bar{C}_1' \cap \bar{C}_1^{j_0}$, which is the image-versus-intersection identity again — available for type-T factors by Step 1 of the proof above and *not* for the others, since the saturation clause of Lemma 7.60 is exactly what the $<_{\mathrm{S}}$ alternative withholds.

Pass to a minimal subfamily. Choosing F' minimal with $\bigcap_{j \in F'} \bar{C}_1^j \cap \bar{B}_2 = \varnothing$ would make minimality supply the leave-one-out hypothesis of Theorem 6.18, whose four conclusions have no room for S — so no member of F' would be S-typed. But no such F' need exist: the whole family gives $\left(\bigcap_j C_1^j\right)/\sigma \cap \bar{B}_2 = \varnothing$, whereas $\left(\bigcap_j C_1^j\right)/\sigma \subseteq \bigcap_j \bar{C}_1^j$ with the inclusion generally strict (§2.1, item viii). This is the route an earlier draft of this document took, and it is invalid.

Descend into a strong subuniverse. The $<_{\mathrm{S}}$ factors do at least never help: a factor with $\bar{C}_1^j <_{\mathrm{S}} \bar{B}_1$ contains a subuniverse $\bar{D}$ of $\bar{B}_1$ that is simultaneously binary absorbing and central, and Lemma 7.24 makes $\bar{D}$ meet $\bar{B}_1 \cap \bar{B}_2$, so $\bar{C}_1^j \cap \bar{B}_2 \neq \varnothing$. But the configuration reproduces itself on $\bar{D}$ — $\bar{D} \lhd\lhd\lhd \bar{A}$, $\bar{C}_1 \cap \bar{D} \lhd\lhd\lhd \bar{D}$, $\bar{B}_2 \cap \bar{D} \lhd\lhd\lhd \bar{D}$, still disjoint, with $\bar{D} \cap \bar{B}_2 \neq \varnothing$ — so no iteration of that fact closes anything, and the descent shrinks the ambient while the conclusion's congruence family is indexed by the ambient.

So the hypothesis stays, and the debt is recorded in §11.

A fourth route is open, and unlike the three above it has not failed — it has not been attempted. Lemma 6.30 has exactly one consumer, the typing step of Lemma 8.21, and the older proof of the dichotomy establishes the parallelogram property of a crucial constraint *without any statement of this shape*: by recursion on a measure over reductions, restricting to the minimal linear reduction containing a solution of the weakened instance, with a fit-in lemma for one-of-four chains through a subdirect relation doing the intersection work. No maximal congruence appears. If Lemma 8.21 is proved along those lines, its typing conclusion recovered separately — irreducibility from Lemma 8.14, the type from the bridge machinery — then the sole call site of Lemma 6.30 disappears, and Hypothesis i with it. The transport cost is not small: the fit-in lemma has to be restated in this document's vocabulary, and the recursion needs the reduction machinery of §8. But it converts an open problem into writing, and it is the first route with that property.

8 The main statements

Everything converges here. Theorem 8.1 is the single induction that carries the whole proof; the three theorems the algorithm consumes are corollaries of it.

8.1 The three claims the algorithm needs

The algorithm of §9 is justified by exactly three facts. Informally:

(IC1) every domain of size ≥ 2 has a strong subset, or an equivalence σ with $D_x/\sigma \cong \mathbb{Z}_p$;
(IC2) reducing a variable's domain to a strong subset of it does not destroy solvability, for a consistent enough instance;
(IC3) for a linked, consistent enough, strong-subset-free instance, the set of "linear cosets" containing a solution is empty, full, or a codimension-one affine subspace.

(IC1) is Lemma 6.2 together with Lemma 4.18. (IC2) is Theorem 8.6. (IC3) is Theorem 8.9.

8.2 The main induction

Theorem 8.1 (Main inductive claim)**.** Let $\mathcal{I}$ be an irreducible cycle-consistent instance and let $D^{(1)}$ be a 1-consistent reduction for $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$.

(1) If $\mathcal{I}$ is crucial in $D^{(1)}$ then (1a) holds, and (1b) or (1c) holds:

- (1a) every constraint of $\mathcal{I}$ has the parallelogram property;
- (1b) $\mathcal{I}$ is a connected linear-type instance whose solution set is subdirect;
- (1c) there is an expanded covering $\mathcal{J}$ of $\mathcal{I}$, crucial in $D^{(1)}$, with a linked connected subinstance Υ whose solution set is not subdirect.

(2) If $D^{(2)} \leq_T D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$ with $T \in \{\text{BA}, \text{C}\}$ — and, when $T = \text{BA}$, uniform: $D^{(2)} \leq_{\text{BA}(t)} D^{(1)}$ for a single binary term t (Caveat 5.7) — and $\mathcal{I}^{(1)}$ has a solution, then $\mathcal{I}^{(2)}$ has a solution.

Caveat 8.2 (The induction, stated precisely)**.** "Induction on the size of $D^{(1)}$" hides the entire well-foundedness argument, and it must be made precise, because *the two halves of the conclusion are proved by mutual induction and each invokes the other.*

The intended measure is

$$\mu(D^{(1)}) = \sum_{x \in \mathrm{Var}(\mathcal{I})} |D_x^{(1)}| \in \mathbb{N},$$

and the statement being proved by strong induction on μ is the conjunction of (1) and (2), *universally quantified over* $\mathcal{I}$:

$$P(k): \quad \forall \mathcal{I},\ \forall D^{(1)} \text{ with } \mu(D^{(1)}) \leq k,\ (1) \text{ and } (2) \text{ hold.}$$

The quantifier over $\mathcal{I}$ must be *inside*, because every use of the inductive hypothesis changes the instance: (2) applies it to a weakening $\mathcal{I}'$ of $\mathcal{I}^{(2)}$; Case 1 of (1) applies it to $\mathcal{I}$ with $D^{(2)}$; Case 2 applies it to weakenings and to expanded coverings. An induction that fixed $\mathcal{I}$ and inducted on $D^{(1)}$ alone does not go through.

Three further points.

(a) *The two halves are established in a fixed order at each level.* Part (2) at a given measure uses (1) only at strictly smaller measure, whereas part (1) at that measure uses *(2) at the same measure* — Case 1 below does exactly that. So the scheme is: strong induction on $k = \mu(D^{(1)})$; at each level establish (2) for all instances, then (1) for all instances. That is acyclic, and it is why part (2) is proved first.

(b) *Which appeals decrease.* (2) invokes the hypothesis at $D^{(2)} \subsetneq D^{(1)}$. The containment is strict in the only case that needs an argument: $D^{(2)} \leq_T D^{(1)}$ permits equality, but if $D^{(2)} = D^{(1)}$ then $\mathcal{I}^{(2)} = \mathcal{I}^{(1)}$ and (2) is its own hypothesis. Case 1 of (1) likewise. Case 2 invokes it at $D^{(B,\perp)} \subseteq D^{(B,\top)}$, which is $D^{(1)}$ with the z-domain cut to $B \lneq D_z^{(1)}$, the properness being that of $B <^{D_z}_{T(\sigma)} D_z^{(1)}$. In every case the decrease is at a single variable and by at least one element. The uniform statement is: *if $D' \leq_T D$ with T any type and $D' \neq D$, then $\mu(D') < \mu(D)$.*

(c) *The measure does survive, because no appeal is made at an expanded covering.* This looks like the weak point of the scheme and is not. An expanded covering has *more* variables than $\mathcal{I}$, and the reduction of a covering is the extension of $D^{(1)}$ along the parent map (Proposition 5.20(h)), which repeats the parent domains; its μ is *larger*, and no observation about one parent variable being cut compares the two sums, since arbitrarily many children can outweigh the decrease. So if the induction did appeal at a covering, μ would not serve.
It does not. Every appeal above is to a *weakening* of the current instance — Var can only shrink — paired with a reduction strictly smaller in μ: part (2) at $D^{(2)}$; Case 1 of (1) at $D^{(2)}$ twice; the proof of (1a) at the minimal $\mathcal{MT}$ reduction; and the three appeals in the $\mathcal{B}_0 \neq \varnothing$ branch at $D^{(B,\perp)}$, whose members $\mathcal{I}'$ come from Ω, every member of which is a weakening of $\mathcal{I}$ by condition 1 of Construction 8.5. Expanded coverings enter only as *outputs* — the $\mathcal{J}$ of (1c), and the auxiliary $\perp(\cdot)$, $\Delta(\cdot)$ and Θ built from them — and an output carries no well-foundedness obligation.
What does apply this theorem to a covering is Theorem 8.7, which is a separate statement proved *after* it and consuming it as a black box; the appeal there is an ordinary application of a proved theorem, not a step in this induction. Confusing the two is the natural mistake, and it is what an earlier draft of this caveat made.

Proof of (2). Suppose $\mathcal{I}^{(2)}$ has no solution. Weaken $\mathcal{I}^{(2)}$ to an instance $\mathcal{I}'$ crucial in $D^{(2)}$ (Remark 5.18). By the inductive hypothesis applied to $\mathcal{I}'$ and $D^{(2)}$, $\mathcal{I}'$ satisfies (1a) and (1b) or (1c).

Case (1c). There is an expanded covering $\mathcal{J}$ of $\mathcal{I}'$, crucial in $D^{(2)}$, with a linked connected subinstance Υ. Let x be a variable of a constraint $C \in \Upsilon$. By Lemma 8.17(p), $\mathrm{Con}_1(C,x)$ is a perfect linear congruence; choose $\zeta \subseteq \mathbf{D}_x \times \mathbf{D}_x \times \mathbf{Z}_p$ with $(y_1,y_2,0) \in \zeta \iff (y_1,y_2) \in \mathrm{Con}_1(C,x)$ and $\mathrm{pr}_{1,2}(\zeta) = \mathrm{Con}_1(C,x)^*$.

Replace the variable x of C in $\mathcal{J}$ by a fresh x' and add the constraint $\zeta(x,x',z)$ with z fresh of domain $\mathbf{Z}_p$; call the result Θ, and extend $D^{(1)}, D^{(2)}$ by $D^{(1)}_{x'} = D^{(2)}_{x'} = D_{x'}$. Let E_i be the set of $b \in \mathbf{Z}_p$ such that Θ has a solution in $D^{(i)}$ with $z = b$. By Lemma 6.12, $E_2 \dot{\leq}_T E_1$.

Since $\mathcal{J}$ is crucial in $D^{(2)}$, $\Theta^{(2)}$ has a solution but none with $z = 0$; since $\mathcal{I}^{(1)}$ has a solution, so does $\mathcal{J}^{(1)}$, hence $\Theta^{(1)}$ has one with $z = 0$. Thus $0 \in E_1$, $0 \notin E_2$, and E_1 has at least two elements. As $\mathbf{Z}_p$ has no proper subalgebra of size > 1, $E_1 = \mathbf{Z}_p$; then $E_2 <_T \mathbf{Z}_p$ with $T \in \{\textsc{ba}, \textsc{c}\}$, contradicting Lemma 6.36.

Case (1b). Corollary 8.27 gives a contradiction directly. □

Caveat 8.3 (Three gaps in the case (1c) argument)**.** (a) "E_1 contains at least two different elements, $0 \in E_1$ and $0 \notin E_2$, hence $E_1 = \mathbf{Z}_p$" uses that E_1 is a subuniverse of $\mathbf{Z}_p$ — true because $E_1 = \Theta^{(1)}(z)$ is a relation defined by an instance (Definition 5.5) — *and* that the subuniverses of $\mathbf{Z}_p$ are the singletons and the whole set. The latter needs $w^{\mathbf{Z}_p}$ to generate: from $a \neq b$ one reaches every element. Prove it; it is one line but it is the crux.

(b) $E_2 \dot{\leq}_T E_1$ is applied with the *dotted* conclusion of Lemma 6.12, and the argument then treats E_2 as a proper subuniverse. The case $E_2 = \varnothing$ must be handled: it is excluded because $\Theta^{(2)}$ has a solution. Say so.

(c) The choice of ζ is by Definition 3.40, which supplies ζ with $\mathrm{pr}_{1,2}\,\zeta = \sigma^*$ and $(a_1, a_2, b) \in \zeta \Rightarrow ((a_1, a_2) \in \sigma \iff b = 0)$. The proof uses the *biconditional* $(y_1, y_2, 0) \in \zeta \iff (y_1, y_2) \in \mathrm{Con}_1(C, x)$, which is stronger: it needs, for each $(y_1, y_2) \in \sigma$, that $(y_1, y_2, 0) \in \zeta$. This follows from $\mathrm{pr}_{1,2}\,\zeta = \sigma^* \supseteq \sigma$ plus the implication, but the step is suppressed.

Proof of (1)*, sketch of the structure.* First, $|D^{(1)}_x| > 1$ for some x: otherwise $\mathcal{I}^{(1)}$ has all domains singletons and 1-consistency makes it solvable, contradicting cruciality.

Case 1: some $D^{(1)}_x$ has a nontrivial BA *or central subuniverse.* Lemma 8.20 yields a 1-consistent reduction $D^{(2)} \leq_T D^{(1)}$, $T \in \{\mathrm{BA}, \mathrm{C}\}$. One checks $\mathcal{I}$ is crucial in $D^{(2)}$: for a weakening $\mathcal{J}$ of a constraint, $\mathcal{J}^{(1)}$ has a solution, so by the inductive hypothesis (part (2) at $D^{(2)}$) $\mathcal{J}^{(2)}$ has one. Applying the inductive hypothesis (part (1)) at $D^{(2)}$ gives the conclusion.

Case 2: otherwise. Lemma 6.2 gives $E <^{D_z}_{T(\sigma)} D^{(1)}_z$ for some z and $T \in \{\mathrm{PC}, \mathrm{L}\}$. Property (1a) is obtained by choosing, for each x, a minimal $D^{(2)}_x \leq_{\mathcal{MT}} D^{(1)}_x$ containing the value at x of a solution produced by weakening a fixed constraint C; Lemma 6.22 gives $D^{(2)} \lhd\!\lhd\!\lhd^D D^{(1)}$ and Lemma 8.12 splits into two subcases, in one of which the inductive hypothesis applies and in the other of which Lemma 8.21 applies directly.

For (1b)/(1c) one forms, for each B in the family $\mathcal{B} = \{B : B <^{D_z}_{T(\sigma)} D^{(1)}_z\}$, the reduction $D^{(B,\top)}$ that equals $D^{(1)}$ off z and B at z, and the maximal 1-consistent reduction $D^{(B,\perp)} \subseteq D^{(B,\top)}$ (possibly empty). Lemma 8.19 supplies tree-coverings $\Upsilon_{B,x}$ computing $D^{(B,\perp)}_x$, and their conjunction Υ_x works uniformly in B. Writing $\mathcal{B}_0$ for the B with $D^{(B,\perp)}$ nonempty, the argument splits:

- $\mathcal{B}_0 = \varnothing$: the type must be L, and Lemma 8.25 applied to a suitably weakened tree-covering produces, for any two constraints sharing a variable, a reflexive bridge between their Con_1's. Hence $\mathcal{I}$ is connected, and (1b) or (1c) holds according as its solution set is subdirect.
- $\mathcal{B}_0 \neq \varnothing$: one constructs a family Ω of weakenings of $\mathcal{I}$ that is *critically* unable to solve all of $\mathcal{B}_0$ (conditions 1–4 of Construction 8.5) and argues by whether some member fails (1b).

□

Caveat 8.4 (What Ω's construction turns on)**.** Condition 3 is a minimality condition, and its statement is where the work is. Read as "if we replace any instance in Ω by *all* weaker instances then 2 is not satisfied" it inherits the defect of Caveat 5.16: the replacement need not make progress, and the iteration that is supposed to produce Ω has no measure. Read as in Proposition 8.5 — one member replaced by the finitely many $\mathcal{J}[C \Uparrow]$, and termination measured on the constraints rather than on the solution set — both problems go away at once, and the derivation of condition 4 becomes exact rather than approximate, because what condition 3 then hands over is verbatim Definition 5.17.

Two things about the construction are worth keeping in view.

(a) *The counting argument is used twice, one level apart.* Lemma 5.15 orders *instances* by counting their constraints by ν-value; Proposition 8.5 orders *sets of instances* by counting their members. Both are the multiset extension of a well-founded order, written as a lexicographic order on a product of copies of $\mathbb{N}$ indexed by a finite set, which is the same argument in a form whose well-foundedness is visible without citing Dershowitz–Manna.

(b) *Condition 4's witness depends on $\mathcal{J}$.* The B produced varies with the member, so condition 4 is a $\Pi\Sigma$ statement: "for every $\mathcal{J}$ there is B" may not be strengthened to a single B serving all of Ω, and the consumers in §8 pick a fresh B for each $\mathcal{J}$ accordingly.

(c) *A second descending sequence sits inside the first.* Subcase 1 of the $\mathcal{B}_0 \neq \varnothing$ branch builds $\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \dots$ by repeatedly applying the inductive hypothesis, terminating because $\mathcal{B}_1$ is finite and the sequence is strictly decreasing. That is a different and much easier well-foundedness argument than Proposition 8.5's, and the two are worth keeping apart: nested constructions of this shape are where informal proofs most often hide a circularity, and neither of these does.

Proposition 8.5 (Construction of Ω)**.** Let $\mathcal{I}$ be crucial in $D^{(1)}$ with $\mathcal{B}_0 \neq \varnothing$, and for a weakening $\mathcal{J}$ of $\mathcal{I}$ put $\mathrm{Sol}(\mathcal{J}) = \{B \in \mathcal{B}_0 : \mathcal{J}^{(B,\perp)} \text{ has a solution}\}$. Then there is a finite set Ω of instances with:

1. every member is a weakening of $\mathcal{I}$;
2. $\bigcap_{\mathcal{J}\in\Omega} \mathrm{Sol}(\mathcal{J}) = \varnothing$;
3. for every $\mathcal{J} \in \Omega$, the set $(\Omega \setminus \{\mathcal{J}\}) \cup \{\mathcal{J}[C \Uparrow] : C \in \mathcal{J}\}$ does *not* satisfy 2;
4. for every $\mathcal{J} \in \Omega$ there is $B \in \mathcal{B}_0$ with (a) $\mathcal{J}$ crucial in $D^{(B,\perp)}$ and (b) $B \in \mathrm{Sol}(\mathcal{J}')$ for every $\mathcal{J}' \in \Omega \setminus \{\mathcal{J}\}$.

Proof. Monotonicity. If $\mathcal{J}'$ is weaker than $\mathcal{J}$ then every solution of $\mathcal{J}^{(B,\perp)}$ restricts to one of $\mathcal{J}'^{(B,\perp)}$, so $\mathrm{Sol}(\mathcal{J}) \subseteq \mathrm{Sol}(\mathcal{J}')$.

Conditions 1–3. Start with $\Omega = \{\mathcal{I}\}$. Condition 1 is clear, and condition 2 holds because $D^{(B,\perp)} \subseteq D^{(1)}$ and $\mathcal{I}^{(1)}$ has no solution, so $\mathrm{Sol}(\mathcal{I}) = \varnothing$. While condition 3 fails at some $\mathcal{J} \in \Omega$ — that is, while the displayed set still satisfies 2 — replace Ω by that set. Condition 1 survives, since a single weakening step applied to a weakening of $\mathcal{I}$ is again one; condition 2 survives by the choice of the step.

Termination. Each $\mathcal{J}[C \Uparrow]$ is one weakening step from $\mathcal{J}$, so by Lemma 5.15(c) it is strictly below $\mathcal{J}$ in the order that lemma provides on the set W of weakenings of $\mathcal{I}$; and W is finite by Lemma 5.15(b). Fix a linear order on W extending it, and for a finite $\Omega \subseteq W$ let $c(\Omega) \in \mathbb{N}^W$ count how many members Ω has at each element. One replacement removes a member and inserts members strictly below it, so the coordinate there drops by one, the coordinates above are untouched, and only coordinates below grow. Ordering $\mathbb{N}^W$ lexicographically from the largest element of W downwards — again a lexicographic order on a finite product of copies of $\mathbb{N}$ — every replacement strictly decreases $c(\Omega)$, so the loop halts at an Ω satisfying 1–3.

This is the same counting argument twice over: once inside Lemma 5.15 to order instances by their constraints, and once here to order sets of instances by their members.

Condition 4. Fix $\mathcal{J} \in \Omega$. By condition 3 the displayed set fails 2, so some $B \in \mathcal{B}_0$ lies in all of its members' Sol. Then $B \in \mathrm{Sol}(\mathcal{J}')$ for every $\mathcal{J}' \in \Omega \setminus \{\mathcal{J}\}$, which is 4(b); and $B \notin \mathrm{Sol}(\mathcal{J})$, since otherwise B would lie in $\bigcap_{\mathcal{K}\in\Omega} \mathrm{Sol}(\mathcal{K})$, empty by condition 2. So $\mathcal{J}^{(B,\perp)}$ has no solution while $\mathcal{J}[C \Uparrow]^{(B,\perp)}$ has one for every constraint C of $\mathcal{J}$ — which is Definition 5.17, once no constraint of $\mathcal{J}$ has a dummy variable.

None does. If C had one, its projection C' off that variable is weaker than C with the same violation set, and C' is among the constraints of $\mathcal{J}[C \Uparrow]$; so every solution of $\mathcal{J}[C \Uparrow]$ satisfies C' and hence C, giving $\mathrm{Sol}(\mathcal{J}[C \Uparrow]) = \mathrm{Sol}(\mathcal{J})$ and so $B \in \mathrm{Sol}(\mathcal{J})$. Finally $\mathcal{J}$ has at least one constraint, since a weakening step replaces a constraint by a nonempty family rather than deleting one, and $\mathcal{I}$ is crucial. ☐

8.3 The three consumed theorems

Theorem 8.6 (Reductions are safe)**.** *Let Θ be a cycle-consistent irreducible instance and $B <_T^{D_x} D_x$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{pc}\}$. Then Θ has a solution if and only if Θ has a solution with $x \in B$.*

Proof. For $T = \textsc{pc}$ this is Theorem 8.7. For $T \in \{\textsc{ba}, \textsc{c}\}$, Lemma 8.20 gives a 1-consistent reduction $D^{(1)} \leq_T D$ with $D^{(1)}_x \subseteq B$; by Theorem 8.1(2), $\Theta^{(1)}$ has a solution, hence Θ has one with $x \in B$. □

Theorem 8.7 (PC reductions do not kill all solutions)**.** *If $\mathcal{I}$ is cycle-consistent and irreducible, $B <^{D_y}_{\mathrm{PC}(\sigma)} D_y$ for some $y \in \mathrm{Var}(\mathcal{I})$, and $\mathcal{I}$ has a solution, then $\mathcal{I}$ has a solution with $y \in B$.*

Lemma 8.8 (Subuniverses of these products are cosets)**.** *Let $q_1, \ldots, q_r$ be primes with $\mathbf{Z}_{q_j} \in \mathcal{V}_n$ and let $\varnothing \neq L \leq \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_r}$. Then $L = g + M$ for some $g \in L$ and some subgroup M of the product.*

Proof. By Proposition 2.7, $q_j \mid n-1$ for every j and w acts on each factor as $x_1+\cdots+x_n$; so w acts on the product as the n-fold sum, and $(n-1)h = 0$ for every h in the product. Fix $g \in L$ and put $M = L - g$. If $\bar{y}_1, \ldots, \bar{y}_n \in M$ then $w(\bar{y}_1 + g, \ldots, \bar{y}_n + g) = \sum_j \bar{y}_j + ng = \left(\sum_j \bar{y}_j\right) + g$, which lies in L; so M is closed under the n-fold sum. Now $0 \in M$, and for $y, z \in M$ taking $(y, z, 0, \ldots, 0)$ — legitimate since $n \geq 3$ — gives $y + z \in M$. A finite nonempty subset of a group closed under addition is a subgroup. □

Theorem 8.9 (Codimension one)**.** Suppose

(i) $\mathcal{I}$ is a linked cycle-consistent irreducible instance with $\mathrm{Var}(\mathcal{I}) = \{x_1, \ldots, x_m\}$;
(ii) $\mathbf{D}_{x_i}$ is S-free for every i;
(iii) $\mathcal{I}[C \Uparrow]$ has subdirect solution set for every constraint C of $\mathcal{I}$;
(iv) σ_{x_i} is the intersection of all linear congruences σ on $\mathbf{D}_{x_i}$ with $\sigma^* = D^2_{x_i}$;
(v) $L_{x_i} = \mathbf{D}_{x_i}/\sigma_{x_i}$;
(vi) $\phi\colon \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \to L_{x_1} \times \cdots \times L_{x_m}$ is a homomorphism, q_i prime;
(vii) $\mathcal{I}[C \Uparrow]$ has a solution in $\phi(\bar{a})$ for every constraint C of $\mathcal{I}$ and every $\bar{a} \in \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k}$ — that is, $\mathcal{I}$ is crucial in every $\phi(\bar{a})$ (Definition 5.17).

Then $\Delta = \{\bar{a} : \mathcal{I}$ has a solution in $\phi(\bar{a})\}$ is empty, full, or a coset of the kernel of a surjective homomorphism $\lambda\colon \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \to \mathbf{Z}_p$ — the solution set of the single equation $\lambda(\bar{a}) = d$. Such a λ vanishes on every factor with $q_i \neq p$, so the equation involves only the coordinates over $\mathbb{F}_p$.

Proof. If Δ is full we are done. Otherwise pick $\bar{b} \notin \Delta$ and regard $\phi(\bar{b})$ as a reduction $D^{(1)}$ for $\mathcal{I}$; condition (vii) says $\mathcal{I}$ is crucial in $D^{(1)}$.

Step 1: some $\mathrm{Con}_1(C, x)$ *is a perfect linear congruence.* By Lemma 8.12, either $C^{(1)}_0 = \varnothing$ for some constraint C_0, or $D^{(1)}$ is 1-consistent.

- If $C^{(1)}_0 = \varnothing$: cruciality forces $\mathcal{I} = \{C_0\}$, say $C_0 = R(y_1, \ldots, y_t)$. By Lemma 8.21, R has the parallelogram property and $\mathrm{Con}_1(R, 1)$ is linear with $\mathrm{Con}_1(R, 1)^* = D^2_{y_1}$. By Lemma 4.18, $\mathbf{D}_{y_1}/\delta \cong \mathbf{Z}_p$; with ψ the quotient map, $\zeta = \{(a_1, a_2, b) : \psi(a_1) - \psi(a_2) = b\}$ witnesses perfection.
- If $D^{(1)}$ is 1-consistent: by Theorem 8.1(1) every constraint has the parallelogram property and (1b) or (1c) holds. In either case Lemma 8.17(p) applies — under (1c) to the linked connected subinstance, whose containing a constraint of $\mathcal{I}$ is forced by condition (iii); under (1b) to $\mathcal{I}$ itself.

Step 2: add a $\mathbf{Z}_p$ *coordinate.* Let ζ witness perfection of $\mathrm{Con}_1(C, x)$. Add a variable z with domain $\mathbf{Z}_p$, replace x in C by a fresh x', and add $\zeta(x, x', z)$, giving $\mathcal{I}'$. Put

$$L = \{(\bar{a}, b) : \mathcal{I}' \text{ has a solution with } z = b \text{ in } \phi(\bar{a})\} \leq \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k} \times \mathbf{Z}_p,$$

and write $G = \mathbf{Z}_{q_1} \times \cdots \times \mathbf{Z}_{q_k}$ and $\pi\colon G \times \mathbf{Z}_p \to G$ for the projection. Condition (vii) gives $\pi(L) = G$; in particular $L \neq \varnothing$. And $(\bar{b}, 0) \notin L$, since a solution of $\mathcal{I}'$ with $z = 0$ in $\phi(\bar{b})$ would, by the defining property of ζ, yield a solution of $\mathcal{I}$ in $\phi(\bar{b})$, and $\bar{b} \notin \Delta$.

Step 3: L is a graph. By Lemma 8.8, $L = (\bar{g}, c) + M$ for some $(\bar{g}, c) \in L$ and some subgroup $M \leq G \times \mathbf{Z}_p$; and $\pi(M) = \pi(L) - \bar{g} = G$. The subgroup $M \cap (\{0\} \times \mathbf{Z}_p)$ is a subgroup of $\mathbf{Z}_p$, so it is trivial or the whole of $\mathbf{Z}_p$. If it were the whole of $\mathbf{Z}_p$ then L would be closed under changing its last coordinate arbitrarily, and $\pi(L) = G$ would then give $(\bar{b}, 0) \in L$, which Step 2 excludes. So $M \cap (\{0\} \times \mathbf{Z}_p) = 0$, $\pi|_M$ is an isomorphism $M \to G$, and L is the graph of the affine map

$$f \colon G \to \mathbf{Z}_p, \qquad f(\bar{a}) = c + \lambda(\bar{a} - \bar{g}),$$

where $\lambda \colon G \to \mathbf{Z}_p$ is the homomorphism reading off the last coordinate of $(\pi|_M)^{-1}$.

Step 4: read off Δ. By the defining property of ζ, an $\bar{a}$ lies in Δ exactly when $\mathcal{I}'$ has a solution with $z = 0$ in $\phi(\bar{a})$, that is, exactly when $(\bar{a}, 0) \in L$; so $\Delta = f^{-1}(0)$. If $\lambda = 0$ then f is constant and Δ is empty or full according as $c - \lambda(\bar{g}) \neq 0$ or $= 0$. Otherwise λ is onto $\mathbf{Z}_p$, since its image is a nontrivial subgroup of $\mathbf{Z}_p$, and Δ is the coset of $\ker \lambda$ on which f vanishes — the solution set of one equation $\lambda(\bar{a}) = d$. Finally λ kills every factor $\mathbf{Z}_{q_i}$ with $q_i \neq p$: an element of that factor has order dividing q_i, and its image has order dividing $\gcd(q_i, p) = 1$. So the equation involves only the coordinates over $\mathbb{F}_p$. $\square$

Caveat 8.10 (Why Step 3 is not one line)**.** An earlier draft compressed Steps 2–4 into "L has dimension k and is the solution set of one linear equation". Two things make that false as stated.

L is a nonempty subuniverse of an idempotent affine algebra, hence in general an *affine coset* and not a subgroup; Lemma 8.8 is what supplies the translation, and it is where $n \geq 3$ and $q \mid n - 1$ are spent. And a product of prime fields with *different* primes is not one vector space, so "dimension k" and "codimension 1" have no meaning until the primary components are separated. What survives is not a dimension count but the statement that Δ is a coset of the kernel of a surjection onto $\mathbf{Z}_p$; that kernel has index p, which is the only sense in which the codimension is one. The footnote in [14] about "J may contain only variables on the same field" is the same issue surfacing in the algorithm, and (p3) of §9 is organised prime by prime for exactly this reason.

What Step 2 still takes from elsewhere is $\pi(L) = G$, which is condition (vii) read through the construction of $\mathcal{I}'$, and the property of ζ that relates solutions of $\mathcal{I}'$ with $z = 0$ to solutions of $\mathcal{I}$. Both belong to the half of [14] this document has not yet read adversarially; see §11.

A further obligation on Step 1, created by the restored hypotheses of Lemma 8.21 (Caveat 8.22): applying that lemma in the $C_0^{(1)} = \varnothing$ branch requires every block $\phi(\bar{b})_x$ to be S-free, which does not follow from hypothesis (ii) of this theorem — S-freeness of $\mathbf{D}_x$ says nothing about its subalgebras — and is not established here.

8.4 The supporting lemmas

The lemmas invoked above, in dependency order.

Lemma 8.11 ([13, Lem. 6.1])**.** *An expanded covering of a cycle-consistent irreducible instance is cycle-consistent and irreducible.*

Lemma 8.12. *Suppose $\mathcal{I}$ is a 1-consistent instance, $D^{(1)}$ is a 1-consistent reduction for $\mathcal{I}$, every $\mathbf{D}_x^{(1)}$ is S-free, $T \in \{\text{PC}, \text{L}, \text{D}\}$, $D^{(1)} \lhd\lhd\lhd D$, and $D_x^{(2)} \leq_{\mathcal{M}T}^{D_x} D_x^{(1)}$ is a minimal $\mathcal{M}T$ subuniverse of $D_x^{(1)}$ for every x — minimal among the $B \leq_{\mathcal{M}T}^{D_x} D_x^{(1)}$. Then either $C^{(2)} = \varnothing$ for some constraint C, or $\mathcal{I}^{(2)}$ is 1-consistent.*

Proof. If some $C^{(2)} = \varnothing$ we are done; suppose none is. Fix a constraint C and $z \in \mathrm{Var}(C)$; by Definitions 5.3 and 5.10 what must be shown is $\mathrm{pr}_z((R^C)^{(2)}) = D_z^{(2)}$, and the containment $\subseteq$ holds because $(R^C)^{(2)}$ lies in the box $\prod_{y \in \mathrm{Var}(C)} D_y^{(2)}$.

For the reverse, assemble the hypotheses of Corollary 6.11(m1) for the relation R^C: it is subdirect in $\prod_{y\in \mathrm{Var}(C)} \mathbf{D}_y$, by 1-consistency of $\mathcal{I}$; each $D_y^{(1)} \lhd\!\lhd\!\lhd D_y$; each $D_y^{(2)} \leq_{\mathcal{M}T}^{D_y} D_y^{(1)}$; and $\mathrm{pr}_z\big((R^C)^{(1)}\big) = D_z^{(1)}$ — 1-consistency of the reduction — which is S-free by hypothesis. The corollary returns

$$\mathrm{pr}_z\big((R^C)^{(2)}\big) \mathrel{\dot{\leq}}_{\mathcal{M}T}^{D_z} D_z^{(1)}.$$

The left side is nonempty, since $C^{(2)} \neq \varnothing$; so it is an $\mathcal{M}T$ subuniverse of $D_z^{(1)}$ contained in $D_z^{(2)}$, and minimality of $D_z^{(2)}$ forces equality. □

Caveat 8.13 (Where each hypothesis is spent)**.** 1-consistency of $\mathcal{I}$ itself — restored here as for Lemma 8.21, and supplied by cycle-consistency at both call sites — is what makes R^C subdirect over the *full* domains, which is the ambient in which the multi-type data lives; without it Corollary 6.11 has no purchase. The S-freeness of $D_z^{(1)}$ is exactly clause (m1)'s hypothesis, blocking the middle alternative of propagation. And minimality must be *outright* minimality among $\mathcal{M}T$ subuniverses, not minimality containing a given point; the call site in Theorem 8.1 produces the latter, and Caveat 6.21 is what converts it to the former.

Lemma 8.14. *Let $R(x_1,\dots,x_n)$ be a* rectangular *constraint of a 1-consistent instance $\mathcal{I}$, crucial in a reduction $D^{(\top)}$. Then* $\mathrm{Con}_1(R,i)$ *is an irreducible congruence for every $i \in [n]$.*

Proof. Take $i = 1$; the other coordinates are the same. Suppose $\mathrm{Con}_1(R,1) = \omega_1 \cap \omega_2$ with $\mathrm{Con}_1(R,1) \lneq \omega_j \leq \mathbf{D}_{x_1}^2$ for $j = 1,2$, and put

$$R_j(x_1,\dots,x_n) = \exists y\ \big(R(y,x_2,\dots,x_n) \wedge \omega_j(y,x_1)\big).$$

Each R_j is a constraint on the same scope with $R_j \supseteq R$, and the containment is proper because $\omega_j \supsetneq \mathrm{Con}_1(R,1)$; so each R_j is weaker than R and is among the constraints of $\mathcal{I}[R \Uparrow]$. And $R_1 \cap R_2 = R$: a tuple in both has, for each j, a witness y_j with $R(y_j,x_2,\dots,x_n)$ and $(y_j,x_1) \in \omega_j$, and rectangularity of the first variable makes any two such witnesses $\mathrm{Con}_1(R,1)$-related to each other, so $(y_1,x_1) \in \omega_1 \cap \omega_2 = \mathrm{Con}_1(R,1)$ and x_1 may replace y_1.

Hence every solution of $\mathcal{I}[R \Uparrow]^{(\top)}$ satisfies both R_1 and R_2, so it satisfies R and solves $\mathcal{I}^{(\top)}$ — which has no solution. That contradicts cruciality. □

Caveat 8.15 (Rectangularity is a hypothesis here, not a conclusion)**.** An earlier draft of this lemma read "a constraint crucial in a reduction of a cycle-consistent irreducible instance is rectangular, and its Con_1's are irreducible". That runs two statements together and gets the second one's logic backwards: rectangularity is what the argument *uses*, to know that two witnesses for the same tuple are $\mathrm{Con}_1(R,1)$-related. Rectangularity of crucial constraints is a separate fact, and it comes from the parallelogram property (Lemma 8.21), not from cruciality alone.

The proof is also the reason Definition 5.17 quantifies over $\mathcal{I}[R \Uparrow]$ rather than over single weaker constraints: what it contradicts is that R_1 and R_2 can be imposed *together*, and each of them alone is a weakening that does gain a solution.

Lemma 8.16 (A bridge from a relation)**.** *Let $R \leq_{\mathrm{sd}} \mathbf{A}_1 \times \dots \times \mathbf{A}_n$ have its first and last variables rectangular, and suppose there are $(b_1,a_2,\dots,a_n) \in R$ and $(a_1,\dots,a_{n-1},b_n) \in R$ with $(a_1,a_2,\dots,a_n) \notin R$. Then*

$$\delta(x_1,x_2,y_1,y_2) = \exists z_2 \dots \exists z_{n-1}\ R(x_1,z_2,\dots,z_{n-1},y_1) \wedge R(x_2,z_2,\dots,z_{n-1},y_2)$$

is a bridge from $\mathrm{Con}_1(R,1)$ *to* $\mathrm{Con}_1(R,n)$ *with $\widetilde{\delta} = \mathrm{pr}_{1,n}(R)$.*

Proof. Clauses (a) and (b) of Definition 3.29 are immediate from the definition of Con_1. For clause (d), rectangularity of the first and last variables says exactly that $(x_1, x_2) \in \mathrm{Con}_1(R, 1)$ if and only if $(y_1, y_2) \in \mathrm{Con}_1(R, n)$ for tuples sharing the middle block $z_2, \ldots, z_{n-1}$. Clause (c) is witnessed by $(b_1, a_1, a_n, b_n) \in \delta$ — take $z_i = a_i$, so that the two conjuncts are the two tuples of the hypothesis — together with $(b_1, a_1) \notin \mathrm{Con}_1(R, 1)$, which is what $(a_1, \ldots, a_n) \notin R$ says. Finally $\widetilde{\delta}$ is $\mathrm{pr}_{1,n}(R)$, since $(x, x, y, y) \in \delta$ asks for a single tuple of R with first entry x and last entry y. □

Lemma 8.17 (Connected instances)**.** *Let $\mathcal{I}$ be a cycle-consistent connected instance. Then*

(a) *any two constraints with a common variable are adjacent;*
(b) *for constraints C_1, C_2, variables $x_i \in \mathrm{Var}(C_i)$, and any path from x_1 to x_2, there is a bridge δ from $\mathrm{Con}_1(C_1, x_1)$ to $\mathrm{Con}_1(C_2, x_2)$ with $\widetilde{\delta}$ containing all pairs connected by that path;*
(p) *if $\mathcal{I}$ is linked then $\mathrm{Con}_1(C, x)$ is a perfect linear congruence for every $C \in \mathcal{I}$, $x \in \mathrm{Var}(C)$.*

Caveat 8.18. Clause (p) is the bridge between the bridge machinery and the linear algebra; it is what makes Theorem 8.9 possible. Its proof is (b) followed by Lemma 6.34: linkedness of the instance makes $\widetilde{\delta}$ linked as a relation, which upgrades the bridge. The two senses of "linked" (Caveat 3.15) meet exactly here, and the step where one becomes the other should be isolated.

Lemma 8.19 ([15, Lem. 5.6])**.** *For a 1-consistent instance and a reduction, there is a tree-covering whose projection onto a given variable computes the maximal 1-consistent subreduction at that variable.*

Lemma 8.20. *Let $D^{(1)}$ be a 1-consistent reduction of a cycle-consistent instance $\mathcal{I}$ with $D^{(1)} \lhd\lhd\lhd D$, and $B <_T^{D_x} D_x^{(1)}$ with $T \in \{\textsc{ba}, \textsc{c}, \textsc{pc}\}$. Then there is a nonempty 1-consistent reduction $D^{(2)} \lhd\lhd\lhd D^{(1)}$ with $D_x^{(2)} \subseteq B$. Moreover, if $T \in \{\textsc{ba}, \textsc{c}\}$ then $D^{(2)} \leq_T D^{(1)}$; and if $T = \textsc{pc}$ and every $D_y^{(1)}$ is $\textsc{s}$-free then $D^{(2)} \leq_{\mathcal{M}\textsc{pc}} D^{(1)}$.*

Lemma 8.21 (Crucial constraints have the parallelogram property)**.** Suppose

(i) the one-constraint instance $R(x_1, \ldots, x_n)$, with $x_1, \ldots, x_n$ pairwise distinct, is 1-consistent — that is, $R \leq_{\mathrm{sd}} \mathbf{D}_{x_1} \times \cdots \times \mathbf{D}_{x_n}$ — and $D^{(1)}$ is a 1-consistent reduction for it;
(ii) every $\mathbf{D}_{x_i}^{(1)}$ is $\textsc{s}$-free;
(iii) $T \in \{\textsc{l}, \textsc{pc}, \textsc{d}\}$ and $D^{(2)} \leq_{\mathcal{M}T}^{D} D^{(1)} \lhd\lhd\lhd D$;
(iv) R is crucial in $D^{(2)}$.

Then R has the parallelogram property, and for every $i \in [n]$ the congruence $\mathrm{Con}_1(R, i)$ is irreducible — linear if $T = \textsc{l}$, PC if $T = \textsc{pc}$ — with $\mathrm{Con}_1(R, i)^* \supseteq (D_{x_i}^{(1)})^2$. If moreover $T = \textsc{pc}$ then $n = 2$.

Caveat 8.22 (Hypotheses (i) and (ii) are load-bearing)**.** Subdirectness over the *unreduced* domains is what makes each $\mathrm{Con}_1(R, i)$ reflexive on all of D_{x_i}, hence eligible to be a congruence on $\mathbf{D}_{x_i}$ at all (Lemma 5.23), and the proof spends it again each time Corollary 6.11 is applied with the full domains as ambient. The $\textsc{s}$-freeness of the $\mathbf{D}_{x_i}^{(1)}$ is what Lemma 6.28 carries.

Both call sites supply the two hypotheses. In Case 2 of the proof of Theorem 8.1(1), no $\mathbf{D}_x^{(1)}$ has a nontrivial binary absorbing or central subuniverse, and a $<_{\textsc{s}}$-witness would be one; and the instance there is cycle-consistent, so each constraint is subdirect. In Step 1 of Theorem 8.9, subdirectness again comes from cycle-consistency, but the $\textsc{s}$-freeness of the blocks $\phi(\bar{b})_x$ does *not* follow from hypothesis (ii) of that theorem and is a new obligation on its outline; it is recorded in Caveat 8.10.

Proof of Lemma 8.21, modulo the claim. Write $\mathbf{A}_i = \mathbf{D}_{x_i}$; hypothesis (i) is used in the form $\mathrm{pr}_i(R) = A_i$. First, $n \geq 2$: a subdirect unary constraint is $R = A_1$, whose reduction $D^{(2)}_{x_1}$ is nonempty, contradicting cruciality. Second, the form in which cruciality is spent throughout: $R^{(2)} = \varnothing$, while any finite set of constraints, *each properly weaker than* R, is *simultaneously* satisfiable in $D^{(2)}$, because a solution of $\mathcal{I}[R \Uparrow]^{(2)}$ satisfies every weaker constraint at once (Definition 5.17). Third, $R^{(1)} \neq \varnothing$: the domains $D^{(1)}_{x_i}$ are nonempty by $D^{(1)} \lhd\!\lhd\!\lhd D$ (Caveat 5.8), and 1-consistency projects $R^{(1)}$ onto them.

Step 1: the parallelogram property. By Caveat 3.20 it suffices to fix a split of $[n]$ into nonempty blocks K and L and prove the parallelogram implication there. Put $E_K = \mathrm{pr}_K(R)$, $E_L = \mathrm{pr}_L(R)$ — subdirect in the corresponding products of full domains — and let $W \leq_{\mathrm{sd}} \mathbf{E}_K \times \mathbf{E}_L$ be R read as a binary relation between the blocks. For $j \in \{1, 2\}$ let $E^{(j)}_K = E_K \cap \prod_{i \in K} D^{(j)}_{x_i}$ and likewise $E^{(j)}_L$.

Corollary 6.11(r), applied to the relation E_K with the chains $D^{(1)}_{x_i} \lhd\!\lhd\!\lhd A_i$, gives $E^{(1)}_K \,\dot{\lhd}\!\dot{\lhd}\!\dot{\lhd}\, E_K$, and $E^{(1)}_K \supseteq \mathrm{pr}_K(R^{(1)}) \neq \varnothing$, so $E^{(1)}_K \lhd\!\lhd\!\lhd E_K$. Corollary 6.11(m) gives $E^{(2)}_K \dot{\leq}^{E_K}_{\mathcal{MT}} E^{(1)}_K$; and $E^{(2)}_K$ is nonempty, because the constraint $\mathrm{pr}_K(R)$ on the variables $\{x_i : i \in K\}$ is properly weaker than R — its variable set is properly smaller — so cruciality gives it a solution in $D^{(2)}$. Every $<_{\mathrm{L}}$ or $<_{\mathrm{PC}}$ step is a $<_{\mathrm{D}}$ step, so $E^{(2)}_K \leq^{E_K}_{\mathcal{M}\mathrm{D}} E^{(1)}_K \lhd\!\lhd\!\lhd E_K$, and the same at L.

Now $W \cap (E^{(2)}_K \times E^{(2)}_L) = \varnothing$, since a pair there assembles to a tuple of $R^{(2)}$, while $W \cap (E^{(1)}_K \times E^{(1)}_L) \neq \varnothing$, since $R^{(1)} \neq \varnothing$. The *contrapositive* of Lemma 6.23 therefore gives $S \cap (E^{(2)}_K \times E^{(2)}_L) = \varnothing$ for S the rectangular closure of W. For a binary relation the rectangular closure is reached by iterating $W \mapsto W \circ W^{-1} \circ W$: each iterate is a subuniverse, positively primitively defined from W, the sequence is increasing, and its stable value is the least rectangular relation containing W (Lemma 3.19(b)); so S is a subuniverse of $\mathbf{E}_K \times \mathbf{E}_L$. Let S_R be S read back as an n-ary relation on the scope of R. Then $S_R \supseteq R$ is a constraint on the same variables, and $S^{(2)}_R = \varnothing$: a tuple of $S_R \cap \prod_i D^{(2)}_{x_i}$ has its K-part in $E^{(2)}_K$ and its L-part in $E^{(2)}_L$, and S misses that box. If $S_R \supsetneq R$ then S_R is properly weaker than R and cruciality makes $S^{(2)}_R$ nonempty — a contradiction. So $S_R = R$, i.e. W is rectangular, which is the parallelogram implication at the split (K, L). As the split was arbitrary, R has the parallelogram property; by Lemma 3.19(a) it is rectangular.

Step 2: the Con_1*'s are irreducible congruences.* Every variable of R is rectangular and $\mathrm{pr}_i(R) = A_i$, so $\sigma_i := \mathrm{Con}_1(R, i)$ is a congruence on $\mathbf{A}_i$ (Lemma 5.23(c)); and the instance $\{R\}$ is 1-consistent with R crucial in $D^{(2)}$, so Lemma 8.14 makes every σ_i irreducible.

Step 3: all σ_i are of one class. For $i \neq j$ there is a bridge from σ_i to σ_j. Permute the coordinates so that $i = 1$, $j = n$. If the witness configuration of Lemma 8.16 fails, then for every $\bar{a}$: if $\bar{a}[1 \mapsto b_1] \in R$ for some b_1 and $\bar{a}[n \mapsto b_n] \in R$ for some b_n, then $\bar{a} \in R$. That says R is the conjunction of the two constraints $\mathrm{pr}_{[n] \setminus \{1\}}(R)$ and $\mathrm{pr}_{[n] \setminus \{n\}}(R)$, each properly weaker than R; by cruciality some assignment in $D^{(2)}$ satisfies both at once, hence satisfies R, contradicting $R^{(2)} = \varnothing$. So the configuration exists, and Lemma 8.16 — its rectangularity hypotheses are Step 1 — returns a bridge from σ_1 to σ_n. Reversing a bridge reverses its ends, so by Lemma 4.15 either every σ_i is linear or none is: *the σ_i are all linear or all PC.*

Step 4: typing at a coordinate. Fix the coordinate; permute so it is the first; write $\mathbf{A} = \mathbf{A}_1$, $\sigma = \sigma_1$, $B = D^{(1)}_{x_1}$, and

$$E = \mathop{\mathrm{pr}}_{1}\bigl(R \cap (A \times D^{(2)}_{x_2} \times \cdots \times D^{(2)}_{x_n})\bigr).$$

Six facts.

(a) $E \neq \varnothing$: the constraint $\mathrm{pr}_{[n] \setminus \{1\}}(R)$ is properly weaker than R, so it has a solution in $D^{(2)}$, and the first entry of a witnessing tuple of R lies in E.

(b) $E \cap D^{(2)}_{x_1} = \varnothing$, since $R^{(2)} = \varnothing$.

(c) $E \circ \sigma = E$: for $a \in E$ with witness $(a, \bar{c}) \in R$ and $(a, a') \in \sigma$, stability of the first variable (Lemma 5.23(b)) gives $(a', \bar{c}) \in R$.
(d) $E \lhd\lhd\lhd A$: Corollary 6.11(r1) with the chains $D^{(2)}_{x_i} \lhd\lhd\lhd A_i$ — Lemma 6.22 composed with $D^{(1)} \lhd\lhd\lhd D$ — gives $E \dot{\lhd}\dot{\lhd}\dot{\lhd} A$, and (a) removes the dot.
(e) σ is *maximal* among congruences δ on $\mathbf{A}$ with $(E \circ \delta) \cap D^{(2)}_{x_1} = \varnothing$; it is one by (b) and (c). Let $\delta \supsetneq \sigma$ and put

$$R_\delta(x_1, \dots, x_n) = \exists z\ R(z, x_2, \dots, x_n) \wedge \delta(z, x_1),$$

a constraint on the scope of R with $R_\delta \supseteq R$, by reflexivity of δ. If $R_\delta = R$, then the first variable of R is stable under δ — for $(u, v) \in \delta$ and $(u, \bar{c}) \in R$ the tuple $(v, \bar{c})$ satisfies R_δ — and then, $\mathrm{pr}_1(R)$ being all of A, every δ-pair is a σ-pair, contradicting $\delta \supsetneq \sigma$. So R_δ is properly weaker; cruciality gives it a solution in $D^{(2)}$, whose witness z lies in E and satisfies $(z, a_1) \in \delta$ for its first entry $a_1 \in D^{(2)}_{x_1}$; so $(E \circ \delta) \cap D^{(2)}_{x_1} \neq \varnothing$.
(f) If $E \cap B \neq \varnothing$ then $B^2 \not\subseteq \sigma$: for $b \in E \cap B$ and $c \in D^{(2)}_{x_1} \subseteq B$, $(b, c) \in \sigma$ would put $c \in E \circ \sigma = E$ against (b).

Case A: $E \cap B \neq \varnothing$. Then $D^{(2)}_{x_1} \neq B$ by (b), so Definition 4.24 writes $D^{(2)}_{x_1} = \bigcap_{j \le t}(B \cap E_j)$, $t \geq 1$, with $B \cap E_j <^{A}_{T(\tau_j)} B$: each τ_j is an irreducible congruence on $\mathbf{A}$ — linear if $T = \textsc{l}$, PC if $T = \textsc{pc}$ — with $\tau_j^* \supseteq B^2$, and E_j is the block of τ_j containing $D^{(2)}_{x_1}$. Note $\tau_j \not\supseteq B^2$: otherwise $B \subseteq E_j$ and the step is not proper. Put $\tau = \tau_1 \cap \dots \cap \tau_t$ and consider R_τ as in (e).

If $R_\tau = R$: as in (e), $\tau \subseteq \sigma$. If moreover $\tau_j \subseteq \sigma$ for some j, then either $\sigma = \tau_j$, or $\sigma \supsetneq \tau_j$ — and σ, a subuniverse of $\mathbf{A}^2$ stable under τ_j and properly containing it, then contains $\tau_j^* \supseteq B^2$ (Proposition 3.25(b)), against (f). So $\sigma = \tau_j$, and the conclusion holds at this coordinate. If no $\tau_j \subseteq \sigma$ — possible only for $t \geq 2$ — then σ is an irreducible congruence with $\tau_1 \cap \dots \cap \tau_t \subseteq \sigma$ and, by (f), $B^2 \not\subseteq \sigma$; Lemma 6.28, whose $\textsc{s}$-freeness hypothesis is (ii), gives the conclusion.

If $R_\tau \supsetneq R$: cruciality gives a solution of R_τ in $D^{(2)}$ whose witness z lies in E with $(z, a_1) \in \tau$ for some $a_1 \in D^{(2)}_{x_1}$; each E_j is a block of $\tau_j \supseteq \tau$ containing a_1, so $z \in \bigcap_j E_j$. Together with (b) this is

$$E \cap \textstyle\bigcap_j E_j \neq \varnothing, \qquad E \cap B \cap \bigcap_j E_j = \varnothing, \qquad E \cap B \neq \varnothing. \tag{α}$$

Case B: $E \cap B = \varnothing$ — which is where the coordinates with $D^{(2)}_{x_1} = D^{(1)}_{x_1}$ always land, by (b). Fix chain data for $B \lhd\lhd\lhd A$ (Definition 4.25): $A = B_0 \supseteq \dots \supseteq B_m = B$ with $B_k <^{A}_{T_k(\xi_k)} B_{k-1}$ and $T_k \in \{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{d}\}$. Since $E \cap B_0 \neq \varnothing$ and $E \cap B_m = \varnothing$, there is a least k with $E \cap B_k = \varnothing$; write $B' = B_{k-1}$, $C' = B_k$, so $E \cap B' \neq \varnothing$, $E \cap C' = \varnothing$, and $D^{(2)}_{x_1} \subseteq B \subseteq C'$. As in (f), $B'^2 \not\subseteq \sigma$.

The step cannot be of type $\textsc{s}$: it would supply a nonempty $D \subseteq C'$ with $D <_{\textsc{ba},\textsc{c}} B'$ (Definition 4.6); $E \cap B'$ is nonempty with $E \cap B' \dot{\lhd}\dot{\lhd}\dot{\lhd}^{A} B'$ by Lemma 6.16(i), so Lemma 7.24 gives $(E \cap B') \cap D \neq \varnothing$, against $E \cap C' = \varnothing$.

If the step is of type $\textsc{d}$, consider R_{ξ_k} as in (e). If $R_{\xi_k} = R$ then $\xi_k \subseteq \sigma$, and as in Case A either $\sigma \supseteq \xi_k^* \supseteq B'^2$ — excluded — or $\sigma = \xi_k$: then σ is irreducible with $\sigma^* \supseteq B'^2 \supseteq B^2$, and for $T = \textsc{d}$ the coordinate is done, while for $T \in \{\textsc{l}, \textsc{pc}\}$ only the *class* of σ remains, which Step 3 settles whenever some coordinate is typed outright. If $R_{\xi_k} \supsetneq R$, cruciality gives $z \in E$ with $(z, a_1) \in \xi_k$, $a_1 \in D^{(2)}_{x_1} \subseteq C'$; writing E' for the block of ξ_k containing a_1, so that $C' = B' \cap E'$, this is

$$E \cap E' \neq \varnothing, \qquad E \cap B' \cap E' = \varnothing, \qquad E \cap B' \neq \varnothing, \tag{β}$$

the shape of (α) one chain step up, with the single block E' of the irreducible ξ_k in place of the type-T blocks.

The remaining possibility is a step of type BA or C:

$$C' <^{A}_{T_k} B', \quad T_k \in \{\text{BA}, \text{C}\}, \qquad E \cap B' \neq \varnothing, \qquad E \cap C' = \varnothing. \tag{γ}$$

Configuration (γ) *does not occur.* Run the construction of Step 1 at the split $K = \{1\}$, $L = \{2, \ldots, n\}$: since $E_K = \mathrm{pr}_1(R) = A$, this reads R as $W \leq_{\mathrm{sd}} \mathbf{A} \times \mathbf{E}_L$, and Step 1 established $E_L^{(2)} \leq^{E_L}_{\mathcal{M}\mathrm{D}} E_L^{(1)} \lhd\lhd\lhd E_L$. By construction $E = \mathrm{pr}_1(W \cap (A \times E_L^{(2)}))$. Both anchor configurations of Lemma 6.26(i) are nonempty: $W \cap (B' \times E_L^{(2)}) \neq \varnothing$ is exactly $E \cap B' \neq \varnothing$, and $W \cap (C' \times E_L^{(1)}) \neq \varnothing$ because every tuple of $R^{(1)}$ has first entry in $D^{(1)}_{x_1} = B \subseteq C'$ and L-part in $E_L^{(1)}$, and $R^{(1)} \neq \varnothing$. With the member $C' <^{A}_{T_k} B' \lhd\lhd\lhd A$, $T_k \in \{\text{BA}, \text{C}\}$, clause (ii) of the lemma holds, and its conclusion $W \cap (C' \times E_L^{(2)}) \neq \varnothing$ says $E \cap C' \neq \varnothing$ — against (γ). So the chain step at the first failure is never of type BA or C, and Case B always lands in $\sigma = \xi_k$ or in (β).

Claim 8.23 (The typing anchor)**.** *In configuration* (α)*,* σ *is linear if* $T = \text{L}$ *and PC if* $T = \text{PC}$*, with* $\sigma^* \supseteq B^2$*. In configuration* (β)*, the class of* $\sigma = \xi_k$ *is that of* T *when* $T \in \{\text{L}, \text{PC}\}$*. And if* $T = \text{PC}$ *then* $n = 2$*.*

Granting the claim, the lemma follows. Step 1 is the parallelogram property. Fix a coordinate: Step 4 runs at it, and every branch — Case A through $\sigma = \tau_j$ or Lemma 6.28, Case A through (α) and the claim, Case B through $\sigma = \xi_k$ with the claim or Step 3 supplying the class, Case B through (β) and the claim — ends with σ_i irreducible, of the class of T when $T \in \{\text{L}, \text{PC}\}$, and with $\sigma_i^* \supseteq (D^{(1)}_{x_i})^2$. The clause $n = 2$ for $T = \text{PC}$ is the claim's last part. □

Caveat 8.24 (What is proved, what the claim must do, and why it carries R)**.** Everything in the proof above is proved except Claim 8.23; Lemma 6.28, which the good branch of Case A consumes, is itself only stated, with its $t = 1$ case immediate from Proposition 3.25(b). So what stands complete is: the parallelogram property (Step 1, resting on the stated Lemma 6.23 and Corollary 6.11); irreducibility (Step 2); the one-class fact (Step 3); the exclusion of configuration (γ) by Lemma 6.26; and, at each coordinate, the reduction of the typing to the two configurations (α) and (β). The claim is the residue, and the class part of (β) is the same type-D-step mismatch as Caveat 7.66(b). Note that the single-type BA/C obstruction of Caveat 7.66(a), though it looks like (γ), is *not* discharged by the same lemma: it arises at the level of subsets of one algebra, where no subdirect relation is available to anchor.

One route is closed, and it is worth recording with the example that closes it. Configuration (α) cannot be *refuted*: it is realizable by strong subsets. In $\mathbf{A} = \mathbf{Z}_p \times \mathbf{Z}_p$ the relations $\delta_U = \{(a, b) : a - b \in U\}$, for U one of the subgroups $\langle(1,0)\rangle$, $\langle(0,1)\rangle$, $\langle(1,1)\rangle$, are linear irreducible congruences with $\delta_U^* = A^2$; take $E = \langle(1,0)\rangle$, $B = \langle(0,1)\rangle$, and for E_1 the block $(1,0) + \langle(1,1)\rangle$ of $\delta_{\langle(1,1)\rangle}$. The three pairwise intersections are nonempty and the triple intersection is empty, so (α) holds with $t = 1$, E is saturated for $\sigma = \delta_{\langle(1,0)\rangle}$, and σ is maximal with $(E \circ \sigma) \cap (B \cap E_1) = \varnothing$. In this configuration the claim's conclusion is true — σ is linear with full cover — but no contradiction is available: any proof of the claim must therefore *derive* the typing inside (α), and must use that E is the trace of the crucial R, which is the one hypothesis the configuration above does not model. This is also why Hypothesis i resisted every attempt at the level of subsets of a single algebra (Caveat 7.67): the statement to be proved is not true at that level.

Neither the claim nor anything else in this proof cites Lemma 6.30, and nothing else consumes it; with this proof in place, Hypothesis i is no longer on any path to the main statements, and the open obligations of this route are Claim 8.23 and Lemma 6.28. See §11.

Lemma 8.25 (Bridges from a subdirect PC/linear instance)**.** Suppose

(i) $\mathcal{I}$ has subdirect solution set;
(ii) $D^{(1)}$ is a reduction with $D_x^{(1)} \lhd\lhd\lhd D_x$ for every x;
(iii) $C \in \mathcal{I}$ is a constraint of type $T \in \{\text{PC}, \text{L}\}$;
(iv) $B <^{D_z}_{\mathcal{T}(\xi)} D_z^{(1)}$ for some variable z, with $\mathcal{T} \in \{\text{BA}, \text{C}, \text{S}, \text{PC}, \text{L}\}$;
(v) if $T = \text{PC}$ then $\mathcal{T} \in \{\text{PC}, \text{L}\}$;
(vi) $\mathcal{I}^{(1)}$ has a solution;
(vii) $\mathcal{I}^{(1)}$ has no solution with $z \in B$;
(viii) $\mathcal{I}[C \Uparrow]$ has a solution in $D^{(1)}$ with $z \in B$.

Then $\mathcal{T} = T$, and for every variable x of C there is a bridge δ from ξ to $\text{Con}_1(C, x)$ with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$.

Caveat 8.26 (Stated only)**.** Lemma 8.25 carries eight numbered hypotheses and is *stated without proof here*. It is the only source of bridges in §8, and every use of connectedness routes through it; stated without proof, likewise, are Lemma 8.20, Corollary 8.27, and clauses (a) and (b) of Lemma 8.17; Lemma 8.21 is proved modulo Claim 8.23 and Lemma 6.28. These are gaps in the logical chain of this document, not imported results, and §11 records them as such.

Its conclusion has two independent parts — "$\mathcal{T} = T$" and "there is a bridge with $\widetilde{\delta} \supseteq \mathcal{I}(z, x)$" — and different call sites use different parts. The dependency order is Theorem 6.18, which it consumes, then this lemma, then Theorem 8.1, which consumes it.

Corollary 8.27. *In the situation of Theorem 8.1(2) with $\mathcal{I}'$ satisfying (1b), the type $\mathcal{T}$ of the reduction and the type of the constraints must agree, which contradicts $\mathcal{T} \in \{\text{BA}, \text{C}\}$.*

9 The algorithm

The algorithm is [13]'s. It is stated here because target (T1) of §1 is a proposition *about* it and cannot be written down otherwise.

9.1 Solve

```
function Solve(I):
    repeat
        I := ForceConsistency(I)
        if I = false: return "No"
        if some D_x has a strong subset B:
            I := ReduceDomain(I, x, B)
    until nothing changed
    return SolveLinear(I)
```

ForceConsistency enforces cycle-consistency and irreducibility, returning `false` if some domain empties. The reduction step is licensed by Theorem 8.6: restricting to a strong subset cannot destroy solvability. When no strong subset remains, Lemma 6.2 together with Lemma 4.18 gives, for each domain of size at least 2, a congruence σ_x with $D_x/\sigma_x \cong \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_{n_x}}$; choose σ_x minimal. That case is SolveLinear.

9.2 SolveLinear

Write $\phi\colon \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k} \to D_{x_1}/\sigma_{x_1} \times \cdots \times D_{x_m}/\sigma_{x_m}$ for a linear map and

$$\phi^{-1}(\mathcal{I}) = \{\bar{a} : \mathcal{I} \text{ has a solution in } \phi(\bar{a})\}.$$

Computing $\phi^{-1}(\mathcal{I})$ would solve $\mathcal{I}$; the algorithm cannot do that, but it can do four things:

- **(p0)** decide $\bar{a} \in \phi^{-1}(\mathcal{I})$, by reducing each domain to the corresponding block and recursing on a smaller domain;
- **(p1)** decide $\phi^{-1}(\mathcal{I}) = \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$, by testing $\bar{0}$ first and then the k unit vectors with (p0). The set $\phi^{-1}(\mathcal{I})$ is in general an affine *coset*, not a subgroup; testing $\bar{0}$ first is what makes it one, after which membership of the coordinate generators implies fullness;
- **(p2)** compute $\phi^{-1}(\mathcal{I})$ when $\mathcal{I}$ is not linked, by splitting into linked pieces, recursing, and taking the union;
- **(p3)** compute $\phi^{-1}(\mathcal{I})$ when it is empty or of codimension one, by finding $\bar{a} \notin \phi^{-1}(\mathcal{I})$, then for each i a b_i with $\bar{a}[i \mapsto b_i] \in \phi^{-1}(\mathcal{I})$ if one exists, and reading off the equation $\sum_{i \in J}(y_i - a_i)/(b_i - a_i) = 1$ over the i for which b_i exists. The displayed quotient is meaningful only within one prime: the codimension-one equation has a single target prime p, and only coordinates over $\mathbb{F}_p$ can carry a nonzero coefficient, so the computation is organised prime by prime (Caveat 8.10).

```
function SolveLinear(I):
    m := dim(D_{x_1}/s_1 x ... x D_{x_n}/s_n)
    phi := a bijective linear map Z_{p_1}x...xZ_{p_m} -> that product
    while phi^-1(I) != Z_{p_1}x...xZ_{p_m}:              // (p1)
        I' := I
        for C in I':                                // drop what can be dropped
            if phi^-1(I' \ {C}) != Z_{p_1}x...xZ_{p_m}:   // (p1)
                I' := I' \ {C}
        F := phi^-1(I')                                // (p2) or (p3)
```

```
        if F = {}: return "No"
        m := dim F ;  psi := a bijective linear map onto F ;  phi := phi o psi
    return "Yes"
```

The inner loop makes $\mathcal{I}'$ minimal with $\phi^{-1}(\mathcal{I}')$ not full. At that point Theorem 8.9 applies — its hypothesis (vii) is exactly minimality — so $\phi^{-1}(\mathcal{I}')$ is empty or of codimension one, and (p3) computes it; or $\mathcal{I}'$ is not linked and (p2) does. Since $\phi^{-1}(\mathcal{I}) \subseteq \phi^{-1}(\mathcal{I}')$, the image of ϕ shrinks while still containing all solutions. The dimension strictly decreases, so the loop terminates.

9.3 What (T1) has to say

Caveat 9.1 (Two obstacles between the pseudocode and a function)**.** (a) *Termination is not structural.* Solve recurses through (p0) on strictly smaller domains, and SolveLinear loops on a strictly decreasing dimension. Neither is a structural recursion; both need an explicit well-founded measure, and the two interleave — SolveLinear calls Solve on a smaller domain, which calls SolveLinear again. The measure is lexicographic: $(\sum_x |D_x|,\ \dim \phi)$.

(b) *Several steps quantify over objects that are only finitely many by accident.* "If some D_x has a strong subset B" quantifies over subsets of D_x, over congruences on D_x, and — through absorption — over *term operations*, of which there are infinitely many. For the types the algorithm actually searches for, the search is finite for a reason that needs no general arity bound: ba has a binary witness by definition, and a central subuniverse has a ternary one by Lemma 4.4.

9.4 Solve as a function

The pseudocode is not a function: it makes unconstrained choices, and its recursion is not structural. Both are fixed here, so that Theorem 9.5 has a subject.

Definition 9.2 (The call set and its measure)**.** A *call* is either $\mathsf{S}(\mathcal{I})$, for $\mathcal{I}$ an instance of CSP(Γ) with domains D_x, or $\mathsf{L}(\mathcal{I}, \phi)$, where in addition ϕ is an injective linear map $\mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k} \to \prod_x D_x/\sigma_x$ with the σ_x the minimal congruences of §9. Put $N(\mathcal{I}) = \sum_x |D_x|$ and

$$\mu(\mathsf{S}(\mathcal{I})) = (N(\mathcal{I}),\, N(\mathcal{I}) + 1), \qquad \mu(\mathsf{L}(\mathcal{I}, \phi)) = (N(\mathcal{I}),\, \dim \phi),$$

ordered lexicographically on $\mathbb{N}^2$. Since $\dim \phi \le N(\mathcal{I})$ always, a call $\mathsf{S}(\mathcal{I})$ is strictly above every $\mathsf{L}(\mathcal{I}, \phi)$ with the same instance.

Definition 9.3 ($\textsc{Solve}_{\text{alg}}$)**.** Fix once and for all a linear order on the finite set of pairs (x, B) with $B \subseteq D_x$, and one on the finite set of candidate maps ϕ; "the least" below means least in these orders, which is what turns each choice in the pseudocode into a value. Define $\textsc{Solve}_{\text{alg}}$ on calls by well-founded recursion on μ:

$\mathsf{S}(\mathcal{I})$**.** Put $\mathcal{I}' = \textsc{ForceConsistency}(\mathcal{I})$. If some domain of $\mathcal{I}'$ is empty, return `false`. If some D'_x has a proper nonempty subset B with $B <_T D'_x$ for some $T \in \{\textsc{ba}, \textsc{c}, \textsc{d}\}$, take the least such (x, B) and return $\textsc{Solve}_{\text{alg}}(\mathsf{S}(\textsc{ReduceDomain}(\mathcal{I}', x, B)))$. Otherwise return $\textsc{Solve}_{\text{alg}}(\mathsf{L}(\mathcal{I}', \phi_0))$ for the least admissible ϕ_0.

$\mathsf{L}(\mathcal{I}, \phi)$**.** Run the body of SolveLinear, reading each test $\bar{a} \in \phi^{-1}(\mathcal{I})$ of (p0) as the value of $\textsc{Solve}_{\text{alg}}$ at S of the instance $\mathcal{I}$ with each domain cut to the block $\phi(\bar{a})_x$, and each iteration of the loop as the value at $\mathsf{L}(\mathcal{I}, \phi \circ \psi)$.

Lemma 9.4 (The recursion is well founded)**.** *Every recursive call made by Definition 9.3 is at a strictly smaller μ, so $\textsc{Solve}_{\text{alg}}$ is a total function into $\{$`true`$,$ `false`$\}$.*

Proof. Four kinds of call. ForceConsistency only shrinks domains, so it does not increase N; the reduction step replaces some D'_x by a proper subset, strictly decreasing N. The call from $\mathsf{S}(\mathcal{I}')$ to $\mathsf{L}(\mathcal{I}', \phi_0)$ leaves N fixed and drops the second component, by the inequality in Definition 9.2. The (p0) calls cut every domain to a block of σ_x; each σ_x is proper at some variable, since L is reached only when some domain has size at least 2, so N strictly decreases. The loop calls leave N fixed and strictly decrease $\dim \phi$, as §9 records. ☐

Theorem 9.5 (Target (T1))**.** *For every instance $\mathcal{I}$ of* CSP(Γ)*,*

$$\textsc{Solve}_{\mathrm{alg}}(\mathsf{S}(\mathcal{I})) = \texttt{true} \iff \mathcal{I} \text{ has a solution.}$$

Caveat 9.6 (What Definition 9.3 still owes)**.** Theorem 9.5 is about *this* function, and that is the whole of its content: the weaker reading — "there exists a Boolean-valued function deciding satisfiability" — follows for a fixed finite template by exhaustive search and says nothing. Three things are needed before the definition is airtight, and none is deep.

(a) ForceConsistency and ReduceDomain are used as given total functions. The first must be exhibited as a fixpoint iteration that terminates because it only shrinks domains; the second is immediate.
(b) The search in S ranges over subsets B and over the witnesses making $B <_T D'_x$. It is finite for the reason in Caveat 9.1(b) — ba has a binary witness by definition and c a ternary one by Lemma 4.4, and d quantifies over congruences — but the bound has to be stated for the "least such (x, B)" to be well defined.
(c) Only ba, c and d are searched for, not s; that is legitimate because $<_{\mathrm{s}}$ is witnessed by a subset carrying the other two types, but the correctness proof must use Theorem 8.6 in the form that covers exactly the types searched.

Theorem 9.5 itself is stated and not proved here; see §11.

Given that, Theorem 9.5 mentions no machine, is provable from §8, and is the honest content of "the algorithm works": that one particular, highly non-obvious algorithm — the one whose existence is the dichotomy — is correct.

Theorem 9.7 (Target (T2); stated, not proved)**.** *For each finite* Γ *there are a program computing* $\textsc{Solve}_{\mathrm{alg}}$ *and a polynomial* p_Γ *bounding its running time on instances of* CSP(Γ) *in the size of the instance.*

Caveat 9.8 ((T2) is not a wrapper)**.** Theorem 9.7 is the only statement of the tractable half that needs a cost model, and it is not a corollary of Theorem 9.5: it requires $\textsc{Solve}_{\mathrm{alg}}$ to be exhibited as a program with a proved step bound. Note also that the statement is non-uniform. Because Γ is fixed, the searches over subsets of A, over congruences on A, and over term operations of bounded arity are constant-time — but "constant" means compiled away into a program whose *size* depends on $|A|$. So $\textsc{Solve}_{\mathrm{alg}}$ is a family with a polynomial for each Γ, and a bound uniform in Γ would be a different statement, and a false one.

10 The hard half

If no weak near-unanimity operation preserves Γ, then $\mathrm{CSP}(\Gamma)$ is NP-complete. This half is older, shorter, and better understood than the tractable half, and it factors cleanly into algebra and complexity.

10.1 The chain

(1) *Reduction to a core, and to an idempotent algebra.* One may assume Γ is a rigid core containing all singleton unary relations, so that $\mathrm{Pol}(\Gamma)$ is idempotent, at the cost of a logspace reduction [2, "Cores and Idempotent Reducts"]. This is Caveat 1.6.

(2) *Maróti–McKenzie.* A finite idempotent algebra has a Taylor term if and only if it has a weak near-unanimity term [10].

(3) *No Taylor term $\Rightarrow$ two disjoint subalgebras.* If a finite idempotent $\mathbf{A}$ has no Taylor term then there are nonempty $B, C \leq \mathbf{A}$ with $B \cap C = \varnothing$ and

$$N = (B \cup C)^3 \setminus (B^3 \cup C^3) \;\leq\; \mathbf{A}^3$$

[2, Cor. 1.5.11]. This replaces the usual formulation — "the variety generated contains a two-element algebra whose only operations are projections, and from there Γ pp-interprets every finite language" — whose "and" hides an HSP representation in a finite power and its translation into an interpretation. The cited corollary does that work and lands on a concrete relation, from which Lemma 10.1 builds the interpretation in four lines.

(3′) *Inv–Pol.* N is a subuniverse of $\mathbf{A}^3$ and so is preserved by every polymorphism of Γ; Theorem 10.8 turns that into pp-definability from Γ. This is the one substantial import of the step, and only the hard direction of it is used.

(4) *Bulatov–Jeavons–Krokhin.* A primitive positive interpretation of Δ in Γ yields a polynomial-time many-one reduction $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$ [1].

(5) *Cook–Levin.* NAE-3-SAT is NP-hard.

Steps (1)–(3) are algebra; step (4) is a syntactic construction plus a complexity-theoretic wrapper; step (5) is pure complexity theory.

10.2 The layered targets

Lemma 10.1 (The interpretation, explicitly)**.** *Let $\mathbf{A}$ be an algebra, $B, C \leq \mathbf{A}$ nonempty and disjoint, and suppose $N = (B \cup C)^3 \setminus (B^3 \cup C^3)$ is a subuniverse of $\mathbf{A}^3$. Then $\{N\}$ pp-interprets $\{\mathrm{NAE}\}$ on $\{0, 1\}$, with $d = 1$.*

Proof. A triple lies in N exactly when its entries lie in $B \cup C$ and are not all on the same side of the partition; in particular, for a repeated entry,

$$(x, z, z) \in N \iff x, z \in B \cup C \text{ and } x, z \text{ lie on opposite sides.} \tag{10.1}$$

Take $F = \mathrm{pr}_1(N)$, which is pp-definable from N and equals $B \cup C$: each $b \in B$ is the first entry of $(b, c, c) \in N$ for any $c \in C$, and symmetrically. Let $\pi \colon F \to \{0, 1\}$ send B to 0 and C to 1; it is surjective because B and C are nonempty. The π-preimage of NAE is N itself, by the description above. The π-preimage of equality on $\{0, 1\}$ is $B^2 \cup C^2$, and by (10.1)

$$B^2 \cup C^2 = \{(x, y) : \exists z\, N(x, z, z) \wedge N(y, z, z)\} :$$

a z opposite to both x and y exists exactly when x and y are on the same side, since there are only two sides and both are nonempty. □

Theorem 10.2 (Target (H0)). *Let* $\mathbf{A}$ *be a finite idempotent algebra with no weak near-unanimity term operation. Then* $\mathrm{Inv}(\mathbf{A})$ *primitively positively interprets the constraint language* $\{\textsc{Nae}\}$ *on* $\{0,1\}$*, where* $\textsc{Nae} = \{0,1\}^3 \setminus \{(0,0,0),(1,1,1)\}$*.*

Proof. By Lemma 6.32's source [10], a finite idempotent algebra has a Taylor term if and only if it has a weak near-unanimity term; so $\mathbf{A}$ has no Taylor term. Step (3) supplies nonempty disjoint $B, C \leq \mathbf{A}$ with $N = (B \cup C)^3 \setminus (B^3 \cup C^3) \leq \mathbf{A}^3$, so $N \in \mathrm{Inv}(\mathbf{A})$, and Lemma 10.1 builds the interpretation. Every relation it uses is pp-definable from N and hence lies in $\mathrm{Inv}(\mathbf{A})$. □

Theorem 10.3 (Target (H1); stated, not proved). *If* Γ *primitively positively interprets* Δ *then* $\mathrm{CSP}(\Delta) \leq_p \mathrm{CSP}(\Gamma)$*. Consequently, under* (H0)*,* $\mathrm{CSP}(\Gamma)$ *is* NP*-hard.*

Caveat 10.4 (Hardness is not completeness). Theorem 10.3 concludes NP-*hardness*. The word *complete* in Theorem 1.2 additionally needs membership, which is Theorem 10.5. Small, but it needs the cost model, so it belongs with (H1) and not with (H0).

Theorem 10.5 (NP membership; stated, not proved). *For each finite* Γ*,* $\mathrm{CSP}(\Gamma) \in \mathrm{NP}$*.*

Proof sketch. A solution is a certificate: an assignment of a value of A to each variable, of size $O(v \log |A|)$ for an instance with v variables, hence linear in the instance. Verifying it evaluates each constraint once, and for fixed Γ that is a table lookup of constant cost. Writing this out needs the encoding and the machine, which is why it is stated with (H1) rather than proved here. □

With Theorems 10.3 and 10.5 together, the hard half of Theorem 1.2 reads NP-*complete* as claimed.

Theorem 10.2 mentions no machine. Theorem 10.3 is where the complexity layer enters, and it is small — the reduction is a syntactic substitution of formulas — but it cannot be stated at all without a precise notion of polynomial-time many-one reduction.

10.3 Definitions

Definition 10.6 (pp-formula, pp-definability). A *primitive positive formula* over Γ is $\exists y_1 \ldots \exists y_s \bigwedge_j R_j(\bar{z}_j)$ with $R_j \in \Gamma \cup \{=\}$. A relation is *pp-definable* from Γ if some pp-formula defines it.

Definition 10.7 (pp-interpretation). Γ on A *pp-interprets* Δ on B if there are $d \geq 1$, a pp-definable $F \subseteq A^d$, and a surjection $\pi \colon F \to B$ such that the π-preimage of every relation of Δ, and of the equality relation on B, is pp-definable from Γ.

Theorem 10.8 (Geiger; Bodnarchuk–Kalužnin–Kotov–Romov). *For a finite* A*, a relation is pp-definable from* Γ *if and only if it is preserved by every polymorphism of* Γ*.*

Caveat 10.9 (The Galois connection is the expensive import). Theorem 10.8 is what converts the algebraic hypothesis ("no WNU polymorphism") into a syntactic conclusion ("pp-defines a hard relation"), and it is the only genuinely substantial import of §10. The easy direction is a one-line calculation. The hard direction — every relation preserved by all polymorphisms is pp-definable — goes through the free algebra on $|A|^k$ generators and an indicator-instance construction; it is roughly four printed pages and is the single largest item in the (H0) budget.

Remark 10.10 (Why the hard half is not the hard half). Despite its name, (H0) is far smaller than (T0): perhaps twenty printed pages against a hundred. The asymmetry is intrinsic. Showing a problem is hard requires exhibiting one construction; showing it is easy requires an algorithm that works on every instance, and the correctness proof must survive every configuration the instance can present.

11 Status and imports

This section says, for every statement in this document, whether it is proved here, imported, or not yet proved. It replaces the running commentary that a partial exposition invites, and it is the only place where completeness claims are made.

11.1 The five verdicts

proved A complete proof is given here, or the statement is an immediate specialisation of one that is.

imported Proved elsewhere and used as a black box. Every such statement carries an exact citation in §11.4, with the hypotheses in the form the source states them.

outlined An argument is given at the level of its case structure, or with steps suppressed. *This is not a proof* and the statement should not be relied on.

stated The statement appears and nothing is offered towards it. These are gaps in the logical chain, not imports.

open Confirmed to need an argument that is not available. There are none left in the table; what remains of that kind is Claim 7.65 and Claim 8.23, both written into proofs rather than left implicit. Hypothesis i was the third; since the proof of Lemma 8.21 no longer routes through Lemma 6.30, it is no longer on any path to the main statements (Caveat 8.24).

Nothing below is marked PROVED on the strength of a machine search. Two claims here rest on finite verification — the counterexamples of Remark 7.45 and of Caveat 6.13 — and both are written so that a reader can check them in a few lines by hand.

One structural property *is* checked mechanically: the proof-dependency graph of this document — an edge from each statement to every statement its proof cites — is acyclic, over 116 statements, 80 of which carry proofs, and 190 edges. That is a check on the arrangement, not on the mathematics, and it has caught three real cycles: Caveats 6.6 and 7.27, and a third that a first draft of `lem:self-intersection-pc` introduced by borrowing four steps from the proof of `lem:intersection-good` — which consumes it. The borrowing was only rhetorical, but the parser is right that a proof may not cite a statement that cites it, and the steps are now written out.

11.2 The table

Statement	Verdict	Note
§2–§4: vocabulary		
`prop:p-divides`, `cor:Zp-in-Vn`	PROVED	
`lem:sg-terms`	IMPORTED	standard; [5]
`lem:rect-basics`, `prop:cover`, `lem:irred-quotient`, `lem:linked-basics`	PROVED	
`lem:bridge-comp`	IMPORTED	[13, Lem. 6.3]
`lem:symmetrise`	PROVED	the collapse identity is proved by unfolding, so no irreducibility hypothesis is borrowed
`lem:central-ternary`	IMPORTED	[15, Cor. 6.11.1]
`lem:sfree-equiv`, `lem:linear-bridge`	PROVED	
`lem:linear-equiv`, `lem:bridge-reflexive`, `lem:rectangularise`	PROVED	
`lem:no-cross-bridge`, `lem:bridge-to-pc`	PROVED	written out in §7.4; the two normalisations they rested on are now `lem:rectangularise` and three lines, Caveat 7.51

Statement	Verdict	Note
lem:pc-bridges-trivial	PROVED	third case not written out
lem:linear-on-top	PROVED	
lem:pc-on-top	STATED	consumed by nothing, and now carrying the BA-and-centre-free hypothesis its proof needs; Caveat 4.20
§5: instances		
prop:covering-basics	PROVED	(g) is the imported Lemma 8.11; (e), (f) carry the renaming proviso of Caveat 5.21
lem:conone-basics	PROVED	transitivity needs only rectangularity; reflexivity on all of A_i is where subdirectness is spent
def:constraint-relation	—	the repeated-variable policy; every variable-named operation below is taken on R^C, Caveat 5.4
def:violation, lem:weakening-terminates, rem:get-crucial	PROVED	the measure counts *excluded* assignments, and cruciality quantifies over *single* weaker constraints; Caveat 5.16
§6–§7: strong subuniverses		
thm:brady-absorption	IMPORTED	[2, Thm. 3.11.1]
lem:linked-implies-abs	PROVED	a corollary of the above
lem:pp-propagation	IMPORTED	[4, Lem. 2.9]; conclusion dotted, call sites audited, Caveat 7.5
lem:barto-kazda	IMPORTED	[4, Prop. 2.14]
lem:factor-type	IMPORTED	C only; [15, Lem. 6.8]
lem:power-to-base	IMPORTED	[15, Lem. 6.24]
thm:central-relation-source	IMPORTED	[15, Thm. 6.15]
lem:hobby-mckenzie	IMPORTED	[8]
lem:possible-intersections	IMPORTED	[15, Lem. 6.25]
lem:absorbing-equality	IMPORTED	[13, Lem. 7.2]
lem:ba-center-implies	IMPORTED	[15, Cor. 6.1.2, 6.9.2]; both hypotheses restored, Caveat 6.13
lem:special-wnu	IMPORTED	[10, Lem. 4.7]
lem:building-perfect	IMPORTED	[13, Cor. 8.17.1]
lem:common-binary-witness	PROVED	the missing hypothesis of lem:ba-center-implies is exactly productivity of binary absorption; Caveat 6.15

Statement	**Verdict**	**Note**
`lem:reverse-hom`, `lem:bacon-left`, `lem:no-central-critical`, `cor:no-central-critical`, `lem:bridge-abelian`, `lem:ba-center-intersect`, `lem:no-absorbing-diagonal`, `lem:absorb-linked-sub`, `lem:no-unary-quotient`, `lem:central-relation`, `lem:strong-nonempty`, `lem:multiset-shift`, `lem:tot-sym-full`, `lem:tot-sym-irred`, `lem:full-fibre`, `cor:intersection-good`, `lem:parallelogram-D`, `lem:nice-bridge-abelian`, `lem:nontrivial-bridge`, `lem:pc-inductive-step`, `lem:bridge-between-congruences`, `lem:two-stable`, `lem:cut-or-nothing`, `lem:main-existence`, `lem:ubiquity`, `lem:intersect-all`, `cor:intersect-all-weak`, `lem:propagation`, `cor:prop-quotient`, `thm:stable-intersection`, `cor:stable-intersection`, `lem:multitype-chain`, `lem:perfect-from-linked`, `lem:no-abs-in-linear`, `lem:ba-central-meets`	PROVED	
`lem:self-intersection-pc`	PROVED	consumed by `lem:intersection-good`, subsubcase 2A3; Caveat 7.38
`lem:intersection-good`	PROVED	both parts written out, with the five properness splits and the two full-fibre steps; Caveat 7.40
`lem:anchored-intersection`	PROVED	the anchored form of stable intersection: leave-one-out is manufactured by first failure along the multi-type chain; closes configuration (γ) in the proof of `lem:parallelogram-crucial` and the mixed-type half of the source's D19; Caveat 6.27
`lem:preserve-linked-anchored`	PROVED	the provable core of `lem:preserve-linked`: with the anchor, the full/cut dichotomy on the linking congruence closes both ways; consumed by nothing yet; Caveat 6.25
`lem:block-good-bridge`, `lem:center-pushed-in`, `lem:leftlinked-stay-full`, `lem:factor-by-delta`	OUTLINED	
`lem:multiply-all-linear`	OUTLINED	Case 1 and the reduction of Case 2 are proved; Claim 7.65, which is the whole difficulty, is not. Caveat 7.66
`cor:prop-from-factor`, `cor:prop-times-cong`, `cor:prop-relations`, `lem:preserve-linked`	STATED	the last two are now consumed by `lem:parallelogram-crucial`'s proof
`lem:typed-above`	STATED	the $t = 1$ case is proved in Caveat 6.29; consumed by the good branch of `lem:parallelogram-crucial`

Statement	Verdict	Note
`lem:maximal-mult`	PROVED	*under the added Hypothesis i*; consumed by nothing since `lem:parallelogram-crucial` acquired its own proof, so the hypothesis is no longer a debt of the chain; Caveats 6.31, 7.67 and 8.24
§8: the main statements		
`thm:main-inductive`(2)	PROVED	modulo the common-term obligation of Caveat 6.13
`thm:main-inductive`(1)	OUTLINED	structure only. The measure *is* well founded, Caveat 8.2(c)
`thm:reductions-safe`	OUTLINED	the BA and C cases follow from `thm:main-inductive`(2); the PC case is `thm:pc-safe`, which is stated only
`thm:pc-safe`	STATED	
`lem:subuniverse-coset`	PROVED	
`thm:codim-one`	OUTLINED	Steps 2–4 are now written out and the linear algebra is correct; what is outlined is Step 1, which rests on `lem:connected`, and two facts taken from the unread half. Caveat 8.10
`lem:expanded-consistency`	IMPORTED	[13, Lem. 6.1]
`lem:tree-coverings`	IMPORTED	[15, Lem. 5.6]
`lem:crucial-irreducible`, `lem:bridge-from-relation`	PROVED	both were paraphrases before; the first had rectangularity as a conclusion where it is a hypothesis, Caveat 8.15
`lem:connected`, `lem:find-consistent`, `lem:bridge-from-instance`, `cor:same-type`	STATED	Caveat 8.26
`lem:minimal-consistent`	PROVED	from Corollary 6.11(m1) and outright minimality; 1-consistency of $\mathcal{I}$ restored as a hypothesis, Caveat 8.13
`lem:parallelogram-crucial`	OUTLINED	the parallelogram property, irreducibility, the one-class fact, the good branch of the typing, and the exclusion of the BA/C corner (via `lem:anchored-intersection`) are proved; what remains is Claim 8.23 and the stated `lem:typed-above`. Two hypotheses restored: subdirectness of R and S-freeness of the $\mathbf{D}^{(1)}_{x_i}$, Caveats 8.22 and 8.24
§9–§10: the algorithm and the hard half		
`def:solve-alg`, `lem:solve-terminates`	PROVED	$\textsc{Solve}_{\mathrm{alg}}$ is now a function, by well-founded recursion on the measure of `def:solve-measure`; the three residual obligations are Caveat 9.6
`thm:t1`	STATED	now has a subject; correctness itself is not proved here
`thm:t2`, `thm:h1`	STATED	need a cost model
`thm:np-membership`	STATED	needs the encoding and the machine; with `thm:h1` it supplies the word *complete*
`lem:nae-interp`	PROVED	the interpretation written out from the two disjoint subalgebras
`thm:h0`	PROVED	from [2, Cor. 1.5.11] and `lem:nae-interp`
`thm:galois`	IMPORTED	Geiger; Bodnarchuk et al.
§1: the statement		
`thm:dichotomy-core`	STATED	the theorem this document is about; what it needs is everything above
`thm:core-reduction`	IMPORTED	[2, Prop. 1.4.3, Thm. 1.4.7, Prop. 1.4.10, 1.4.11]
`lem:wnu-descends`	PROVED	
`thm:dichotomy`	PROVED	from `thm:dichotomy-core`, `thm:core-reduction` and `lem:wnu-descends`; Caveat 1.6

11.3 What this adds up to

The algebraic core of the strong-subuniverse theory (§6–§7) is the part in the best shape: of its sixty-three statements, forty-one are proved here — one of them, `lem:maximal-mult`, only under Hypothesis i — twelve are imported with exact citations, five are outlined, and five are stated only. The two ends of the document, §1 and §§9–10, are now stated rather than gestured at: the theorem is stated for a rigid core and derived for an arbitrary Γ, $\textsc{Solve}_{\mathrm{alg}}$ is a function, `thm:h0` is proved, and *complete* has a membership statement behind it. What is left there is the two complexity wrappers and the correctness of the algorithm itself. §8 is in much worse shape: its spine — `lem:bridge-from-instance`, and with it most of Lemma 8.17 — is stated without proof, and part (1) of the main induction is an outline whose measure is not yet well founded. `lem:parallelogram-crucial` is the exception: its parallelogram half is proved outright, and its typing half is proved up to Claim 8.23 and Lemma 6.28.

Three items block the rest, and they are the honest reading of the table.

1. Claim 8.23 and Lemma 6.28, the residue of the typing half of `lem:parallelogram-crucial`. They replace Hypothesis i in this list: the new proof does not cite `lem:maximal-mult`, so that hypothesis — to which three routes failed, Caveat 7.67 — is no longer on any path to the main statements. The claim cannot be settled at the level of subsets of a single algebra (the three-lines example of Caveat 8.24), which is also why the failed routes failed.
2. The rewrite of `lem:multiply-all-linear`, which is not yet a proof. With `lem:intersection-good` and `lem:self-intersection-pc` both written out, this is what is left of the intersection-and-factorisation layer.
3. Step 1 of `thm:codim-one` and the two facts its Steps 2–4 take from the unread half — the linear algebra itself is now correct (Caveat 8.10); and, for the common binary absorption witness, the check that every producer of a type-BA reduction produces a *uniform* one, now that `thm:main-inductive`(2) hypothesises it (Caveats 5.7 and 6.15). *Three entries stood here and no longer do.* The rectangularity normalisation is Lemma 3.33, and the second normalisation beside it is three lines. The induction measure across expanded coverings is not an open item — no appeal is made at a covering, and Caveat 8.2(c) enumerates the seven that are. The set-valued iteration behind Ω is Proposition 8.5, proved once condition 3 is stated with single weakening steps.

The dependency order among the three large statements is

`thm:stable-intersection` $\longrightarrow$ `lem:bridge-from-instance` $\longrightarrow$ `thm:main-inductive`.

Stable intersection is the sole source of bridges; bridge-from-instance is the only thing that turns them into instance-level structure; the main induction consumes that. Nothing in the reverse direction is used.

11.4 The import ledger

Every statement used and not proved here, with its source and the hypotheses in the form the source states them. Where our statement differs from the printed one — because a hypothesis has been restored, or an alternative added — the difference is noted, and the reasons are in the companion audit document.

Here	**Source**	**Hypotheses, and any difference**
`thm:brady-absorption`	[2, Thm. 3.11.1]	$R \leq_{\text{sd}} \mathbf{A} \times \mathbf{B}$ subdirect and linked, $\mathbf{A}, \mathbf{B}$ finite idempotent Taylor. Stated here in the source's three-alternative form; `lem:linked-implies-abs` is our weaker symmetric *corollary* under properness, and is not Brady's theorem.
`lem:pp-propagation`	[4, Lem. 2.9], [15, Lem. 6.1, Thm. 6.9]	R pp-defined by Φ in which S occurs; $S' <_T S$ with $T \in \{\text{BA}, \text{C}\}$ replaces *every* occurrence. Note the conclusion is *dotted*: replacing a relation by a proper strong subrelation can make the pp-defined result empty, so $R' \dot{\leq}_T R$ unless nonemptiness of R' is separately known. Every use below is in dotted form.
`lem:barto-kazda`	[4, Prop. 2.14], [15, Lem. 3.2]	$B \leq \mathbf{A}$, $n \geq 2$. Stated as printed.
`lem:factor-type`	[15, Lem. 6.8]	$B \leq_{\text{C}} \mathbf{A}$, σ a congruence on $\mathbf{A}$. Only the C case is imported; BA and S are proved here (Caveat 7.8).
`lem:power-to-base`	[15, Lem. 6.24]	R a nontrivial strong subuniverse of $\mathbf{A}_1 \times \cdots \times \mathbf{A}_n$ of type $T \neq \text{PC}$. Already general in T; the case $T = \text{S}$ needs no separate argument (Caveat 7.10).
`thm:central-relation-source`	[15, Thm. 6.15]	$R \leq_{\text{sd}} \mathbf{A} \times \mathbf{B}$, $\mathbf{A}, \mathbf{B}$ finite idempotent. Three alternatives, the third being a nontrivial projective subuniverse of $\mathbf{B}$; `lem:central-relation` eliminates it under Convention 2.1, using also [15, Lem. 3.4].
`lem:possible-intersections`	[15, Lem. 6.25]	$B <_{T_1} \mathbf{A}$, $C <_{T_2} \mathbf{A}$, $B \cap C = \varnothing$, $T_1, T_2 \in \{\text{BA}, \text{C}, \text{S}\}$. Stated as printed.
`lem:ba-center-implies`	[15, Cor. 6.1.2, 6.9.2]	$R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ with $\text{pr}_1(R) = A_1$, and $C_i \leq_{\text{BA}(t)} \mathbf{A}_i$ for a *single* binary t. Both hypotheses are as in [15] and both are needed (Caveat 6.13).
`lem:hobby-mckenzie`	[8]	Abelian iff some congruence of the square has the diagonal as a block; Abelian iff affine, for a finite algebra with a WNU term.
`lem:absorbing-equality`	[13, Lem. 7.2]	$0_{\mathbf{A}} \subseteq \sigma \leq \mathbf{A}^2$, $\omega <_{\text{BA}} \sigma$ *proper.* The equality case needs separate treatment and gets it at the one call site.
`lem:bridge-comp`	[13, Lem. 6.3]	$\sigma_0, \sigma_1, \sigma_2$ *all* irreducible. The hypothesis is easy to drop by accident; `lem:symmetrise` avoids the citation altogether.
`lem:building-perfect`	[13, Cor. 8.17.1]	σ irreducible on $\mathbf{A} \in \mathcal{V}_n$, δ a bridge from σ to σ with $\widetilde{\delta} = A^2$. Used only by `lem:perfect-from-linked`.
`lem:special-wnu`	[10, Lem. 4.7]	w an idempotent WNU of arity n on a finite A. We use only "Clo(w) contains a special idempotent WNU of some arity N", not the printed bound.
`lem:expanded-consistency`	[13, Lem. 6.1]	an expanded covering of a cycle-consistent irreducible instance.
`lem:tree-coverings`	[15, Lem. 5.6]	a 1-consistent instance and a reduction.
`lem:sg-terms`	[5]	standard; the "fixed generator list" phrasing is the one used (Caveat 3.9).
`lem:central-ternary`	[15, Cor. 6.11.1]	a central subuniverse is ternary absorbing.
`thm:galois`	Geiger; Bodnarchuk–Kalužnin–Kotov–Romov	A finite. See Caveat 10.9.

Here	**Source**	**Hypotheses, and any difference**
Essential relations for $n \geq 3$	[15, Lem. 6.11]	$C_i \leq_{\mathrm{C}} \mathbf{A}_i$, $n \geq 3$: no $(C_1, \ldots, C_n)$-essential R. Used once, in `thm:stable-intersection`; its printed proof cites itself where it means the preceding lemma (Caveat 7.63).
Cores and idempotent reducts	[2]	used by Caveat 1.6 and §10.

That is nineteen imported statements. The two that are real work are `thm:brady-absorption` and `thm:central-relation-source`; the rest are short or routine.

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